Simulations of laser-ultrasonic imaging of full and partial welds using COMSOL and SAFT

1.1 Research question and approach

Laser generation and optical reception offer a way to inspect a weld without a contacting ultrasonic probe. The relevant measurement is the surface motion produced by a laser-generated elastic wave after propagation and scattering. Laser synthetic arrays and laser-ultrasonic weld imaging provide experimental precedents for this approach [1, 2]. Here the question is narrower and testable: do physically plausible laser sources and normal-motion receivers contain enough acoustic information to localize an embedded LOF in either a completed or a partly filled weld?

The two weld states differ in more than the amount of material. In the full weld, all acquisition positions lie on a continuous outer surface. In the partial weld, the exposed weld surface lies inside an open groove. It brings some sources close to the flaw, but blocks some direct routes to the outer receivers. A propagation path that is useful in one state may therefore be weak, poorly timed or physically impossible in the other. The study examines source locations, receiving surfaces and imaging paths together.

The workflow has three distinct stages. First, a small thermoelastic COMSOL model calibrates an equivalent laser source without using the flaw. This is done to reduce computation time. Second, COMSOL models propagate that source in flawed and pristine specimens. Third, a separate C++ SAFT calculation focuses the exported recordings according to declared paths and selection rules.

This chapter explains the shared model and construction, then compares the resulting acquisition strategies. The complete simulation descriptions, path comparisons, filter comparisons, settings and numerical controls are provided in appendix A a fully worked out SAFT workflow example is in XX, and can be useful to understand the analysis.

1.2 Geometries, laser acquisition and COMSOL model

1.2.1 Partial and full weld geometry

Figure 1.1 compares the two geometries, which are replicas of the physically acquired samples. Each is 150 mm wide and 25.4 mm thick. Depth z is measured downwards from the original outer surface; $x = 0$ is the weld centre. The J preparation has a 7.1068 mm opening, a 4.7 mm entrance section, shallow 3° flanks and a rounded root reaching approximately 24.061 mm depth. The internal weld outline is bonded to the surrounding material except at the prescribed separation. The full weld is filled to $z = 0$. The 9 mm partial fill leaves an exposed weld surface at $z = 16.4$ mm, between approximately $x = \pm 2.940$ mm.

The LOF lies on the right flank. Its upper and lower tips are at $(x, z) = (2.831875, 18.467475)$ and $(2.700855, 20.967475)$ mm. Its vertical extent is 2.5 mm and its length along the flank is 2.5034 mm. The lower tip remains 4.4325 mm above the backwall. The flaw is represented by traction-free faces, meaning no contact closure or friction is included. This same segment is used in both weld states, which isolates the effect of the acquisition geometry and fill state rather than introducing a different defect.

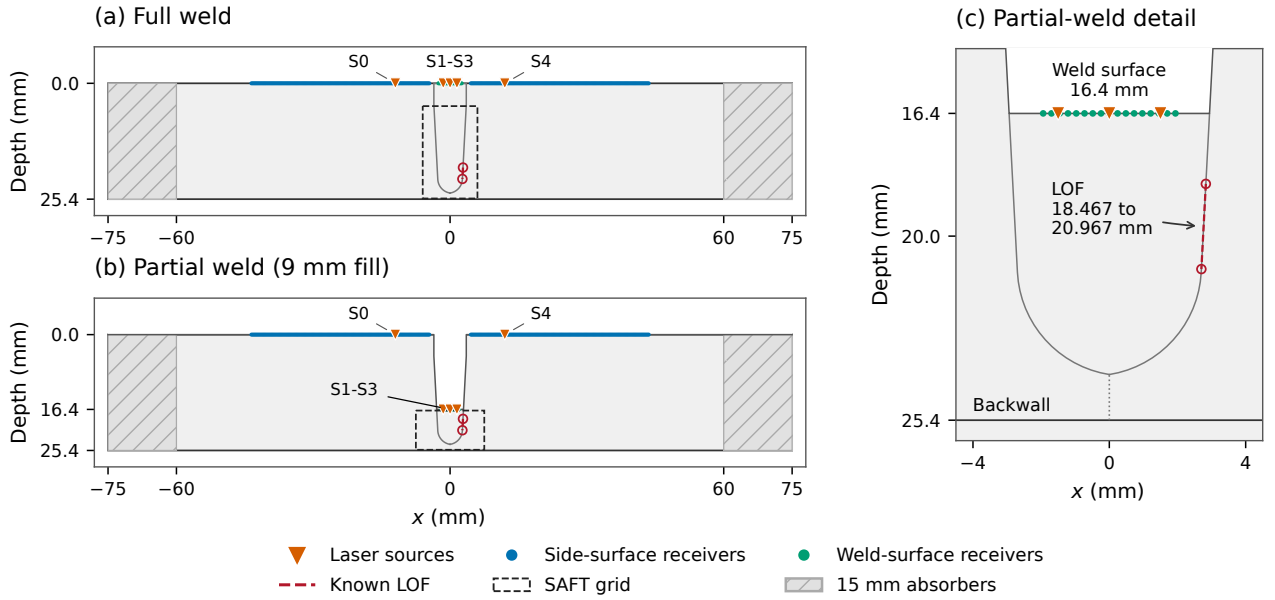


Figure 1.1: Specimen geometry and acquisition. Orange triangles identify the two shoulder sources S0/S4 and three weld-surface sources S1–S3. Blue and green points denote side-surface and weld-surface receivers. The specimens include 15 mm absorbing strips at each lateral end. Separate dashed boxes show the actual full- and partial-weld reconstruction fields. The enlarged partial-weld detail shows the exposed floor, continuous J root and prescribed LOF. The grey internal outline is bonded except along the red flaw segment; it is not an open groove boundary below the fill.

The five source coordinates are $x = -12, -1.5, 0, 1.5, 12$ mm. S0 and S4 are shoulder sources at the original outer surface. S1–S3 lie on the exposed weld surface: $z = 0$ in the full weld and $z = 16.4$ mm in the partial weld. Both specimens have 157 receivers on each side surface at 0.25 mm pitch, extending to approximately $x = \pm 43.5534$ mm. There are also 23 full-weld or 17 partial-weld receivers on the central weld surface, giving 337 and 331 receivers respectively. Every receiver measures the vertical velocity v_y.

Five flawed and five pristine source simulations are made for each weld state: twenty production solves in total. This means each source is separately modelled and they can't affect each other. Every recording contains 5001 samples over 50 μs at 10 ns output spacing. A pristine simulation has the same geometry and source but no LOF. It provides control and allows for signed waveform subtraction. It is also not an input to the LOF-only defect imaging algorithm. This is important because experimentally a pristine sample analysis will be very hard to require. Therefore abusing this information in the simulations is not useful.

1.2.2 Elastic propagation

Simulations are done in COMSOL Multiphysics 6.4, with the Elastic Waves, Time Explicit interface. Steel is homogeneous, isotropic and linearly elastic, under two-dimensional plane strain. The model uses $E = 212.443$ GPa, $\nu = 0.288978$ and $\rho = 7859.975$ kg m^{-3}, giving longitudinal and in-plane

shear speeds of $c_L = 5943.60$ and $c_S = 3237.97\ \mathrm{m\,s^{-1}}$. The corresponding Rayleigh speed used for arrival screening is $c_R = 2997.33\ \mathrm{m\,s^{-1}}$. Longitudinal motion is primarily parallel to propagation; shear motion is transverse. Both can contribute to measured normal surface velocity after interaction with a free surface.

The elastic field satisfies

$$\rho\,\ddot{\mathbf{u}} = \nabla\cdot\boldsymbol{\sigma}, \qquad \boldsymbol{\sigma} = \lambda\,\mathrm{tr}(\boldsymbol{\varepsilon})\mathbf{I} + 2\mu\boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} = \tfrac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^{\top}). \tag{1.1}$$

Here $\mathbf{u}$ is displacement and λ, μ are the Lamé constants. Exposed surfaces and the LOF are traction-free except where the equivalent source acts. Bonded internal boundaries transmit displacement and traction. Weld and surrounding steel have identical properties. Absorbing strips at the lateral ends reduce artificial returns from the truncated specimen, that would not be present in a round pipe.

The numerical discretization uses quartic velocity and strain fields, a discontinuous Galerkin formulation and explicit fourth-order Runge–Kutta integration. Maximum element sizes are approximately 0.32380 mm in the weld and 0.48570 mm in the surrounding material, chosen using a 5 MHz simulation resolve frequency. Higher frequency could contain more information, however experimentally they attenuate faster [3]. Simulation runtime is also an important consideration, higher frequencies require smaller meshes and can cause smaller stable time steps [4]. For our models runtime scales as

$$T(f) \approx T(5\,\mathrm{MHz})\left(\frac{f}{5\,\mathrm{MHz}}\right)^{3}. \tag{1.2}$$

This means increasing the frequency to 10 MHz leads to approximately eight times the runtime.

1.2.3 Laser-equivalent source

The laser source is represented by an equivalent tangential traction that approximates thermoelastic expansion from an absorbed laser line, following the surface-force-dipole approach described by Tu et al. [5]. Its amplitude is calibrated against a thermoelastic calculation. The calibration uses a flat domain measuring 8 mm in width and 4 mm in depth. COMSOL Heat Transfer in Solids connects the temperature field to Solid Mechanics through thermal expansion. A second mechanical field on the same mesh receives the equivalent traction.

For the Gaussian spatial profile, define $\xi = (x - x_s)/w$, where x_s is the source centre and $w = 0.5$ mm is the prescribed $1/e^2$ half-width. The temporal profile uses the 6.5 ns pulse duration reported for the laser system of Yu et al. [6]. Here, this duration is interpreted as the Gaussian full width at half maximum. The absorbed heat flux is

$$q(x,t) = \frac{F_0}{\sqrt{2\pi}\sigma_t}\exp(-2\xi^2)\exp\!\left[-\frac{(t-t_0)^2}{2\sigma_t^2}\right], \qquad \sigma_t = \frac{6.5\ \mathrm{ns}}{2\sqrt{2\ln 2}}. \tag{1.3}$$

The pulse centre is prescribed at $t_0 = 30$ ns so that the short Gaussian pulse lies inside the simulated time window. The parameter F_0 denotes the peak absorbed fluence, set to $100\ \mathrm{J\,m^{-2}}$. The source half-width and absorbed fluence are choices.

The equivalent surface load is a tangential dipole,

$$t_x(x,t) = 4A\xi e^{-2\xi^2} H(t), \qquad t_y = 0, \qquad H(t) = \tfrac{1}{2}\left[1 + \operatorname{erf}\left(\frac{t - t_0}{\sqrt{2}\sigma_t}\right)\right], \tag{1.4}$$

with

$$A = \gamma \frac{E\alpha F_0}{(1-\nu)\rho c_p w}. \tag{1.5}$$

The opposing tangential lobes represent the lateral expansion of a thin heated region. The function $H(t)$ is the time integral of the normalized Gaussian power pulse, consistent with the equivalent-source formulation [5]. Retaining the finite source width accounts for its influence on the angular and frequency response.

The thermal calculation starts from $T_0 = 293.15$ K. Thermal conductivity is 52 $\mathrm{W\,m^{-1}\,K^{-1}}$, specific heat capacity is $c_p = 480$ $\mathrm{J\,kg^{-1}\,K^{-1}}$, and the linear thermal expansion coefficient is $\alpha = 11 \times 10^{-6}$ $\mathrm{K^{-1}}$. These three material properties are taken from the X65 data reported by Dölz et al. [7]. The elastic modulus E, Poisson ratio ν and density ρ are retained from the weld model.

Only the scalar gain γ is fitted. Signed horizontal and vertical velocity histories at five points on a 2 mm radius arc, from 15° to 75°, are compared over 0.12–0.82 μs. Both fields receive the same forward–backward fourth-order 5 MHz Butterworth filter for this calibration comparison. If $\mathbf{d}$ is the vector of filtered unit-gain dipole histories and $\mathbf{v}$ the corresponding thermoelastic histories, the least-squares result is

$$\gamma = \frac{\mathbf{d}^\top \mathbf{v}}{\mathbf{d}^\top \mathbf{d}} = 1.084434. \tag{1.6}$$

Figure 1.2 compares the dipole source with the thermoelastic simulation. After fitting a single amplitude factor, the relative difference between their velocity waveforms is 0.931%. At two additional receiver positions that were not used in the fit, the differences are 1.09% for the axial receiver and 1.27% for the surface receiver. This agreement supports using the simpler dipole source within the tested frequency band.

At the prescribed absorbed fluence of 100 $\mathrm{J\,m^{-2}}$, the calibrated amplitude coefficient is $A = 188.939$ kPa. This gives a maximum force per unit surface area of 229.195 kPa.

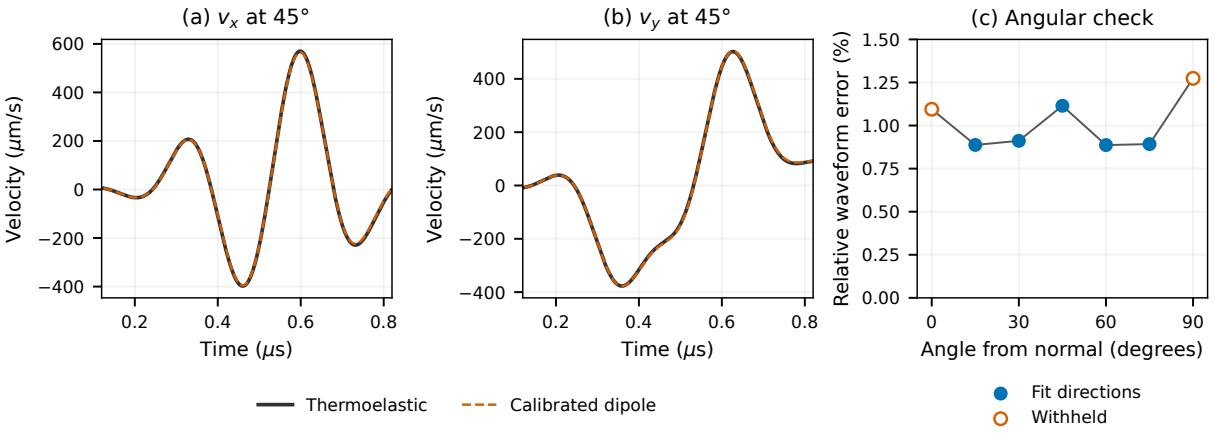


Figure 1.2: Independent source calibration. The equivalent traction is assessed through comparisons of signed outgoing velocity histories and angular response with the thermoelastic calculation. A single gain scales the dipole, without using a reconstructed flaw image. Agreement supports using this dipole as a substitute laser generation

The thermal mesh resolves the thin heated layer with nine layers totalling 25.55 μm. The calibration uses implicit generalized-α integration, with maximum steps of 0.5 ns initially and 2 ns after. Its absorbed energy agrees with the Gaussian input within 0.000639%, and the temperature rise at 100 J m^{-2} is 78.34 K. These checks and the outgoing waveform comparison support transferring the equivalent source to the larger weld model. Calibration of a physical laser instrument remains a separate requirement and could not be done since a generation laser was not obtained.

1.3 SAFT imaging

SAFT asks whether different receiver histories contain motion at the times expected if ultrasound scattered at a trial pixel $\mathbf{p}$. For each source s, receiver r and chosen path P, it predicts

$$T_{sr,P}(\mathbf{p}) = t_0 + \sum_{j \in P} \frac{\ell_j(\mathbf{p})}{c_{m_j}}, \tag{1.7}$$

where each leg has length ℓ_j and mode $m_j \in \{L, S\}$. The path notation places a vertical bar at the trial scattering point. Thus $S \mid S$ is direct shear illumination and direct shear reception; $S \mid SS$ adds one backwall reflection on the receiving route; $SS \mid S$ adds it on the incident route. $L \mid L$ is the direct longitudinal view.

Every segment must remain in steel and every reflection must lie on an actual boundary. In the partial weld, a ray cannot cross the open groove. Valid routes use the original surface, the exposed fill floor or the backwall. The source and receiver sets are evaluated at every image pixel in the SAFT grid.

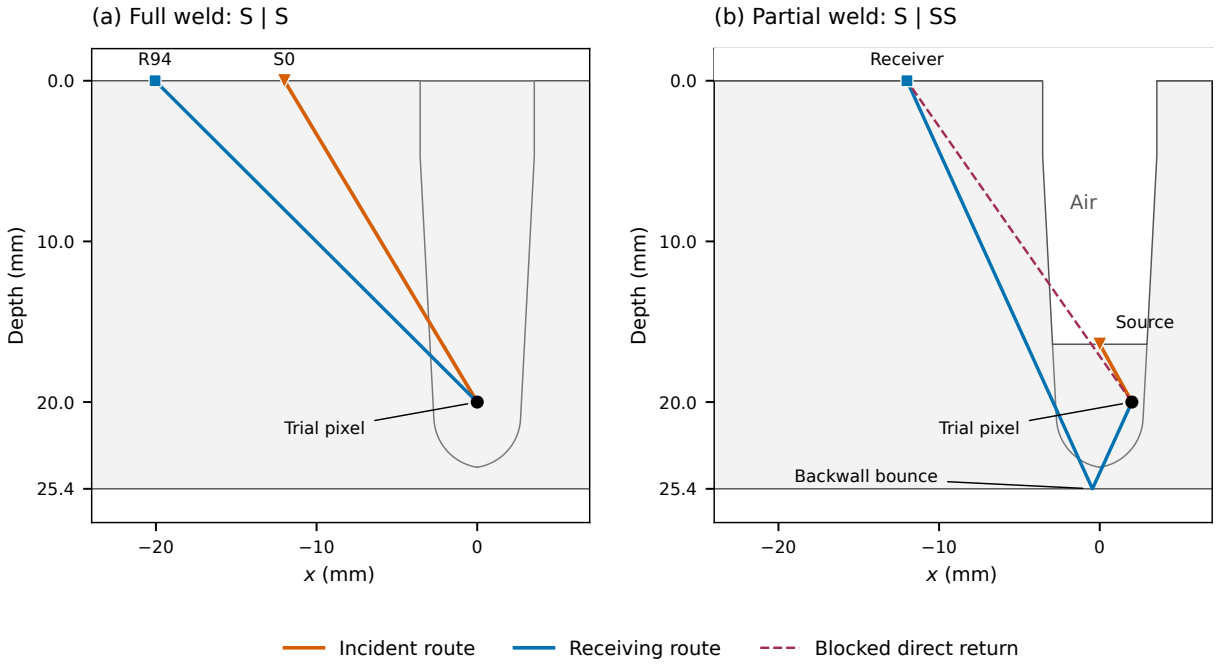


Figure 1.3: Illustrative paths. (a) A full-weld $S \mid S$ route from a shoulder source to a trial pixel and side receiver. (b) In the partial weld, direct illumination from the weld surface and a backwall-reflected return can remain inside steel even when a direct return crosses the open groove.

For example, a full-weld source at $(-12, 0)$ mm, receiver at $(-20.0534, 0)$ mm and pixel at $(0, 20)$ mm give a direct-S delay of about 15.98 μs; the same lengths give about 8.72 μs for direct L. In the partial-weld example, a source at $(0, 16.4)$ mm illuminates $(2, 20)$ mm and returns through a backwall bounce to an illustrative receiver at $(-12, 0)$ mm. The incident length is 4.118 mm and the reflected receiving length is 33.833 mm, giving $T = 11.751$ μs. The direct return would cross the fill depth at $x = -0.52$ mm, inside the open groove, and is invalid. The reflected return crosses that depth at about $x = -7.91$ mm, in steel.

1.3.1 Generation and detection angles

A laser dipole does not launch all modes equally in all directions. A normal-motion receiver also does not observe every incoming mode equally. Free-surface reflection and conversion contribute to its response, so normal sensitivity cannot be estimated from the incident displacement component alone. Therefore, we use the angular magnitudes in Figure 1.4 to admit plausible paths.

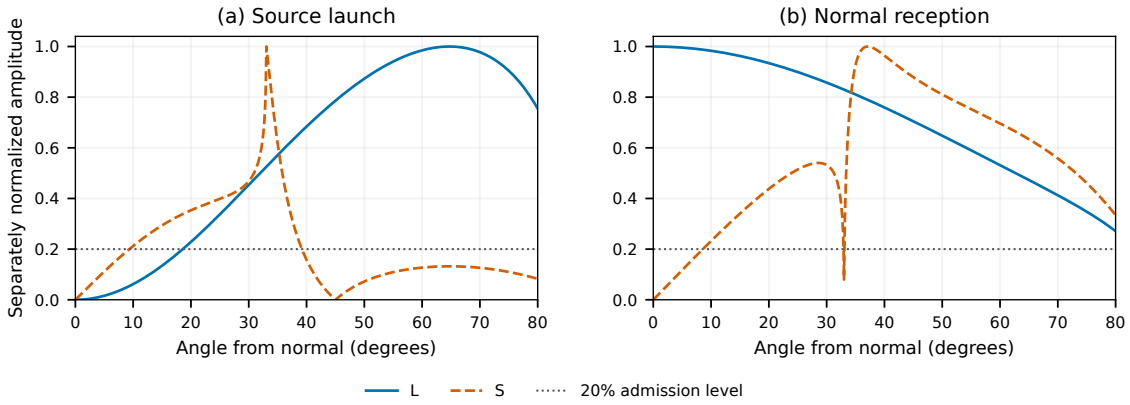


Figure 1.4: Angular launch and normal-reception magnitudes used for channel admission. L and S are normalized separately, so the curves do not compare absolute longitudinal and shear energy. The 20% level is a selection threshold: a channel must satisfy both its source and receiver tests. Admitted receivers are subsequently averaged equally. These curves explain why a large spatial aperture can still contain unsuitable mode/path combinations.

For a path to qualify, both normalized launch and receive magnitudes must reach 20%, and receive angle must not exceed 80° from the normal. A reflected route must also pass a 5% wall-response-product gate. These are modelling choices, not amplitude weights or phase corrections. A 1.5 mm source–receiver separation exclusion keeps receivers outside the approximately 1.425 mm source-footprint half-width.

1.3.2 Frequency filtering and pulse-aware screening

The recorded field includes direct bulk waves, surface motion, reflected packets and flaw scattering. A strong direct packet can be sampled at a delay assigned to a deep trial pixel and create an apparent indication. Its coherence across receivers means that ordinary averaging, or even coherence weighting, may preserve it. This contamination is therefore a mismatch between the packet's actual route and the route assumed in focusing.

The basic SAFT uses an analytic FIR filter over 0.5–5 MHz. It reduces slowly varying motion and high-frequency content while retaining amplitude and phase for delay-and-sum processing. The upper cutoff follows the frequency range used for mesh design and source calibration. The Kaiser design targets 55 dB stopband attenuation with a 0.5 MHz transition width, giving 657 taps and

$\beta = 5.10226$ [8, 9]. At 10 ns sample spacing, filtering requires 3.28 μs of data on either side of each sample. Therefore, usable time ranges are 3.28–46.71 μs. This filter is again a design choice and not established as the optimal, however different filters were also investigated.

Direct L, S and Rayleigh arrivals are then screened per source–receiver pair. A geometry-valid prediction locates an envelope peak within ± 0.25 μs. The packet window extends until the envelope is below -30 dB relative to that peak for at least 0.15 μs, with a further 0.1 μs margin. An imaging sample falling within the excluded window is not admitted. This avoids a single fixed mute duration for packets with different widths. Rayleigh screening requires a common unobstructed flat surface; it does not assume a surface wave crosses the empty groove.

These rules affect the geometries differently. All 1685 full-weld source–receiver pairs have direct bulk paths. Only 365 of 1655 partial-weld pairs have direct paths: shoulder-to-side pairs on their own side and weld-surface pairs on the floor.

The screening windows for an LOF-only image are obtained from its own recording. A pristine image uses its own windows; the waveform-difference control uses the LOF-derived windows. The procedure therefore requires no reference recording to suppress the direct field. It can still reject real flaw information when arrivals overlap, especially for a source close to the partial-weld flaw.

1.3.3 Coherence and source coverage

Let $a_{sr}(\mathbf{p})$ be the complex filtered history interpolated at Equation 1.7, after the channel and timing tests. For N_s admitted receivers, the equal mean and receiver coherence factor are

$$M_s(\mathbf{p}) = \frac{1}{N_s} \sum_r a_{sr}(\mathbf{p}), \qquad \mathrm{CF}_s(\mathbf{p}) = \frac{|\sum_r a_{sr}(\mathbf{p})|^2}{N_s \sum_r |a_{sr}(\mathbf{p})|^2}. \tag{1.8}$$

The coherence factor is zero for zero signal energy and otherwise lies between zero and one. Aligned phases reinforce in the numerator; inconsistent phases reduce it. It encourages phase consistency under the selected delay law, but a coherent structural echo can also pass that test.

Each source must retain at least eight receivers and at least 25% of its aperture eligible before the time-support and pulse tests, again these are choices. Furthermore, the image requires at least two qualifying sources at a pixel. If $\mathcal{S}(\mathbf{p})$ is that set, source results combine as

$$I(\mathbf{p}) = \left[\frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} |\mathrm{CF}_s M_s|^2 \right]^{1/2}. \tag{1.9}$$

This root-mean-square combination avoids cancellation between separately illuminated source views. The output retains velocity units but is a processed focusing score, not acoustic intensity or flaw size. A pixel failing the required source count is unsupported and is coloured grey.

Allowing one qualifying source changes which pixels can be displayed. It does not mean acquiring only one laser shot for the whole image: all selected source shots remain available wherever they qualify, and Equation 1.9 still uses them. This distinction is essential to the partial-weld coverage comparison below.

1.3.4 SAFT images and controls

The full-weld SAFT grid spans $x = \pm 6$ mm and depths 5–25.25 mm, the partial-weld grid spans $x = \pm 7.5$ mm and 16.65–25.25 mm depth. Images use 0.1 mm pixels. At 5 MHz the shear wavelength is approximately 0.648 mm, so the pixel grid samples the image finely but does not establish 0.1 mm physical resolution. Finer 0.05 mm grids and wider $x = \pm 45$ mm fields check peak stability and competing indications.

The images show $20 \log_{10}[I(\mathbf{p})/I_0]$ over a declared display range: 45 dB for the principal path and source-threshold panels, and 40 dB for the partial-weld acquisition comparison. A shared reference is used within each comparison. Black denotes low displayed amplitude, grey denotes unsupported material, and white denotes air. External LOF guides identify the flaw location.

Define a near-flaw region $\mathcal{N}$ within 0.5 mm of the flaw and a background region $\mathcal{B}$ more than 1 mm from it. Over supported pixels,

$$C_{\max} = 20 \log_{10} \frac{\max_{\mathcal{N}} I}{\max_{\mathcal{B}} I}, \qquad C_{99} = 20 \log_{10} \frac{\max_{\mathcal{N}} I}{Q_{99,\mathcal{B}}(I)}. \tag{1.10}$$

Positive $C_{\max}$ means the strongest near-flaw response exceeds every assessed background pixel. Changing an aperture changes both the signal and the assessed region; comparisons therefore report support and, where needed, restrict both images to a common mask.

The LOF-only image is the main result. Pristine images test whether the reference produces a similar indication. Difference controls subtract signed LOF and pristine waveforms before filtering and focusing. Since interference and coherence change, a difference image can have a larger peak than the LOF-only image. These controls support attribution to the flaw.

1.4 Full weld results

Figure 1.5 shows SAFT images for different paths with the same settings. All four paths produce a global maximum within 0.37 mm of the prescribed LOF and support the entire evaluated near-flaw region. Their $C_{\max}$ values are 8.85 dB for $L \mid L$, 6.47 dB for $S \mid S$, 4.65 dB for $S \mid SS$ and 4.37 dB for $SS \mid S$. The images provide complementary views of the defect. Tweaking of the background controls and combining these images could allow for defect sizing, but this is not done in this analysis.

The direct images mainly emphasize the tips. A nearly vertical defect does not necessarily return a strong reflection to the generating surface, as tip scattering and oblique illumination matter. More extended reflected-path indications are consistent with different angular access, as in backwall-assisted imaging strategies [10].

1.4.1 Direct-shear interpretation

For direct S, changing the 657-tap common bandpass to a 257-tap bandpass with the same nominal edges increases $C_{\max}$ from 6.47 to 10.26 dB. The 257-tap 0–5 MHz low-pass gives 12.79 dB. All use the same 3.28 μs admission guard, so this gain is not obtained by simply allowing earlier samples. Filter duration and retained frequency content change the waveforms being summed and their overlap with unwanted arrivals. The choice is path-specific: direct L gives only 3.24 dB with the short low-pass, compared with 8.85 dB under the FIR bandpass.

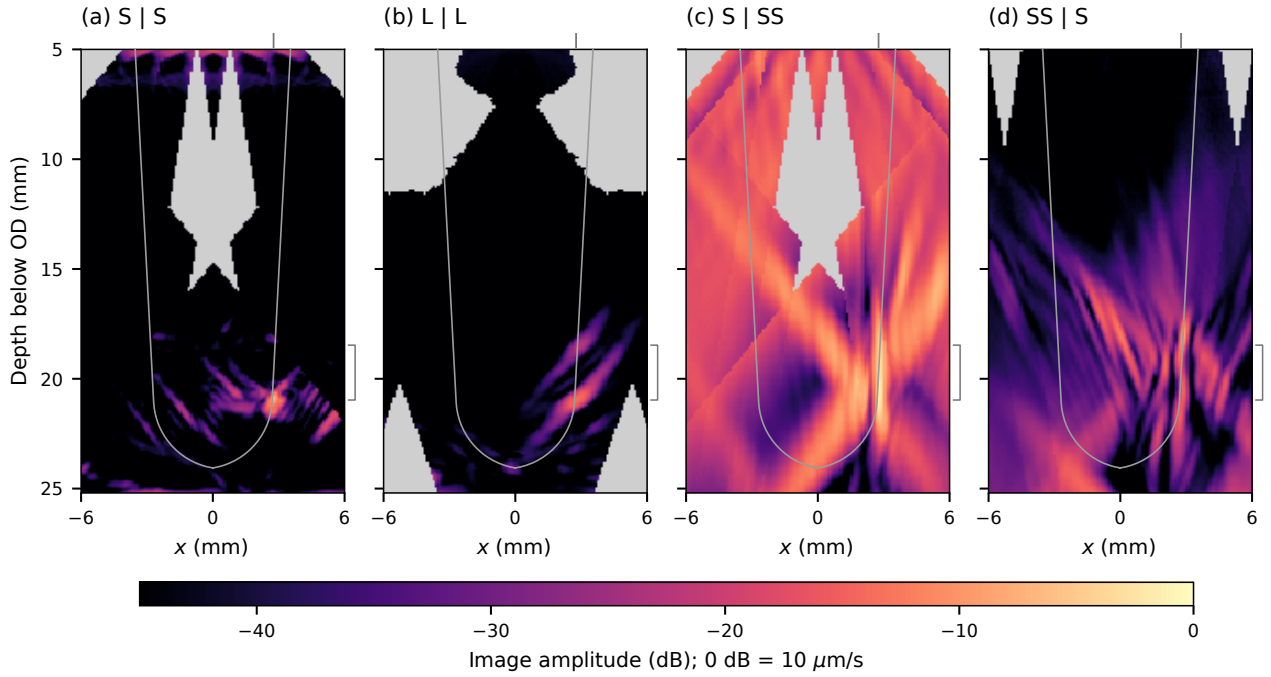


Figure 1.5: Full-weld LOF-only images for four paths using the same 657-tap 0.5–5 MHz filter, screening, coupling and support rules. All panels use 0.1 mm pixels, 0 dB = 10 μm/s and a 45 dB range. Direct paths emphasize the lower-tip vicinity; reflected paths give more extended indications. Grey pixels fail the declared coverage rules. Different paths admit different rays even when their selection rules are identical.

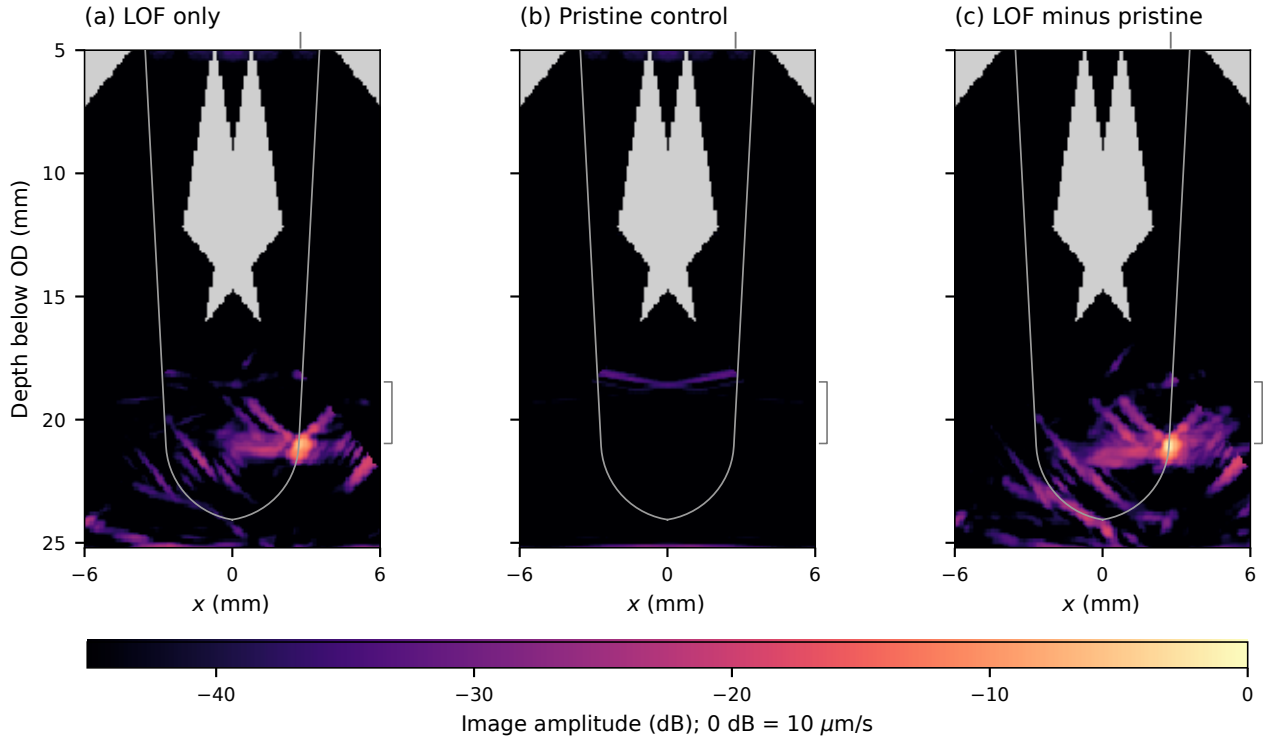


Figure 1.6: Selected full-weld $S \mid S$ image, pristine control and signed-waveform-difference control. The short 0–5 MHz analytic FIR is used with the retained 3.28 μs guard. All panels have 0 dB = 10 μm/s and a 45 dB range. The LOF-only image localizes near the lower tip without reference subtraction; the pristine panel has no physical flaw. External marks show LOF location.

Figure 1.6 compares the direct S SAFT images. The LOF-only peak is 5.184 μm/s at $(2.7, 21.0)$ mm, 0.0325 mm from the finite LOF segment. $C_{\max} = 12.79$ dB, $C_{99} = 21.28$ dB and near-flaw support is 100%. The pristine maximum is 0.485 μm/s and not near the defect. The difference maximum is 8.791 μm/s near the lower tip. The three panels together support a defect localisation in the LOF only image.

At a smaller pixel space of 0.05 mm spacing, contrast is 12.63 dB. Over a wider SAFT grid of $x = \pm 45$ mm field with 0.25 mm spacing, it is 10.62 dB and the global maximum remains near the flaw. The selected direct-S interpretation is therefore not restricted to one SAFT grid or pixel interval. This does not hold for every path. The $SS \mid S$ path has a stronger lateral artefact in its wider field, giving -17.80 dB contrast. Additional paths should be retained for useful geometric information, rather than combined them without care.

1.4.2 Source and receiver sensitivity

Table 1.1 summarizes tests that remove whole source or receiver groups with the selected processing rules. These tests ask which surfaces carry useful information, rather than finding individual combinations that maximize the known-flaw score.

Table 1.1: Full-weld acquisition reductions for selected direct S. Side receivers comprise the two 157-position outer groups. The contrast uses each configuration's supported field, so support must be read alongside it.

Sources	Receivers	$C_{\max}$ (dB)	Near support (%)
All five	All 337	12.79	100.0
Two shoulders	All 337	10.57	92.4
All five	Side surfaces	11.83	100.0
Two shoulders	Side surfaces	9.61	92.4
Three weld-surface	All 337	-20.66	9.8

Shoulder generation and side-surface reception retain a useful indication at 9.61 dB with 92.4% near-flaw support. Generation or reception directly over the full weld is therefore not essential for this defect. Conversely, the three weld-surface sources alone provide poor direct-S coverage and negative contrast. Suitable oblique launch and receive angles, rather than source proximity alone, determine the useful aperture.

1.5 Partial weld results

Direct shear illumination followed by a backwall-reflected shear return, so $S \mid SS$, provides a useful view of the partial-weld defect. As shown in Figure 1.3, sources on the weld surface can illuminate the nearby LOF, while the reflected return reaches outer receivers without crossing the empty groove. The results first establish the effect of source and receiver selection using the original equal-averaging method. They then show how revised processing improves contrast while retaining a good near-flaw support.

1.5.1 Path and filter comparison

Our analysis compared eighteen direct and reflected paths. With all five sources and all 331 receivers and the common 0.5–5 MHz filter, $S \mid SS$ image gives $C_{\max} = 4.36$ dB and 95.8% near-flaw support. Its peak is 16.568 μm/s at $(2.8, 19.95)$ mm, 0.046 mm from the LOF. The pristine maximum lies

elsewhere, while the signed-waveform-difference image also has a strong response near the defect. Changing only the filter gives similar results in the LOF-only images: 4.20 dB for 1–5 MHz, 4.29 dB for a shorter bandpass and 4.34 dB for a shorter low-pass. Shoulder sources alone provide no supported near-flaw image under the rules. With weld-surface sources and both receiving sides, $S \mid SS$ is the only one of the eighteen paths with positive contrast and a maximum near the flaw.

1.5.2 Source and receiver sensitivity

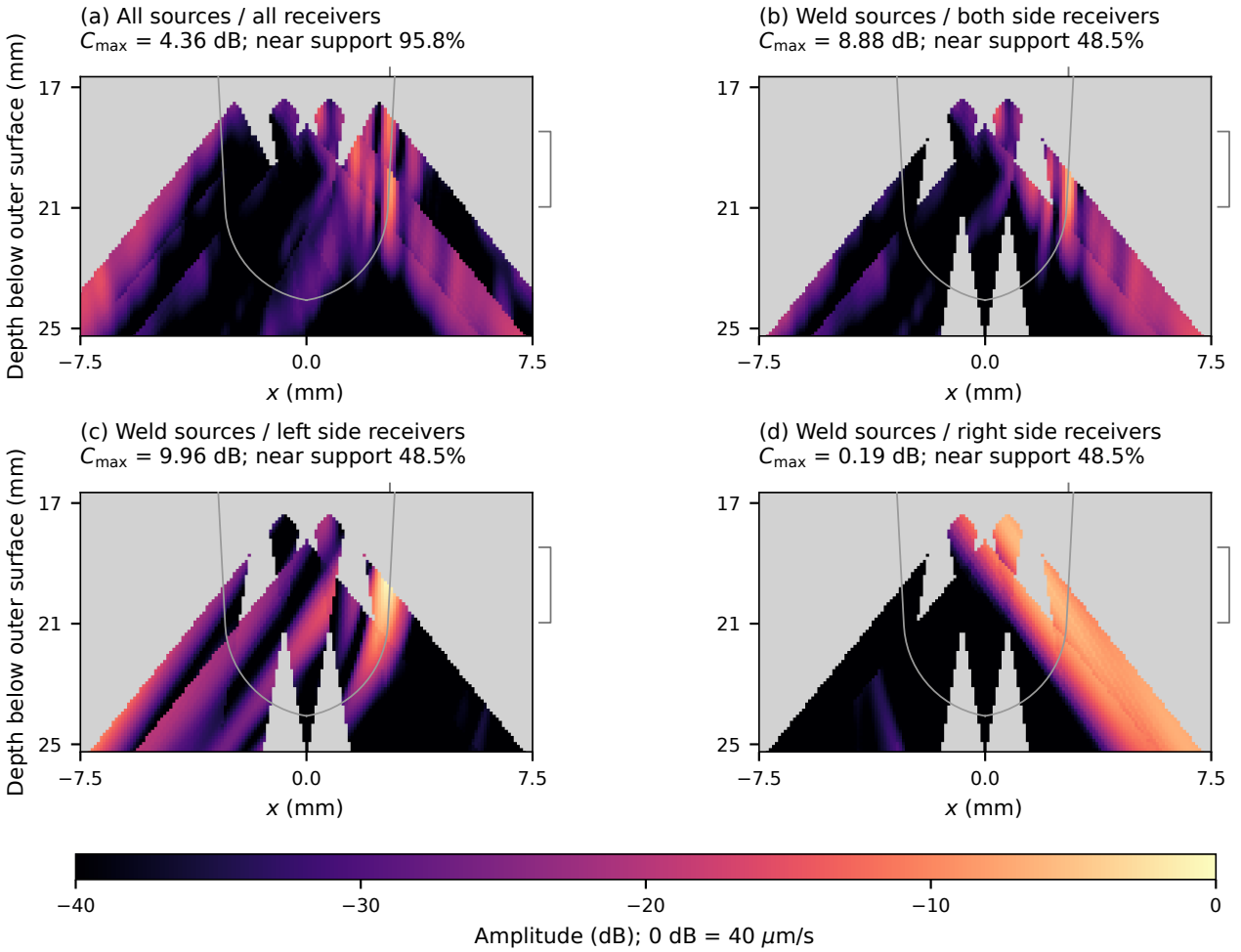


Figure 1.7: $S \mid SS$ acquisition comparison with equal averaging and a two-source minimum. All panels use 0 dB = 40 μm/s and a 40 dB range. Restricting generation to the three weld-surface sources reduces the supported region, shown in grey.

With the three weld-surface sources, left-side reception gives 9.96 dB contrast, both sides 8.88 dB and the right side 0.19 dB. All three support only 48.5% of the near-flaw region (Figure 1.7 and Table 1.2). The right-side maximum is 0.664 mm from the flaw, outside the defined near region, so its slightly positive contrast is not sufficient evidence of useful localization.

Some of the apparent benefit of left-only reception comes from leaving difficult pixels unassessed. On the same supported pixels, the all-source image gives 7.02 dB, making the left-side advantage 2.94 dB rather than the 5.60 dB difference between the unrestricted scores. Left-side contrast remains 10.26 dB at 0.05 mm spacing and 9.11 dB in the wider 0.25 mm field; both-side reception gives 9.20 dB in that wider field. Left reception is therefore useful for this acquisition, but is not generally superior to the other side.

Table 1.2: Original partial-weld $S \mid SS$ results with equal receiver averaging, the common filter and 0.1 mm pixels. "Weld" means the three exposed weld-surface sources. Near support measures the assessed fraction of the region within 0.5 mm of the flaw, not the fraction of flaw length detected.

Sources	Receivers	Minimum sources	$C_{\max}$ (dB)	Near support (%)
All five	All 331	2	4.36	95.8
All five	All 331	1	2.15	100.0
Weld	Both sides	2	8.88	48.5
Weld	Both sides	1	3.06	95.8
Weld	Left side	2	9.96	48.5
Weld	Left side	1	3.72	95.8
Weld	Right side	2	0.19	48.5

Lowering the required number of qualifying sources from two to one increases left-side near support from 48.5% to 95.8%, but lowers contrast from 9.96 to 3.72 dB. The acquisition and all other processing settings remain fixed. Previously supported pixels are unchanged, and all new supported pixel have exactly one qualifying source. All three weld-source shots are still used wherever they qualify, so it's not that only one source is used, all of them are used but only one contribution is required.

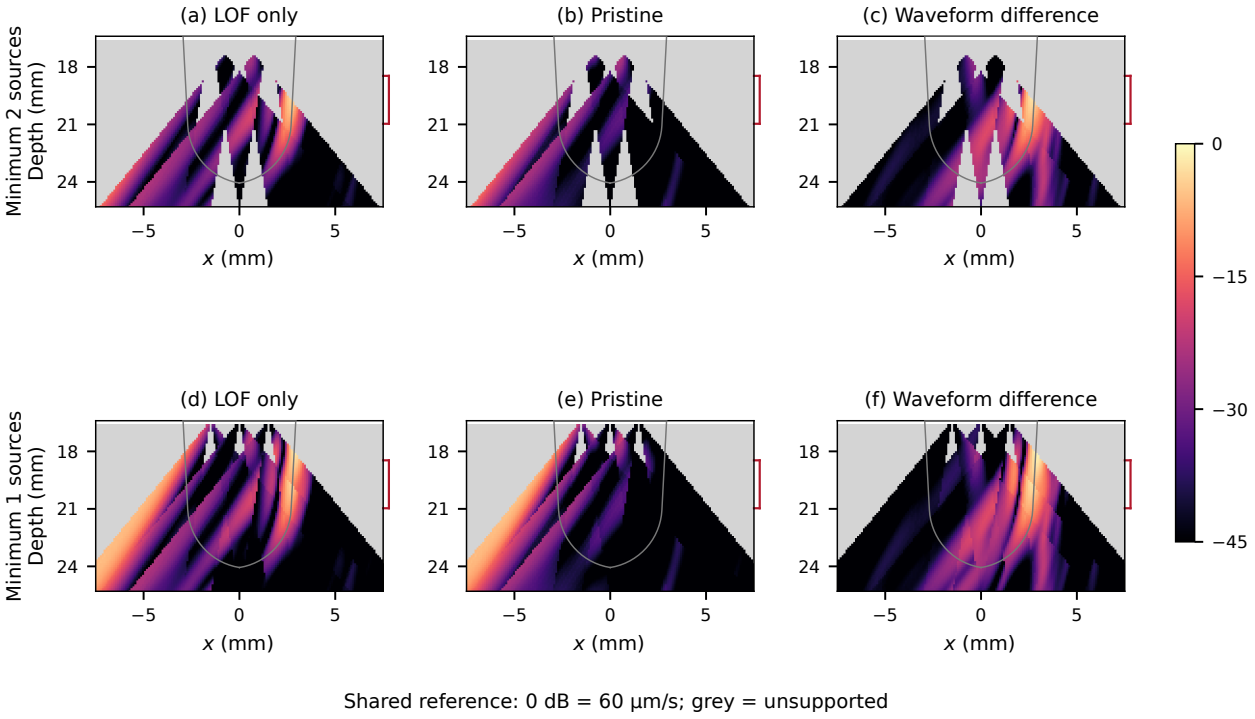


Figure 1.8: Two-source and one-source requirements for the original $S \mid SS$ image, using three weld-surface sources and 157 left receivers. Columns show defect, pristine and signed-waveform-difference images. All panels use 0 dB = 60 μm/s and a 45 dB range. The one-source rule adds coverage and background without changing previously supported pixels.

The one-source image reveals a stronger response near the upper tip, peaking at 46.152 μm/s at $(2.9, 18.15)$ mm. It also admits a lateral background peak of 30.066 μm/s at $(-7.0, 23.45)$ mm. The pristine maximum lies away from the flaw, with -37.53 dB contrast, while the difference image gives 10.02 dB. These controls support a flaw-related contribution, but the brighter near-flaw response does not make the image cleaner.

The left-side one-source contrast remains 3.73 dB at 0.05 mm spacing and 3.22 dB in the wider field, where near support is 94.3%. Both-side reception gives 4.87 dB in that wider field. With all five sources and all receivers, allowing one source fills the near region but lowers contrast from 4.36 to 2.15 dB. Relaxing the source rule therefore trades a cleaner, smaller image for broader coverage. Near support is not the fraction of flaw length detected.

1.5.3 Weighted receiver averaging

Figure 1.9 shows an improved $S \mid SS$ image that uses all five sources and all receivers. Instead of averaging equally, it weights and phase-corrects the receiver samples using the predicted wall and normal detection values. The filter remains 657 taps long, but its band changes to 1–5 MHz and the angular threshold decreases from 20% to 15%. The two-source minimum, receiver-retention rules, coherence exponent of one, RMS source combination and direct-arrival screening are still the same as before.

Using these settings, contrast reaches 9.09 dB with 96.9% near-flaw support on a 0.025 mm grid. A further improvement comes from a narrow mute around the predicted source–backwall–receiver SS path, which does not pass through the trial pixel. Samples within ± 0.225 μs of this arrival are removed, with a smooth transition to full weight at ± 0.30 μs. Only this additional fixed mute is enabled.

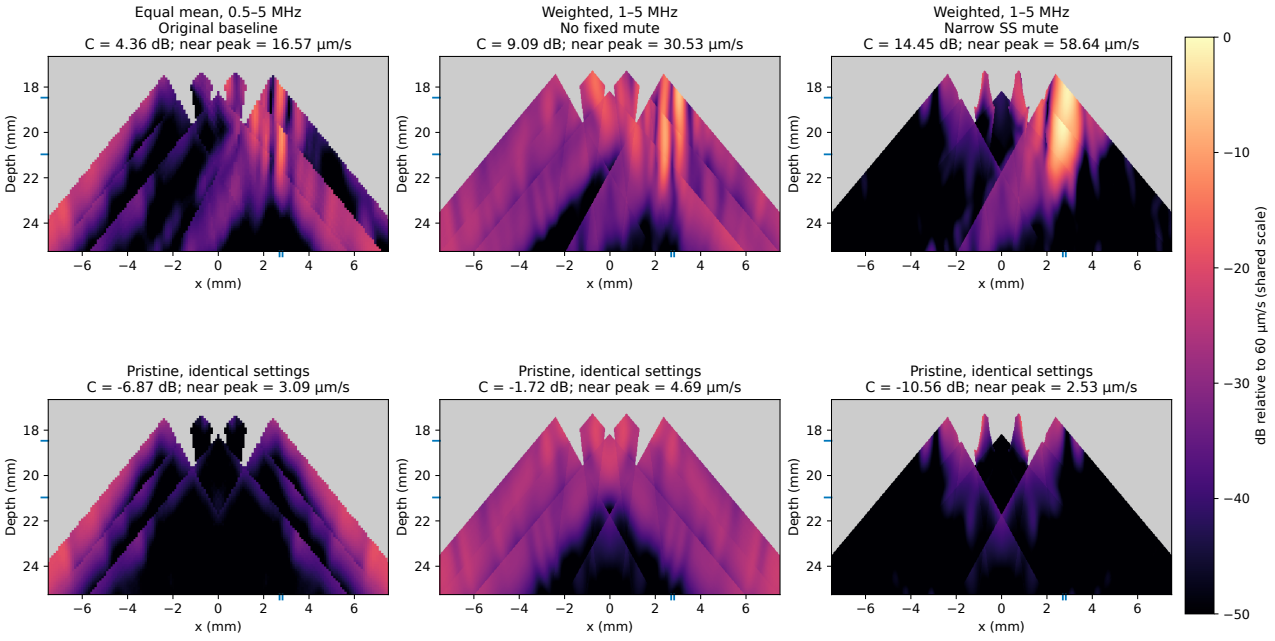


Figure 1.9: Partial-weld $S \mid SS$ improvement with pristine controls on a shared amplitude scale. Left: original equal-average image at 0.1 mm spacing. Centre: response weighting, 1–5 MHz and a 15% angular threshold at 0.025 mm. Right: the same settings with the narrow SS mute. Grey regions are unsupported.

The mute raises contrast to 14.45 dB while retaining 96.9% near support. The maximum is 58.64 μm/s at $(2.700, 19.625)$ mm, 0.071 mm from the flaw. The pristine near-region maximum is 2.53 μm/s and its global maximum lies elsewhere. Muting changes phase cancellation and coherence, therefore the stronger image peak is an analysis improvement and not an increase in physical scattered energy. The mute was selected using this known flaw and does not establish that the entire reflected packet has been removed.

1.5.4 Other paths

The full analysis examined 324 paths across fixed source and receiver groups. Right-side reception does provide a useful image: the right weld-surface source S3, at $(1.5, 16.4)$ mm, with the 157 right receivers supports $S \mid LL$. This path uses direct shear illumination and a backwall-reflected longitudinal return. With response weighting, 1–5 MHz and a 15% angular threshold, it gives 3.02 dB at 0.05 mm spacing. A narrow predicted-LL mute raises this to 5.93 dB with 96.8% near support.

Other useful views include S3 with left receivers and unmuted $S \mid SS$, giving 11.49 dB locally and 11.03 dB in the wider field with about 96–97% near support. S1 with left and weld-surface receivers gives $SLS \mid S$ contrasts of 15.80 and 4.01 dB, with 98.6% and 100% support. These local and wider checks use 0.05 and 0.25 mm grids respectively.

Not every promising local image survives these checks. S1 with right receivers and $SSL \mid SS$ reaches 9.24 dB locally using 1.5–5 MHz, but falls to -25.24 dB in the wider field, where the maximum is 21.14 mm from the flaw. Screening the full reflected packet also removes all near support for the complementary $S \mid LL$ view. These outcomes show why contrast, support, pristine controls and wider-field behaviour must be assessed together. The full results make the partial weld more promising, but it also shows that detection of the defect is dependent on all the different settings we use. More research into these settings and their justification must be done.

1.6 Discussion

Both models provide LOF-only images from a calibrated laser-equivalent source and the resulting vertical surface velocity that an optical device such as an LDV could observe. No exact reference waveforms or response-weighted receiver mean is needed to obtain these main results. Considering these, both full and partial weld inspection are feasible for this modelled case and defects can be localized. What can't be shown is defect sizing, false negatives and spatial resolution of defect localization. It also does not prove these results translate to different geometries and defects.

Table 1.3 summarizes the acquisition choices. In the full weld, direct shear propagation and oblique shoulder illumination provide a strong lower-tip indication; side-surface reception can retain it. In the partial weld, the exposed weld surface provides useful generation and the backwall supplies a return route to outer receivers. Geometry, coupling and arrival timing explain this difference together. An aperture should therefore be selected for the intended mode and path, rather than specified only by its total number of sources or receivers.

Table 1.3: Interpretation of the used acquisition strategies.

Question	Full weld	Partial weld
Principal route	Direct $S \mid S$	Direct illumination, backwall return $S \mid SS$
Useful reduced generation	Two shoulders	Three exposed weld-surface positions
Useful reception	Outer side surfaces	Left side for local view; both sides remain useful
Main coverage issue	Shoulder reduction retains 92.4% near support	Two-source weld-only view retains 48.5%; one source gives 95.8%
Interpretation	Strong lower-tip localization	Local indication with explicit support/background trade-off

1.6.1 Source and receiver sensitivity

Adding receivers can introduce stronger structural echoes as well as useful defect signals. These echoes may reduce contrast, even when the assumed wave paths are geometrically valid. Removing sources can improve the image but also leave more pixels unsupported. Image quality and coverage must therefore be assessed together.

A smaller measurement setup should still provide useful illumination and reception angles. The group comparisons identify useful source and receiver surfaces, but they do not find the minimum number of measurements needed. Further reductions should be tested over the same supported area, wider image fields, different flaws and realistic noise.

1.6.2 Filtering and image interpretation

Filtering affects which signals reinforce during focusing. The common bandpass allows a consistent comparison between paths. A shorter low-pass improves the selected full-weld image, while the partial-weld improvement combines a different band with response weighting, a lower angular threshold and a narrow reflected-arrival mute. These settings work for the studied case, but have not been shown to be optimal for other flaws or measurements.

A shared colour scale makes image amplitudes comparable, but a bright region alone is not sufficient evidence of a flaw. The size of a bright feature depends on the wave field, measurement geometry, filtering and focusing, so it cannot be read directly as flaw size.

1.6.3 Limitations

The models assume homogeneous, isotropic steel and omit weld texture, residual stresses and temperature gradients, which is especially important considering the use case for Allseas. The two-dimensional approximation also excludes finite laser-line length, three-dimensional spreading and out-of-plane scattering. Optical access, surface reflectivity, signal dropout, positioning errors and detector noise are not included. Also the source calibration supports the mechanical approximation of laser generation, but does not validate that it corresponds to the real spectrum of a laser generation inside actual samples. Only one prescribed defect is studied, and its known position is used to calculate contrast and support. Detection probability and false-alarm rates therefore remain untested.

The next step is to test the proposed source positions, receiving surfaces and paths experimentally.

The processing settings should then be kept fixed while testing different flaw positions, sizes and noise levels, with unsupported regions clearly shown.

1.7 Conclusions

The simulations show that the prescribed LOF can be localized in both weld states without subtracting a reference recording. Direct shear imaging works well in the full weld, while the partial weld benefits from weld-surface generation and a backwall-reflected return.

The selected full-weld image gives 12.79 dB contrast with complete near-flaw support. Using only shoulder sources and side receivers retains 9.61 dB with 92.4% support. The revised partial-weld image gives 14.45 dB with 96.9% support, compared with 9.09 dB for the same settings without the narrow reflected-arrival mute.

These results support the acoustic feasibility of laser-ultrasonic localization for the LOF defect. They also show that useful images require suitable wave paths, acquisition positions and processing settings. Experimental reliability and accurate flaw sizing still require further validation.

Detailed simulation settings and SAFT results

A.1 Scope and guide to the evidence

This appendix supplies the detailed model, reconstruction settings and numerical comparisons behind the report chapter on laser-ultrasonic imaging of full and partial welds. It combines the two weld states in one reference chapter. The worked SAFT example is provided in a separate appendix and is not repeated here.

The results come from the same twenty production COMSOL solves: five source positions, each simulated with and without the prescribed flaw, in two weld geometries. Changing a filter, path, receiver group or source requirement reconstructs those recordings; it does not create another independent defect test. The distinction between the original equal-average comparisons and the later response-weighted partial-weld review is retained throughout.

The selected full-weld result uses direct shear imaging and gives 12.79 dB maximum-background contrast with complete near-flaw support. The updated partial-weld result uses $S \mid SS$, response weighting, a 1–5 MHz band, a 15% angular threshold and a narrow predicted backwall-SS mute. It gives 14.45 dB with 96.9% near-flaw support on the 0.025 mm grid, compared with 9.09 dB for its matched unmuted control. Earlier partial-weld scores, including the 9.96 dB left-side result with 48.5% support, remain valid comparisons under their stated settings.

Table A.1: Where to find the evidence supporting the main chapter. The printed registers include unsuccessful and unsupported comparisons.

Question	Evidence in this appendix
What was simulated?	Geometry, material, source calibration and discretization in Section A.2.
How were recordings focused?	Travel times, angular tests, equal/weighted averages, coherence, screening and support in Section A.4.
Which filters were compared?	Exact filter definitions and fourteen baseline variants in Section A.5.1.
What supports the full-weld result?	Selected paths and acquisitions in Section A.6; complete controls and confirmation results in Section A.8.
What supports the original partial-weld result?	Path and acquisition comparisons in Section A.9; complete baseline results in Section A.13.
Why did the partial-weld image improve?	Matched mute and grid checks in Section A.14; settings, path and source investigations in Section A.15.
Where are all individual survey results?	Electronic CSV/JSON records and reusable settings described in Section A.16.

The broad review evaluates 324 implemented paths, seven receiver-group combinations and nine fixed source views: 20,412 acquisition/path combinations, each examined with flawed, pristine and signed-

difference recordings. Printing every image or every signal-case row would obscure the comparisons. The appendix therefore prints the complete path catalogue with its stated best-view summaries, the native settings matrix, source-subset register and confirmation tables. The electronic supplement retains every survey row, including views that were not selected. These summaries are retrospective comparisons on one prescribed flaw, not a pixelwise rule for selecting the brightest result.

A.1.1 Settings used for the main results

Table A.2: Selected reconstruction settings. Differences between recipes are explicit; each reported contrast belongs to its own stated field and supported pixels.

Setting	Full weld	Updated partial weld
Path	$S \mid S$	$S \mid SS$
Sources / receivers	All five / all 337	All five / all 331
Measured component	Normal velocity v_y	Normal velocity v_y
Analytic FIR	257 taps, 0–5 MHz	657 taps, 1–5 MHz
Kaiser parameter	5.10226	5.10226
Retained time guard	3.28 μs	3.28 μs
Receiver mean	Equal	Complex response-weighted
Angular threshold	20%	15%
Wall-response threshold	5%	5%
Source–receiver spacing	At least 1.5 mm	At least 1.5 mm
Receiver requirements	At least 8 actual/effective and 25% of eligible	At least 8 actual/effective and 25% of eligible
Qualifying sources	At least two per pixel	At least two per pixel
Coherence / combination	Exponent one / source RMS	Exponent one / source RMS
Direct-packet screening	L, S and Rayleigh	L, S and Rayleigh
Full reflected-packet screens	Off	Off
Additional fixed mute	None	Only predicted backwall SS; 0.30 μs half-width
Reported local pixel spacing	0.1 mm	0.025 mm

The direct-packet screens use a 30 dB envelope-tail criterion and a 0.1 μs margin. The partial-weld fixed mute is zero within ± 0.225 μs of the predicted specimen-only arrival and rises smoothly to one at ± 0.30 μs. It is distinct from excluding an entire measured reflected packet. Exact definitions and the controls used to assess these choices follow below.

A.2 Model, acquisition and source calibration

A.2.1 Two weld states with the same prescribed flaw

Figure A.1 compares the two specimens. Each is 150 mm wide and 25.4 mm thick. Depth z is measured downwards from the original outer surface; $x = 0$ is the weld centre. The J preparation has a 7.1068 mm opening, a 4.7 mm entrance section, shallow 3° flanks and a rounded root reaching approximately 24.061 mm depth. The internal weld outline is bonded to the surrounding material except at the prescribed separation. The full weld is filled to $z = 0$. The 9 mm partial fill leaves an exposed weld surface at $z = 16.4$ mm, between approximately $x = \pm 2.940$ mm.

The LOF lies on the right flank. Its upper and lower tips are at $(x, z) = (2.831875, 18.467475)$ and $(2.700855, 20.967475)$ mm. Its vertical extent is 2.5 mm and its length along the flank is 2.5034 mm. The lower tip remains 4.4325 mm above the backwall. The flaw is represented by traction-free faces; no contact closure or friction is included. The same finite segment is used in both weld states, which isolates the effect of the acquisition geometry and fill state rather than introducing a different defect.

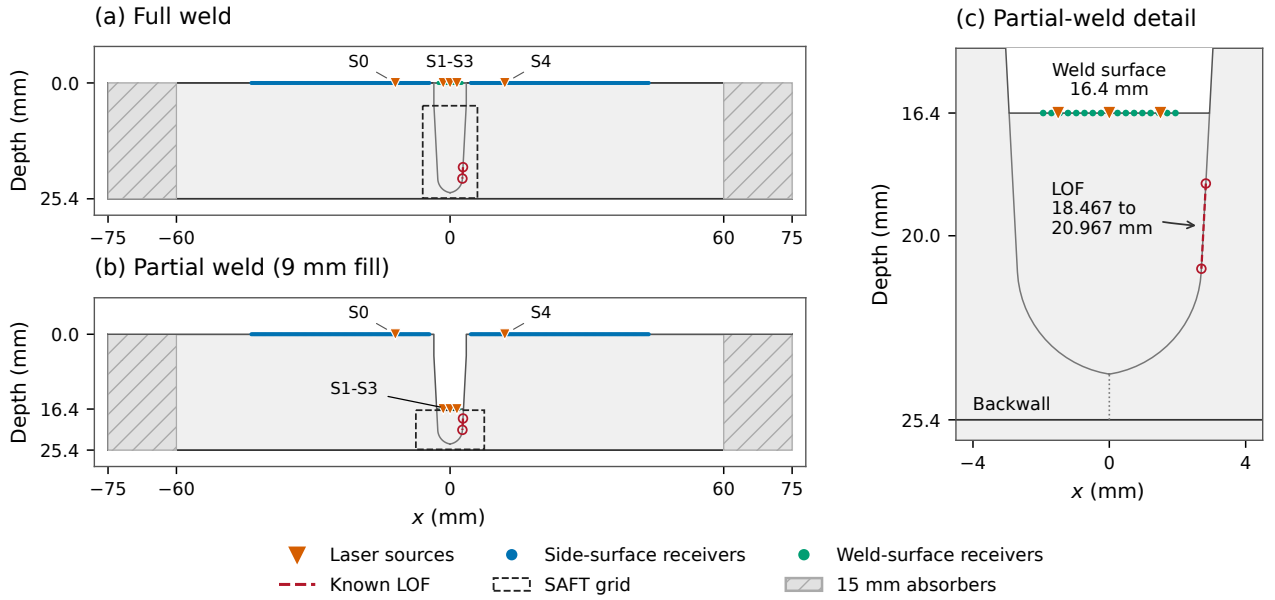


Figure A.1: Specimen geometry and acquisition. Orange triangles identify the two shoulder sources S0/S4 and three weld-surface sources S1–S3. Blue and green points denote side-surface and weld-surface receivers. The specimens include 15 mm absorbing strips at each lateral end. Separate dashed boxes show the actual full- and partial-weld reconstruction fields. The enlarged partial-weld detail shows the exposed floor, continuous J root and prescribed LOF. The grey internal outline is bonded except along the red flaw segment; it is not an open groove boundary below the fill.

The five source coordinates are $x = -12, -1.5, 0, 1.5, 12$ mm. S0 and S4 are shoulder sources at the original outer surface. S1–S3 lie on the exposed weld surface: $z = 0$ in the full weld and $z = 16.4$ mm in the partial weld. Both specimens have 157 receivers on each side surface at 0.25 mm pitch, extending to approximately $x = \pm 43.5534$ mm. There are also 23 full-weld or 17 partial-weld receivers on the central weld surface, giving 337 and 331 receivers respectively. Every receiver measures vertical velocity v_y, which is normal motion on these horizontal receiving surfaces.

Five flawed and five pristine source simulations are available for each weld state: twenty production solves in total. Every recording contains 5001 samples over 50 μs at 10 ns output spacing. A pristine simulation has the same geometry and source but no separation. It provides an interpretation control and permits signed waveform subtraction; it is not an input to the LOF-only imaging recipe.

A.2.2 Elastic propagation and numerical scope

Production calculations use COMSOL Multiphysics 6.4, build 429, with the Elastic Waves, Time Explicit interface. Steel is homogeneous, isotropic and linearly elastic, under two-dimensional plane strain. The model uses $E = 212.443$ GPa, $\nu = 0.288978$ and $\rho = 7859.975\ \mathrm{kg\,m^{-3}}$, giving longitudinal and in-plane shear speeds of $c_L = 5943.60$ and $c_S = 3237.97\ \mathrm{m\,s^{-1}}$. The corresponding Rayleigh speed used for arrival screening is $c_R = 2997.33\ \mathrm{m\,s^{-1}}$. Longitudinal motion is primarily parallel to propagation; shear motion is transverse. Both can contribute to measured normal surface velocity after interaction with a free surface.

The elastic field satisfies

$$\rho\,\ddot{\mathbf{u}} = \nabla\cdot\boldsymbol{\sigma}, \qquad \boldsymbol{\sigma} = \lambda\,\mathrm{tr}(\boldsymbol{\varepsilon})\mathbf{I} + 2\mu\boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} = \tfrac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^{\top}). \tag{A.1}$$

Here $\mathbf{u}$ is displacement and λ, μ are the Lamé constants. Exposed surfaces and the LOF are traction-

free except where the equivalent source acts. Bonded internal boundaries transmit displacement and traction. Weld and surrounding steel have identical properties, so the bonded fusion outline is not itself an impedance reflector. Absorbing strips at the lateral ends reduce artificial returns from the truncated specimen.

The production discretization uses quartic velocity and strain fields, a discontinuous Galerkin formulation and explicit fourth-order Runge–Kutta integration. Maximum element sizes are approximately 0.32380 mm in the weld and 0.48570 mm in the surrounding material, chosen using a 5 MHz design frequency. That choice does not establish a sharp accuracy cutoff at 5 MHz; receiver-level mesh and time-step convergence would be needed to establish a validated bandwidth. Likewise, the 50 MHz Nyquist frequency of the stored histories describes sampling, not the accuracy of the elastic solution. The 10 ns output interval must not be confused with the internal solver step.

A.2.3 Calibrating the laser-equivalent source

The production source is an equivalent tangential traction representing thermoelastic expansion from an absorbed laser line. It is not an arbitrary point force adjusted to brighten the LOF. The calibration uses an independent 8 mm by 4 mm flat specimen, implemented with a 4 mm symmetric half-width. COMSOL Heat Transfer in Solids supplies the temperature field to Solid Mechanics through thermal expansion; a second mechanical field on the same mesh receives the candidate equivalent traction.

For a Gaussian laser profile, define $\xi = (x - x_s)/w$ with $w = 0.5$ mm the $1/e^2$ radius. The absorbed heat flux is

$$q(x,t) = \frac{F_0}{\sqrt{2\pi}\sigma_t} \exp(-2\xi^2) \exp\left[-\frac{(t-t_0)^2}{2\sigma_t^2}\right], \qquad \sigma_t = \frac{6.5 \text{ ns}}{2\sqrt{2\ln 2}}. \tag{A.2}$$

The pulse centre is $t_0 = 30$ ns, or 0.030 μs. It is prescribed so the short Gaussian pulse lies inside the simulated time window; it is not a fitted imaging delay. F_0 is peak absorbed fluence. A 100 $\mathrm{J\,m^{-2}}$ absorbed input is used for the production source, without assuming an optical absorptivity or equating it to incident laser energy.

The equivalent surface load is a tangential dipole,

$$t_x(x,t) = 4A\xi e^{-2\xi^2} H(t), \qquad t_y = 0, \qquad H(t) = \tfrac{1}{2}\left[1 + \operatorname{erf}\left(\frac{t-t_0}{\sqrt{2}\sigma_t}\right)\right], \tag{A.3}$$

with

$$A = \gamma \frac{E\alpha F_0}{(1-\nu)\rho c_p w}. \tag{A.4}$$

The opposing tangential lobes represent the lateral effect of a thin heated region. The finite source width is important to its angular and frequency response [5]. Thermal inputs are $T_0 = 293.15$ K, conductivity 52 $\mathrm{W\,m^{-1}\,K^{-1}}$, $c_p = 480$ $\mathrm{J\,kg^{-1}\,K^{-1}}$ and expansion coefficient $\alpha = 11 \times 10^{-6}$ $\mathrm{K^{-1}}$, following the material benchmark used for the calibration [7].

Only the scalar gain γ is fitted. Signed horizontal and vertical velocity histories at five points on a 2 mm radius arc, from 15° to 75°, are compared over 0.12–0.82 μs. Both fields receive the same forward–backward fourth-order 5 MHz Butterworth filter for this calibration comparison. If $\mathbf{d}$ is the

unit-gain dipole history and $\mathbf{v}$ the thermoelastic history, the least-squares result is

$$\gamma = \frac{\mathbf{d}^\top \mathbf{v}}{\mathbf{d}^\top \mathbf{d}} = 1.084434. \tag{A.5}$$

Figure A.2 shows the calibration comparison. The relative waveform error is 0.931%; held-out axial and surface checks give approximately 1.09% and 1.27%. No angle-dependent gain, time shift or flaw response is fitted. The 10 and 100 J m^{-2} calibrations give the same gain. At the production fluence, $A = 188.939$ kPa and the maximum dipole traction is 229.195 kPa.

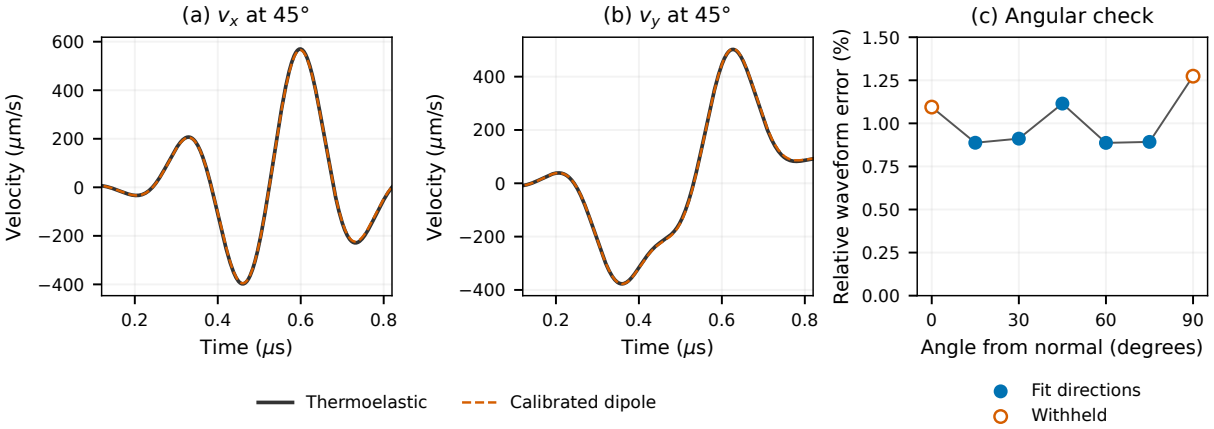


Figure A.2: Independent source calibration. The equivalent traction is compared with the thermoelastic calculation through signed outgoing velocity histories and angular response, rather than through a reconstructed flaw image. A single gain scales the dipole. Agreement supports this equivalent source over the tested calibration band and observation region; it does not establish production-mesh convergence or optical receiver sensitivity.

A.2.4 Source-calibration discretization and independent checks

The small thermoelastic benchmark resolves the thin heated region with nine surface-normal layers of thickness 0.05, 0.10, 0.20, 0.40, 0.80, 1.60, 3.20, 6.40 and 12.80 μm, totalling 25.55 μm. Eighty elements span the 4 mm half-width and the bulk maximum size is 0.08 mm. Implicit generalized-α integration uses a maximum step of 0.5 ns during the first 60 ns and 2 ns thereafter. The exported comparison contains 451 samples from 0 to 0.9 μs at 2 ns spacing. These are calibration-model settings, distinct from the explicit production-weld calculation and its 10 ns output clock.

At 100 J m^{-2} absorbed peak fluence, the integrated deposited energy per out-of-plane length is 0.0626657 J/m, agreeing with the prescribed Gaussian input within 0.000639%. The temperature rise is 78.34 K; at 10 J m^{-2} it is 7.83 K. The thermal field agrees with the independent one-dimensional heat-flow check to 0.0693% relative L_2, and differentiating displacement agrees with exported velocity to 0.0481%. These checks address input energy, thermal consistency and mechanical output consistency. The sub-percent outgoing waveform fit then addresses the particular equivalent-source approximation. They are separate from production-mesh convergence and from any reconstructed-image metric.

A.3 Recorded wave fields in the two welds

A.3.1 What the full-weld recordings contain

Read Figure A.3 from the recording to the geometry. The A-scan shows that large motion occurs even before an identifiable flaw-scattering interpretation is available. The B-scan shows that these

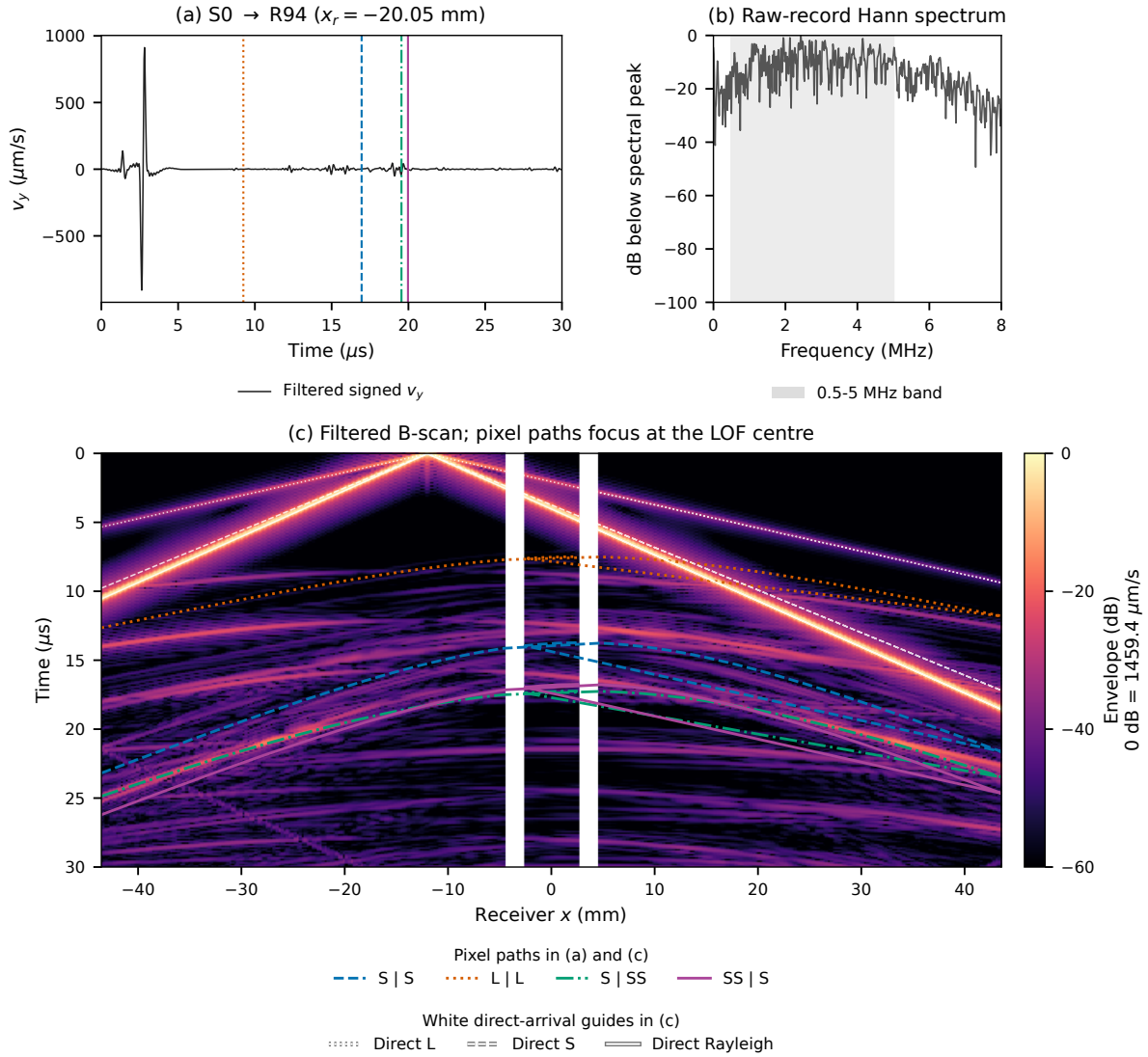


Figure A.3: Recording-level context with the common 0.5–5 MHz filter. The S0/R94 A-scan is one receiver history; the S0 B-scan stacks histories by receiver position. Coloured travel-time guides assume scattering at the known LOF centre: $S \mid S$ blue dashed, $L \mid L$ orange dotted, $S \mid SS$ green dash-dot and $SS \mid S$ purple solid. White direct-arrival guides are L dotted, S dashed and Rayleigh solid. These guides label timing hypotheses, not measured mode identities. The raw Hann-window spectrum has its own peak normalization. Filter-edge samples appear in this explanatory plot but are excluded where reconstruction lacks complete input support.

arrivals have systematic moveout across receivers. The spectrum shows available frequency content, but does not identify which frequencies belong to the flaw. The travel-time guides add a predicted interpretation to the data; a bright band meeting a guide is suggestive, not proof.

A pristine specimen still launches and reflects waves. Much of the prominent source-associated field is therefore expected in both simulations. Its persistence in an LOF-only image is a focusing ambiguity, not evidence that the imposed source has somehow moved into the weld. Increasing the minimum image depth only hides shallow pixels; it does not remove a source packet from the recorded histories. The following reconstruction instead checks path geometry and arrival time before combining samples.

A.3.2 Why the source-associated field is particularly important here

The LOF tips lie only 2.067–4.567 mm below the exposed weld surface. At the shear speed of 3.238 mm/μs, a short local round trip can occupy only a few microseconds. The generated field, its surface motion and early structural returns can therefore overlap the time range that would contain a direct flaw echo. A large recorded amplitude in that interval is not uniquely attributable to the LOF.

Figure A.4 should be read as evidence of superposition. Flawed and pristine histories both contain the early packet. Later bands can include floor/backwall reverberation as well as flaw scattering. The white or coloured time guides are predictions superimposed on this field; they are not a decomposition of the measured signal into modes.

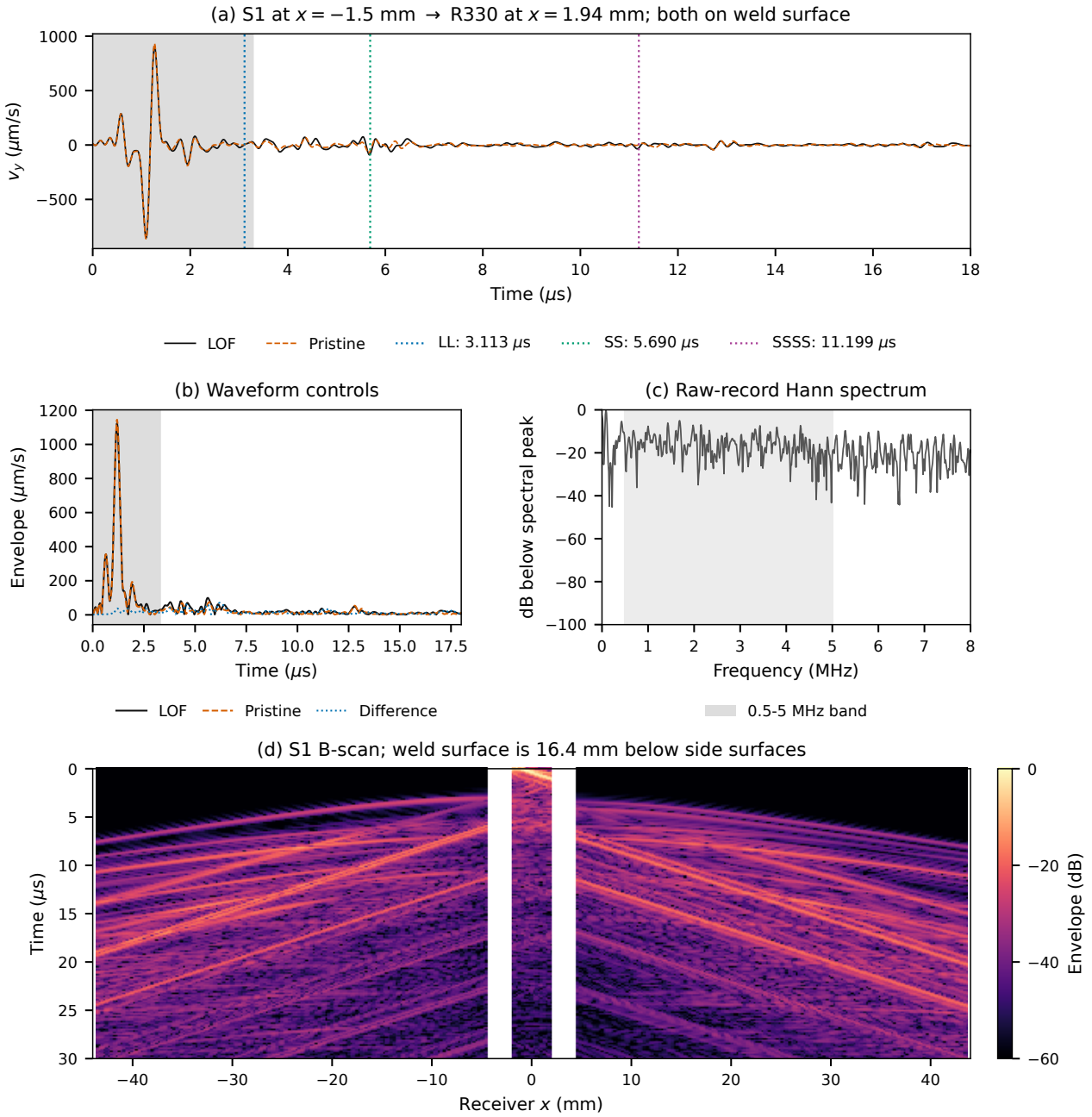


Figure A.4: Partial-weld recording context for weld-surface source S1 and receiver R330. Traces and envelopes use the common 0.5–5 MHz processing; the spectrum uses the raw recording with a Hann window. LL at 3.113 μs, SS at 5.690 μs and SSSS at 11.199 μs denote specimen-only returns with no trial scattering point. Grey early-time shading indicates incomplete filter support. The B-scan has a separate reference of 1449.18 μm/s. The strong early field is present in both flawed and pristine recordings.

A.4 Detailed reconstruction method

A.4.1 A pixel represents a scattering hypothesis

SAFT asks whether different receiver histories contain motion at the times expected if ultrasound scattered at a trial pixel $\mathbf{p}$. For each source s, receiver r and chosen path P, it predicts

$$T_{sr,P}(\mathbf{p}) = t_0 + \sum_{j \in P} \frac{\ell_j(\mathbf{p})}{c_{m_j}}, \tag{A.6}$$

where each leg has length ℓ_j and mode $m_j \in \{L, S\}$. The path notation places a vertical bar at the trial scattering point. Thus $S \mid S$ is direct shear illumination and direct shear reception; $S \mid SS$ adds one backwall reflection on the receiving route; $SS \mid S$ adds it on the incident route. $L \mid L$ is the direct longitudinal view. The notation describes the focusing hypothesis, not a measured separation of modes in the recordings.

Every segment must remain in steel and every reflection must lie on an actual boundary. In the partial weld, a ray cannot cross the open groove merely because its straight-line length is convenient. Valid routes may use the original surface, the exposed fill floor or the backwall where those surfaces exist. For a specified family, the implementation uses the earliest geometrically valid branch. The source and receiver sets are evaluated at every image pixel; a single inspection point does not define the whole image aperture.

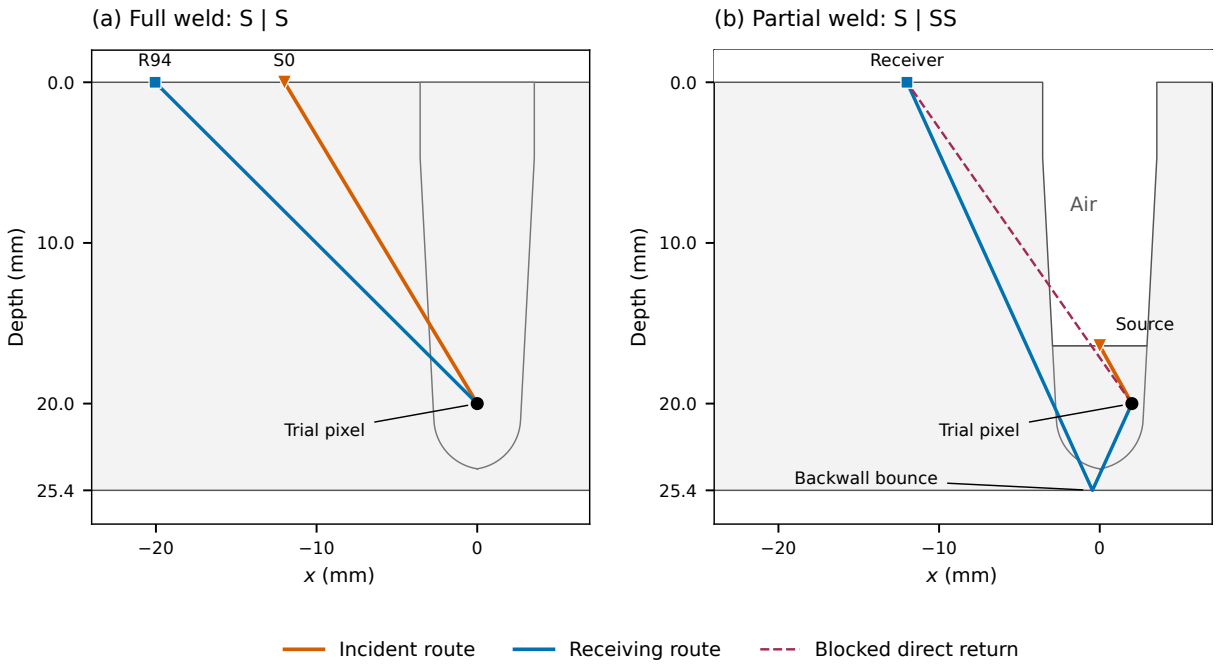


Figure A.5: Illustrative focusing routes, with equal horizontal and depth scales. (a) A full-weld $S \mid S$ route from a shoulder source to a trial pixel and side receiver. (b) In the partial weld, direct illumination from the weld surface and a backwall-reflected return can remain inside steel even when a direct return crosses the open groove. These are worked geometric examples, not identified wave packets or additional measurement positions. The pale internal J outline is bonded; the white groove is air.

For example, a full-weld source at $(-12, 0)$ mm, receiver at $(-20.0534, 0)$ mm and pixel at $(0, 20)$

mm give a direct-S delay of about 15.98 μs; the same lengths give about 8.72 μs for direct L. In the partial-weld example, a source at $(0, 16.4)$ mm illuminates $(2, 20)$ mm and returns through a backwall bounce to an illustrative receiver at $(-12, 0)$ mm. The incident length is 4.118 mm and the reflected receiving length is 33.833 mm, giving $T = 11.751$ μs. The direct return would cross the fill depth at $x = -0.52$ mm, inside the open groove, and is invalid. The reflected return crosses that depth at about $x = -7.91$ mm, in steel. This is why a reflected view can be necessary, rather than merely an optional extra image.

A.4.2 Admitting useful launch and receive angles

A laser dipole does not launch all modes equally in all directions. A normal-motion receiver also does not observe every incoming mode equally. Free-surface reflection and conversion contribute to its response, so normal sensitivity cannot be estimated from the incident displacement component alone. The baseline study uses the angular magnitudes in Figure A.6 to admit plausible channels.

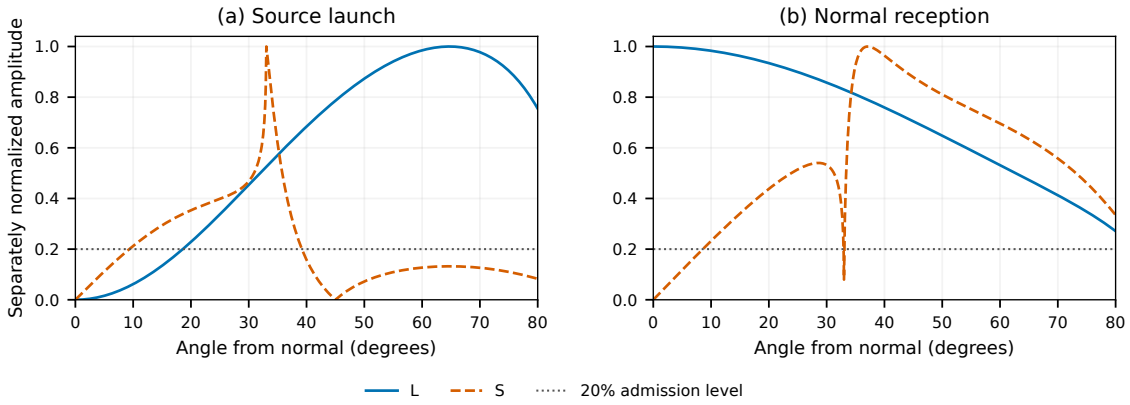


Figure A.6: Angular launch and normal-reception magnitudes used for channel admission. L and S are normalized separately, so the curves do not compare absolute longitudinal and shear energy. The 20% level is a selection threshold: a channel must satisfy both its source and receiver tests. The baseline averages admitted receivers equally; the updated partial-weld recipe also uses their complex response. These curves explain why a large spatial aperture can still contain unsuitable mode/path combinations.

In the baseline, for a channel to qualify, both normalized launch and receive magnitudes must reach 20%, and receive angle must not exceed 80° from the normal. A reflected route must also pass a 5% wall-response-product gate. These are admission tests, not amplitude weights or phase corrections. A 1.5 mm source–receiver separation exclusion keeps receivers outside the approximately 1.425 mm source-footprint half-width. The wall gate is inactive for the principal images but affects some longer partial-weld paths. Removing the separation exclusion changes the principal images measurably. Both controls are therefore retained as declared physical rules; detailed ablations are in this appendix.

A.4.3 Frequency filtering and pulse-aware screening

The recorded field includes direct bulk waves, surface motion, reflected packets and flaw scattering. A strong direct packet can be sampled at a delay assigned to a deep trial pixel and create an apparent indication. Its coherence across receivers means that ordinary averaging, or even coherence weighting, may preserve it. The contamination is therefore a mismatch between the packet's actual route and the route assumed in focusing. Cropping shallow pixels does not remove that packet from the histories.

The common comparison uses a 657-tap analytic finite impulse response (FIR) filter over 0.5–5 MHz,

with Kaiser parameter 5.10226. It supplies a complex signal with amplitude and phase for delay-and-sum processing. At 10 ns spacing its half-support is 3.28 μs. Reconstruction requires complete filtering and interpolation support, giving a usable delay interval of approximately 3.28–46.71 μs. The filter suppresses components outside a declared analysis band; it is not justified by the stored sampling limit, nor does it prove the mesh resolves all retained frequencies accurately.

Direct L, S and Rayleigh arrivals are then screened per source–receiver pair. A geometry-valid prediction locates an envelope peak within ± 0.25 μs. The packet window extends until the envelope is below -30 dB relative to that peak for at least 0.15 μs, with a further 0.1 μs margin. An imaging sample falling within the excluded window is not admitted. This avoids a single fixed mute duration for packets with different widths. The finite search reach is tied to the source/filter pulse. Rayleigh screening requires a common unobstructed flat surface; it does not assume a surface wave crosses the empty groove.

The same rule acts differently in the two geometries. All 1685 full-weld source–receiver pairs have direct bulk paths. Only 365 of 1655 partial-weld pairs have direct paths: shoulder-to-side pairs on their own side and weld-surface pairs on the floor. Applying flat-surface direct-arrival predictions to the remaining pairs would exclude samples using physically invalid paths. The baseline images use this direct-arrival screening with fixed mutes and reflected-arrival screening off. The updated partial-weld recipe adds the narrow fixed mute defined below.

The screening windows for an LOF-only image are obtained from its own recording. A pristine image uses its own windows; the waveform-difference control uses the LOF-derived windows. The procedure therefore requires no reference recording to suppress the direct field. It can still reject real flaw information when arrivals overlap, especially for a source close to the partial-weld flaw. This cost appears as lost support and must be reported alongside contrast.

Fourteen filter alternatives are documented in this appendix, including unfiltered and unrestricted-band controls. For the selected full-weld direct-S image, a shorter 257-tap 0–5 MHz low-pass is used, while retaining the same 3.28 μs admission guard. The historical partial-weld results retain the common 657-tap bandpass; the revised weighted recipe uses 1–5 MHz with the same tap count and guard. These are explicitly different selected recipes; the common four-path full-weld comparison retains identical filtering. The source-calibration Butterworth filter is a separate operation and is not the reconstruction FIR.

A.4.4 Baseline receiver averages, coherence and source coverage

Let $a_{sr}(\mathbf{p})$ be the complex filtered history interpolated at Equation A.6, after the channel and timing tests. For N_s admitted receivers, the equal mean and receiver coherence factor are

$$M_s(\mathbf{p}) = \frac{1}{N_s} \sum_r a_{sr}(\mathbf{p}), \qquad \mathrm{CF}_s(\mathbf{p}) = \frac{|\sum_r a_{sr}(\mathbf{p})|^2}{N_s \sum_r |a_{sr}(\mathbf{p})|^2}. \tag{A.7}$$

The coherence factor is zero for zero signal energy and otherwise lies between zero and one. Aligned phases reinforce in the numerator; inconsistent phases reduce it. The factor has exponent one here. It encourages phase consistency under the selected delay law, but a coherent structural echo can also

pass that test. This equation describes the equal-mean baseline. Response weighting for the updated partial-weld result is defined below.

Each source must retain at least eight receivers and at least 25% of its aperture eligible before the time-support and pulse tests. With equal receiver weights, effective receiver count equals actual count. The nominal image requires at least two qualifying sources at a pixel. If $\mathcal{S}(\mathbf{p})$ is that set, source results combine as

$$I(\mathbf{p}) = \left[\frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} |\mathrm{CF}_s M_s|^2 \right]^{1/2}. \tag{A.8}$$

This root-mean-square combination avoids cancellation between separately illuminated source views. The output retains velocity units but is a processed focusing score, not acoustic intensity or flaw size. A pixel failing the required source count is unsupported; it is not assigned a physical zero.

Allowing one qualifying source changes which pixels can be displayed. It does not mean acquiring only one laser shot for the whole image: all selected source shots remain available wherever they qualify, and Equation A.8 still uses them. This distinction is essential to the partial-weld coverage comparison below.

A.4.5 Response weighting and a narrow reflected-arrival mute

The revised partial-weld comparison uses the existing response-weighted option. Let b_{sr} be the complex product of the predicted wall and normal-reception responses and let $w_{sr} = |b_{sr}|$. After interpolation and any mute, the sample is a_{sr}. The implemented sums are

$$Q_s = \sum_r a_{sr} b_{sr}^*, \quad W_s = \sum_r w_{sr}, \quad P_s = \sum_r w_{sr} |a_{sr}|^2, \qquad M_s = Q_s / W_s, \quad \mathrm{CF}_s = \frac{|Q_s|^2}{W_s P_s}. \tag{A.9}$$

Thus the response supplies a phase correction as well as an amplitude weight. It is not inverse-amplitude compensation. Source directivity remains an admission test; an unknown flaw-scattering coefficient is not fitted. The effective receiver count is $N_{\mathrm{eff}} = W_s^2 / \sum_r w_{sr}^2$. Setting every $b_{sr} = 1$ recovers the equal-mean baseline. The revised images retain CF[1], RMS source combination, eight actual and eight effective receivers, the 25% retention rule and a two-source minimum unless stated otherwise.

The revised partial-weld working choice uses a 657-tap 1–5 MHz band, a 15% angular gate and an additional smooth mute about the geometry-valid, specimen-only backwall SS arrival. This arrival travels from source to backwall to receiver without the trial scattering point. For its predicted time t_{SS}, set $d = |T - t_{SS}|/h$ and

$$g(T) = \tfrac{1}{2}\{1 - \cos[\pi \ \mathrm{clip}((d - 0.75)/0.25, 0, 1)]\}, \qquad a_{sr} = g(T)\, a_{sr}^{\mathrm{filtered}}, \quad h = 0.30\ \mu\mathrm{s}. \tag{A.10}$$

Samples within $\pm 0.225\ \mu$s are rejected; the taper reaches full weight at $\pm 0.30\ \mu$s. The original direct L/S/R pulse screens and complete FIR support remain active. This short, predicted-arrival mute is a separate hypothesis from rejecting the entire measured reflected packet. It removes only its central timing region and cannot establish that all reflected contamination has been removed. Its width is selected retrospectively from the sensitivity study, not measured as a universal pulse duration.

A.4.6 Image fields, controls and numerical comparisons

The full-weld grid spans $x = \pm 6$ mm and requested depths 5–25.25 mm; the partial-weld grid spans $x = \pm 7.5$ mm and 16.65–25.25 mm. Baseline principal images use 0.1 mm pixels; the revised partial-weld result is also checked at 0.025 mm and with a shifted grid. Sampling from the full-weld start depth places its last row at 25.2 mm. At 5 MHz the shear wavelength is approximately 0.648 mm, so the pixel grid samples the image finely but does not establish 0.1 mm physical resolution. Finer 0.05 mm grids and wider $x = \pm 45$ mm fields check peak stability and competing indications.

The images show $20 \log_{10}[I(\mathbf{p})/I_0]$ over a declared display range: 45 dB for the principal path and source-threshold panels, and 40 dB for the partial-weld acquisition comparison. A shared reference is used within each comparison. Black denotes low displayed amplitude, grey denotes unsupported material, and white denotes air. Thin external LOF guides identify the prescribed location without covering the image. These marks are ground-truth annotations, including in pristine panels, not a reconstructed flaw outline.

Define a near-flaw region $\mathcal{N}$ within 0.5 mm of the finite prescribed segment and a background region $\mathcal{B}$ more than 1 mm from it. The intervening buffer is omitted. Over supported pixels,

$$C_{\max} = 20 \log_{10} \frac{\max_{\mathcal{N}} I}{\max_{\mathcal{B}} I}, \qquad C_{99} = 20 \log_{10} \frac{\max_{\mathcal{N}} I}{Q_{99,\mathcal{B}}(I)}. \tag{A.11}$$

Positive $C_{\max}$ means the strongest near-flaw response exceeds every assessed background pixel. Near-flaw support is the fraction of $\mathcal{N}$ with a valid image value. It is not the fraction of flaw length detected. Changing an aperture changes both the signal and the assessed region; comparisons therefore report support and, where needed, restrict both images to a common mask.

The LOF-only image is the principal result. Pristine controls test whether the intact structure produces a similar indication. Difference controls subtract signed LOF and pristine waveforms before filtering and focusing; they do not subtract displayed image amplitudes. Since interference and coherence change, a difference image can have a larger peak than the LOF-only image. These controls support attribution to the flaw but do not establish a detection threshold from one simulated defect.

A.4.7 Implementation verification and reproduction scope

The archived verification checks dataset fingerprints, receiver coordinates and the time clock before reconstruction. An independent bandpass calculation for four paths, five points and three signal cases in each geometry agrees with the native C++ engine within 5.4×10^{-19} m s^{-1}. A sinusoidal input checks gain and phase. The archived laser regression run contains 242 passing tests. Image checks independently recompute coordinate bounds, amplitude/support consistency, contrasts and peak locations. These are implementation checks, not a substitute for experimental validation.

The primary reconstruction archive contains 104 full-weld and 280 partial-weld configurations, each with LOF, pristine and waveform-difference images. The joint-acquisition supplement adds 25 scientific partial-weld configurations and one numerical replay control. Thus the primary and joint studies contain 104 full-weld and 305 partial-weld scientific configurations; gate and additional-geometry audits are separate. These are reconstructions of the production recordings, not additional COMSOL solves. The repository retains protocols, model/dataset metadata, numerical arrays, settings and

verification scripts. The portable LaTeX edition contains the complete explanation and figures; it does not embed the raw simulation recordings.

A.5 Filter definitions and baseline variants

A.5.1 Filter definitions and controlled processing variants

The purpose of this comparison is to distinguish three effects: which frequencies are retained, how long the temporal averaging kernel is, and which arrival times are admitted. These can change independently. Holding the admission guard fixed while shortening a filter tests its waveform effect without automatically gaining earlier samples.

The survey holds geometry, source/receiver lists, physical gates, coherence, source qualification and combination fixed. Four full-weld and eighteen partial-weld paths are processed with fourteen variants and three signal cases. This gives 168 full-weld and 756 partial-weld survey images. Finer-grid, wider-field, coupling and acquisition checks bring this primary archive to 384 configurations and 1152 signal-case images. The separately documented joint-acquisition supplement contributes 25 scientific configurations and one replay control. These counts refer to reconstructions of the existing production recordings, not new finite-element solutions.

For reproducibility, the filter coefficients are specified below. A finite sequence of coefficients weights neighbouring samples; the complex exponential shifts the selected band, and the Kaiser window limits sidelobes. For the FIR variants the analytic kernel is

$$h_n = \frac{2b_n}{\sum_k b_k} \exp(2\pi i f_c n \Delta t), \quad b_n = 2B\Delta t \ \mathrm{sinc}(Bn\Delta t) K_\beta(n), \tag{A.12}$$

where $B = f_H - f_L$, $f_c = (f_H + f_L)/2$, $\mathrm{sinc}(u) = \sin(\pi u)/(\pi u)$ and K_β is a Kaiser window with $\beta = 5.10226$. The ordinary positive-frequency gain is approximately two. Nominal edges fall in finite transition bands, rather than defining brick-wall cutoffs. At low frequencies, including DC, the response differs from an ideal analytic transform; Figure A.7 displays that response explicitly.

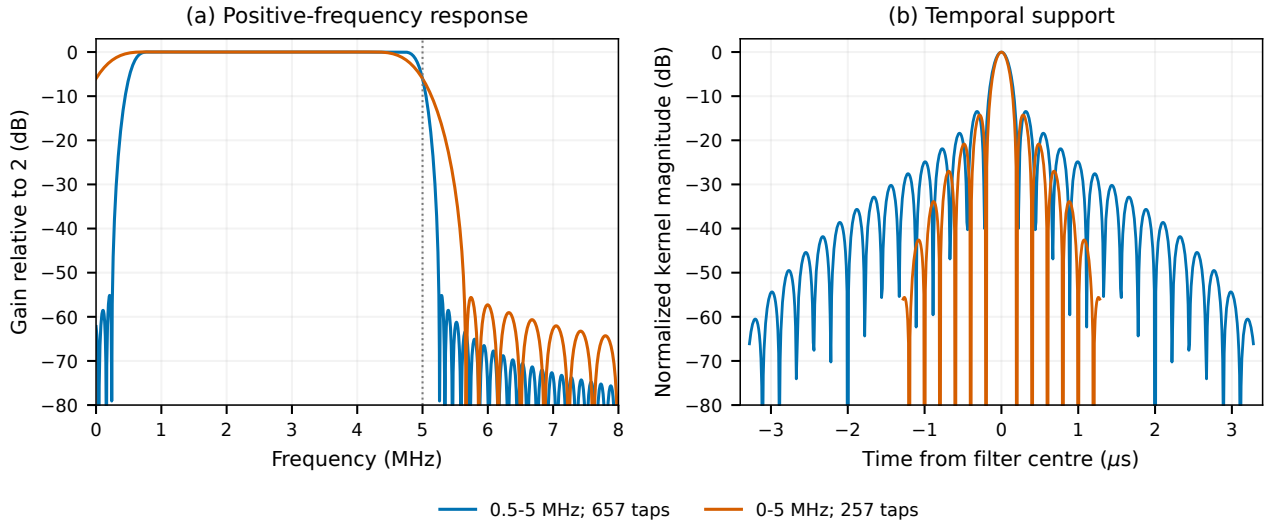


Figure A.7: Frequency response and temporal kernel magnitude for the two selected filters. Both selected images retain a 3.28 μs admission guard even though the 257-tap kernel itself has 1.28 μs half-support. This separates shorter temporal averaging from recovery of earlier samples.

Table A.3: All processing variants. The initial-rest extension admits known zero prehistory but retains rejection of the unknown post-record tail. A stated guard is independent of kernel duration.

Processing	Taps	Guard (μs)
0.5–5, 1–5, 0.5–3, or 0–5 MHz analytic FIR	657	3.28
0.5–5 MHz analytic FIR, matched long guard	257	3.28
0.5–5 MHz analytic FIR, native guard	257	1.28
0–5 MHz analytic FIR, matched long guard	257	3.28
0–5 MHz analytic FIR, native guard	257	1.28
0–5 MHz analytic FIR, shorter kernel	129	0.64
Original bandpass with known initial rest	657	3.28
Short low-pass with known initial rest	257	1.28
5 MHz forward–backward Butterworth, then analytic transform	–	3.28
Unrestricted-band analytic transform	–	3.28
Signed unfiltered samples	–	0

The signed unfiltered case changes phase representation as well as bandwidth, so it is not an envelope-equivalent replacement. The unrestricted analytic case retains broad spectral content without claiming that the production mesh resolves all of it. Zero initial velocity is verified in every record before initial-rest continuation is used. For the noncompact analytic/Butterworth controls, 4096 zero samples precede the 5001-sample recording in a 32768-point transform. Doubling the transform length to 65536 changes usable-window waveforms by at most 0.0171% relative L_2 across sources and signal cases. This checks sensitivity to the finite transform extension, not ideal finite-time analytic support.

The shorter selected full-weld filter leaves approximately 0.040–0.044% of total processed-record energy above 5 MHz. This is a descriptive global energy fraction; it neither proves mesh accuracy nor bounds the error of a focused pixel. The survey is exploratory on a known simulated flaw. Confirmation checks test the stability of its interpretation, rather than converting the selected result into a universally optimal filter.

A.6 Full weld: selected paths and signal controls

A.6.1 Four paths with the same processing rules

The common comparison keeps the 657-tap 0.5–5 MHz filter, timing guard, physical gates, screening, coverage thresholds and equal receiver averaging fixed. Only the focusing path changes. Its eligible aperture still changes, because each path uses different angles and delays. This is a common-rule comparison, not a requirement that every path average identical receiver IDs.

All four global maxima lie within 0.37 mm of the prescribed finite segment. Direct L gives $C_{\max} = 8.85$ dB, direct S 6.47 dB, $S \mid SS$ 4.65 dB and $SS \mid S$ 4.37 dB. Each has 100% near-flaw support. These results answer the first question: several path hypotheses yield a local indication with no pristine subtraction.

The location and shape carry different information. The direct views mainly emphasize the lower-tip vicinity. The broader reflected views sample a different angular relationship to the flaw, but their elongated bright regions can include the imaging point-spread function and structural interference. An image is a map of delay consistency; its bright extent cannot simply be read as physical flaw length.

The agreement of several views strengthens the interpretation, but they reuse the same simulations

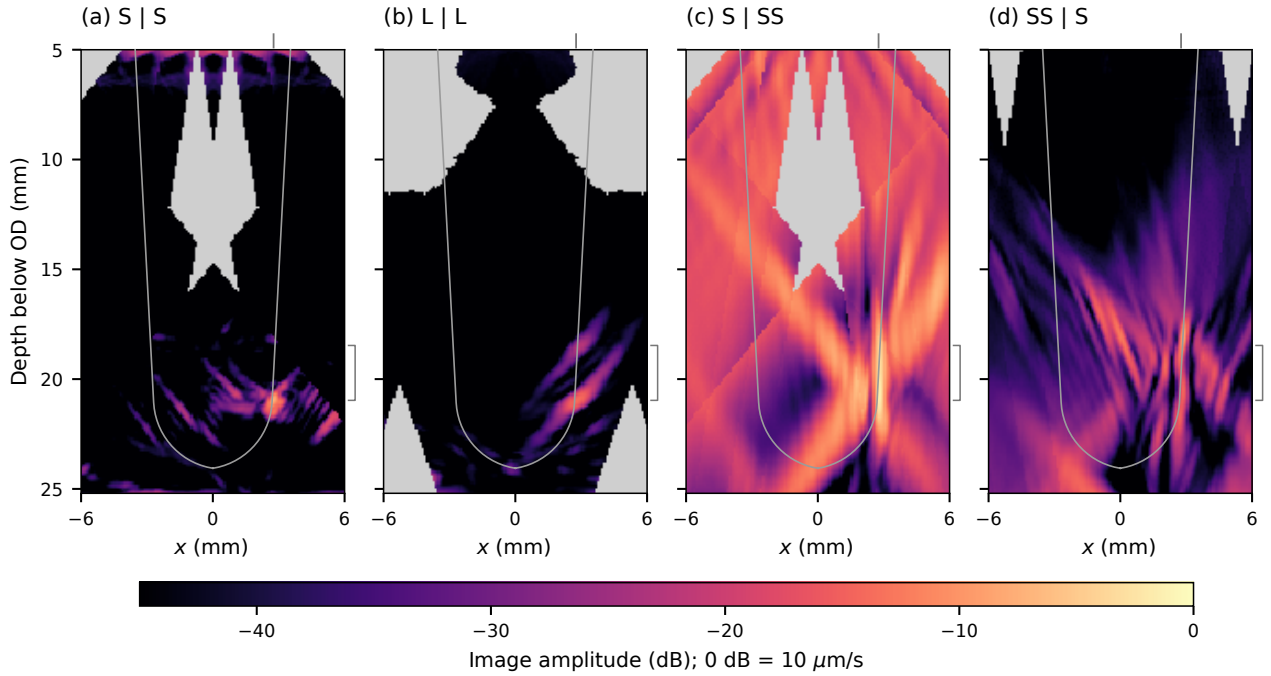


Figure A.8: LOF-only full-weld images for the four tested paths with common processing and 0.1 mm pixels. All panels use 0 dB = 10 μm/s and a 45 dB range. Thin external marks identify the prescribed LOF; grey is unsupported. Direct views concentrate near the lower tip, while the reflected views are more extended. The result supports complementary localization, not a measurement of the full flaw outline.

and are not independent detection trials. Their complete pristine and difference controls are retained in Section A.8. A colour blend may help display complementary responses, but adding arbitrary path colours does not define a calibrated probability or improve the underlying acoustic information.

A.6.2 Separate the filter question from the path question

For direct S, the 657-tap common bandpass gives 6.47 dB contrast. A 257-tap filter with the same nominal 0.5–5 MHz edges gives 10.26 dB. A 257-tap 0–5 MHz low-pass gives 12.79 dB. All three retain the same 3.28 μs admission guard. The difference therefore involves the waveform being averaged, including temporal smearing and frequency content, rather than simply recovering previously excluded early samples.

The selected direct-S result uses the shorter low-pass. It should not be described as unfiltered, since high frequencies are still attenuated and the analytic phase representation is still used. Nor is it a universal best filter: direct L gives only 3.24 dB with the same shorter low-pass, compared with 8.85 dB under the common bandpass. A frequency choice is coupled to the scattering and timing hypothesis.

Fourteen alternatives, including unrestricted-band analytic and signed unfiltered reconstructions, are documented in the supporting comparison. Their role is to test the processing assumption, not to infer a validated frequency limit from the strongest known-flaw score. The selected filter is an exploratory choice on this modeled case.

A.6.3 Selected direct-S image and its controls

The LOF-only image peaks at 5.184 μm/s at $(2.7, 21.0)$ mm, 0.0325 mm from the finite LOF segment. Its maximum-background contrast is 12.79 dB, its percentile contrast is 21.28 dB, and its near-flaw

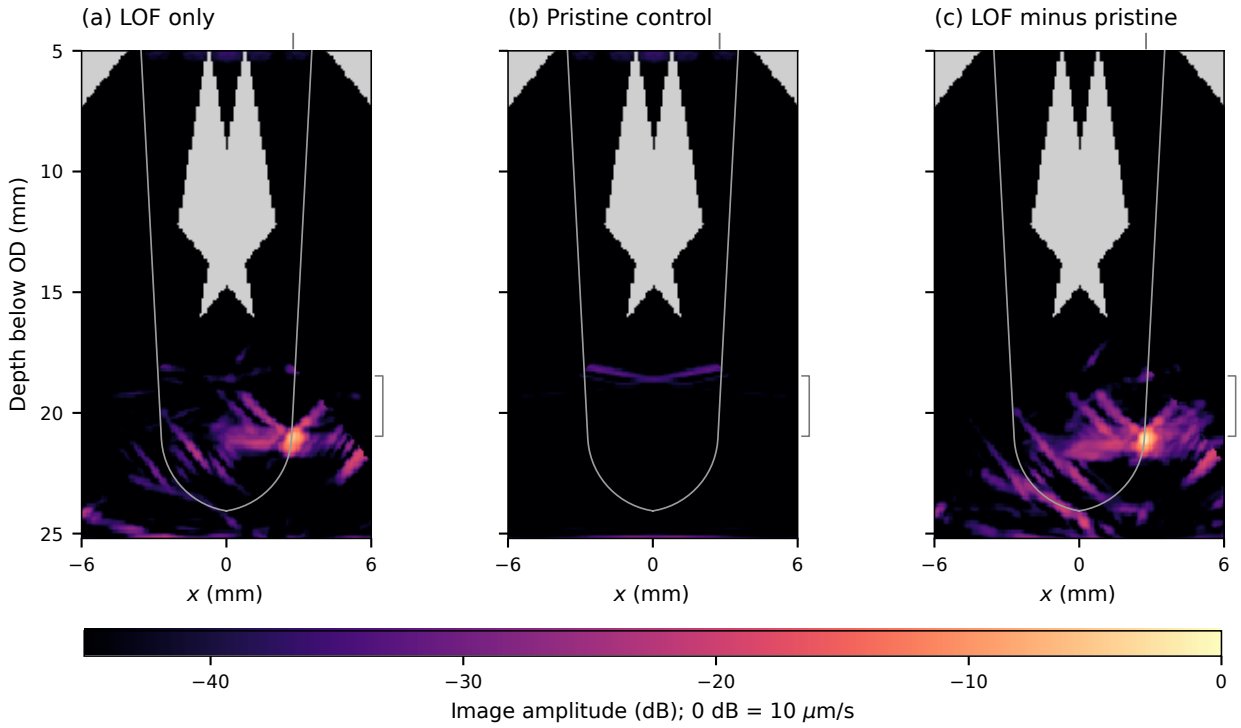


Figure A.9: Selected $S \mid S$ reconstruction and its controls. Processing uses a 257-tap 0–5 MHz analytic FIR filter, the retained 3.28 μs admission guard and the common selection/coherence rules. All panels show the complete requested field at 0.1 mm spacing with 0 dB = 10 μm/s over 45 dB. The pristine panel has no physical flaw; its external marks identify the comparison coordinates.

support is 100%. The dominant indication lies at the lower-tip vicinity. The reported 0.0325 mm distance is a coordinate comparison, not an experimentally demonstrated localization accuracy finer than a pixel.

The pristine global maximum is 0.485 μm/s and occurs away from the flaw. The waveform-difference maximum is 8.791 μm/s at $(2.7, 21.1)$ mm. Taken together, the strong flawed response, weak pristine response and local difference response support an association with introducing the separation. The difference peak is allowed to exceed the LOF-only peak: subtraction changes signed interference, source means and coherence. It is not equivalent to subtracting the two displayed amplitudes.

A.6.4 Does the indication depend on a convenient crop or pixel size?

At 0.05 mm spacing the selected direct-S contrast is 12.63 dB. In the wider $x = \pm 45$ mm field at 0.25 mm spacing it is 10.62 dB and the global maximum remains near the flaw. This supports localization beyond one grid and one crop, while the changed support and background set explain why the numerical contrast need not remain identical.

Changing the angular admission level to 15% or 25% gives 10.16 and 13.79 dB with complete near-flaw support. Recomputing the launch curve for the selected low-pass leaves that image unchanged. These checks support stability of the selected interpretation, but not every path behaves as well. The common-protocol $SS \mid S$ wider-field control has $C_{\max} = -17.80$ dB: a much stronger lateral artefact appears when the field is expanded. Local success must therefore be checked against a suitably broad background region.

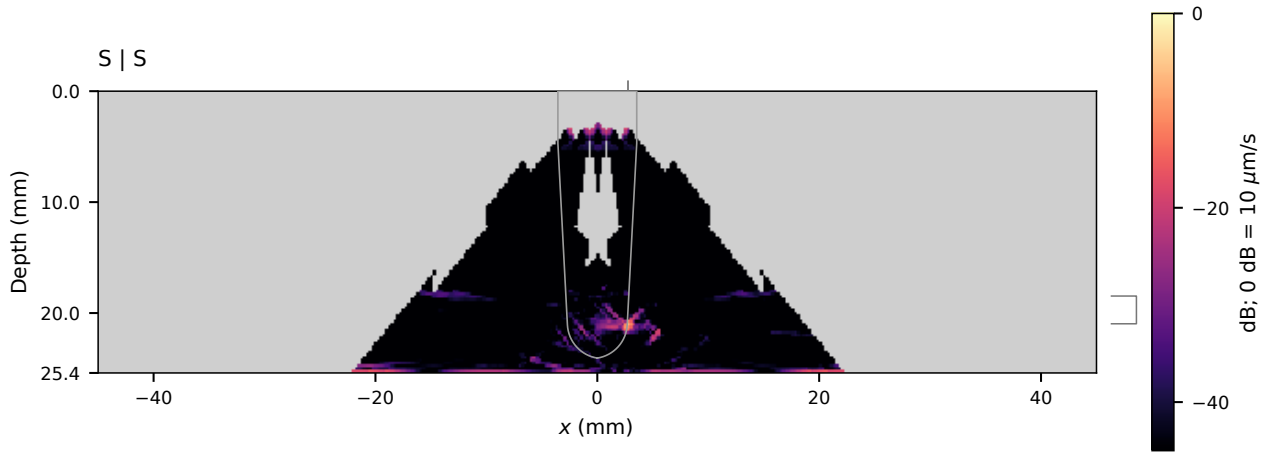


Figure A.10: Selected direct-S image over the expanded field, at 0.25 mm spacing. The amplitude reference and 45 dB range match Figure A.9. Grey areas are unassessed by the declared aperture rules. A wider image checks for competing supported maxima; it does not establish whole-specimen coverage.

A.7 Full weld: source and receiver comparisons

Whole physical groups are removed while holding the selected processing fixed. This asks where useful acoustic information is available. It does not search for an arbitrary subset of individual channels that happens to maximize the known-flaw metric.

Table A.4: Source and receiver group reductions at 0.1 mm. Direct S uses the selected short low-pass; direct L uses the common bandpass. Near support measures available grid pixels, not detected flaw length.

Path	Acquisition	$C_{\max}$ (dB)	Near support (%)
$S \mid S$	All sources and receivers	12.79	100.0
$S \mid S$	Shoulder sources only	10.57	92.4
$S \mid S$	Weld-surface sources only	-20.66	9.8
$S \mid S$	Side-surface receivers only	11.83	100.0
$S \mid S$	Weld-surface receivers only	-2.64	92.4
$S \mid S$	Shoulders and side-surface receivers	9.61	92.4
$L \mid L$	All sources and receivers	8.85	100.0
$L \mid L$	Shoulder sources only	8.85	100.0
$L \mid L$	Weld-surface sources only	–	0.0
$L \mid L$	Side-surface receivers only	6.43	100.0
$L \mid L$	Weld-surface receivers only	-11.77	100.0
$L \mid L$	Shoulders and side-surface receivers	6.43	100.0

For direct S, using only the two shoulder sources with all receivers gives 10.57 dB and 92.4% near support. Retaining all sources but receiving only on the side surfaces gives 11.83 dB and complete near support. Combining shoulder generation with side-surface reception gives 9.61 dB and 92.4%. This is a useful reduced physical arrangement: generation directly over the weld and reception on its narrow central surface are not essential to a local indication for this flaw.

Weld-surface generation alone is much less effective for direct S, giving negative contrast and only 9.8% near support. This agrees with the need for suitable oblique illumination and a sufficiently large qualified aperture. It does not prove that these sources radiate no waves toward the flaw; it means that the complete reconstruction and qualification rules do not retain a useful local image from that group alone.

Direct L is retained in the group comparison because changing mode also changes launch and receive sensitivity. One cannot choose a receiver distribution solely from spatial coverage and then assume it is optimal for every path. The relevant spread is a spread of useful ray angles and delays, evaluated per source and per path.

These group tests justify an acquisition direction, not the smallest receiver count. The side groups still contain densely spaced positions. A subsequent sparse design should preserve angular and timing diversity within each useful source/path aperture, compare results on common support, and be tested on withheld flaw cases and noise. The present result does not justify deleting the remaining channels from a validated instrument design.

A.8 Full weld: complete numerical comparisons

This section retains the complete evidence behind the preceding argument. First, the four-path atlas and table compare LOF, pristine and difference cases under the common protocol at 0.1 mm spacing. The filter definitions and matrix then compare fourteen processing variants on the 0.2 mm survey grid. Finally, the explicit ray expressions and confirmation register document the finer grids, wider fields and physical aperture tests. Values from different grids or fields should not be interpreted as changes in processing alone.

A.8.1 Wall-response gate and source–receiver exclusion

The wall gate (5%) and source–receiver exclusion (1.5 mm) are disabled separately and jointly, with all other controls fixed. The audit includes the common-band path surveys and all confirmations: 52 full-weld and 46 partial-weld configurations. Three perturbations per configuration and three signal cases give 294 reconstructions and 882 paired image comparisons. Repeating the principal nominal settings checks the archived arrays.

Image change is measured before display clipping, over the intersection Ω of supported pixels:

$$\delta_I = 100 \frac{\|I_{\text{test}} - I_{\text{nominal}}\|_{2,\Omega}}{\|I_{\text{nominal}}\|_{2,\Omega}} \ \%. \tag{A.13}$$

Support changes are assessed separately. Table A.5 gives the principal LOF-only view; its support mask is identical in all four cases.

Table A.5: Independent and joint gate checks for the principal full-weld $S \mid S$ image at 0.1 mm spacing. Zero disables the corresponding gate. δ_I measures whole-image amplitude change on common support; it is not a change in detection probability.

Wall gate / exclusion	$C_{\max}$ (dB)	δ_I (%)	Near support (%)
5% / 1.5 mm	12.79	0.00	100.0
0% / 1.5 mm	12.79	0.00	100.0
5% / 0 mm	13.12	6.56	100.0
0% / 0 mm	13.12	6.56	100.0

Disabling the wall gate alone leaves all 52 full-weld configurations exactly unchanged in all three signal cases. Direct paths have no intermediate wall reflection; the tested reflected paths also give identical images. The 5% threshold is inactive here but retained in the shared protocol because multiple-reflection partial-weld routes are sensitive to it.

Disabling separation changes the principal image by 6.56% and raises $C_{\max}$ from 12.79 to 13.12 dB without changing support. Disabling both controls gives the same image. This modest contrast increase for one known flaw does not establish that sampling within the source footprint is preferable.

The 1.5 mm exclusion avoids the 1.425 mm effective source footprint, where directly applied loading complicates interpretation as a remotely received wave. It does not guarantee background removal. The exclusion distance and 5% wall threshold are declared engineering safeguards; neither has been established as a universal optimum.

A.8.2 Complete common-protocol path and control comparison

All four LOF-only maxima lie within 0.37 mm of the prescribed segment. Direct S and L emphasize the lower-tip vicinity; reflected shear views give broader and less selective indications. Near support is 100% in each full-weld field. Table A.6 retains the complete three-case numerical comparison independently of display clipping.

Table A.6: Common-protocol controls at 0.1 mm. I_{pk} is global peak velocity score; d_{pk} is distance from the finite prescribed LOF. N denotes near-flaw support.

Path	Case	I_{pk} (μm/s)	$C_{\max}$ (dB)	C_{99} (dB)	N (%)	d_{pk} (mm)
$S \mid S$	LOF	4.347	6.47	15.57	100.0	0.004
$S \mid S$	Pristine	2.061	-21.39	-10.22	100.0	13.805
$S \mid S$	Difference	8.304	16.13	22.48	100.0	0.133
$L \mid L$	LOF	2.081	8.85	19.11	100.0	0.095
$L \mid L$	Pristine	0.343	-21.94	-12.20	100.0	7.508
$L \mid L$	Difference	2.220	9.10	15.14	100.0	0.033
$S \mid SS$	LOF	8.731	4.65	7.64	100.0	0.080
$S \mid SS$	Pristine	2.320	-4.38	-3.04	100.0	13.175
$S \mid SS$	Difference	18.451	8.34	14.16	100.0	0.269
$SS \mid S$	LOF	3.957	4.37	8.27	100.0	0.369
$SS \mid S$	Pristine	0.840	-6.52	-5.82	100.0	3.321
$SS \mid S$	Difference	4.707	4.49	11.08	100.0	0.030

The pristine and difference panels help distinguish structural response from changes caused by the flaw. They do not establish that pristine subtraction would be available in a real inspection. Different source means, cancellation and qualification can move the peak after subtraction. A blend of paths is a useful display of complementary responses; it is not a statistically justified fusion score or a direct measurement of LOF length.

See the shared filter definitions and variants in Section A.5.1.

A.8.3 Complete filter/path outcomes

The ranking is path dependent. At 0.1 mm, direct-S contrast rises from 6.47 dB for the common bandpass to 10.26 dB for the shorter bandpass with matched guard, then 12.79 dB for the shorter low-pass with that guard. The improvement depends on both shorter temporal averaging and retained low-frequency content. It is not caused by admitting earlier times. Direct L instead falls from 8.85 to 3.24 dB under the same shorter low-pass, so a universal filter replacement is unsupported.

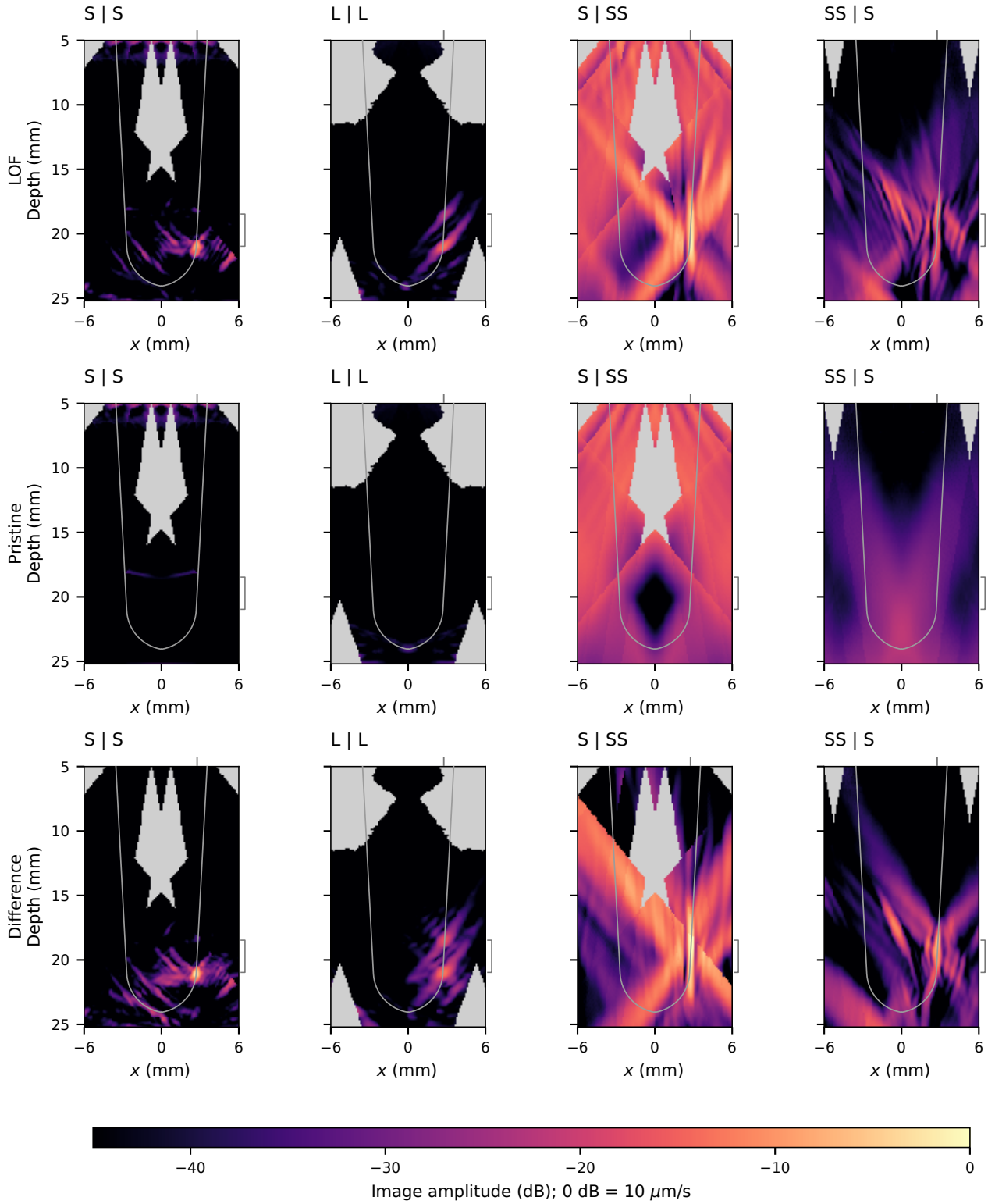


Figure A.11: All four full-weld paths at 0.1 mm spacing under the common 0.5–5 MHz protocol. Rows show LOF, pristine and signed-waveform-difference reconstructions. All panels share a 10 μm/s reference and 45 dB range; grey is unsupported. External LOF guides indicate a coordinate on pristine panels, whose models contain no flaw.

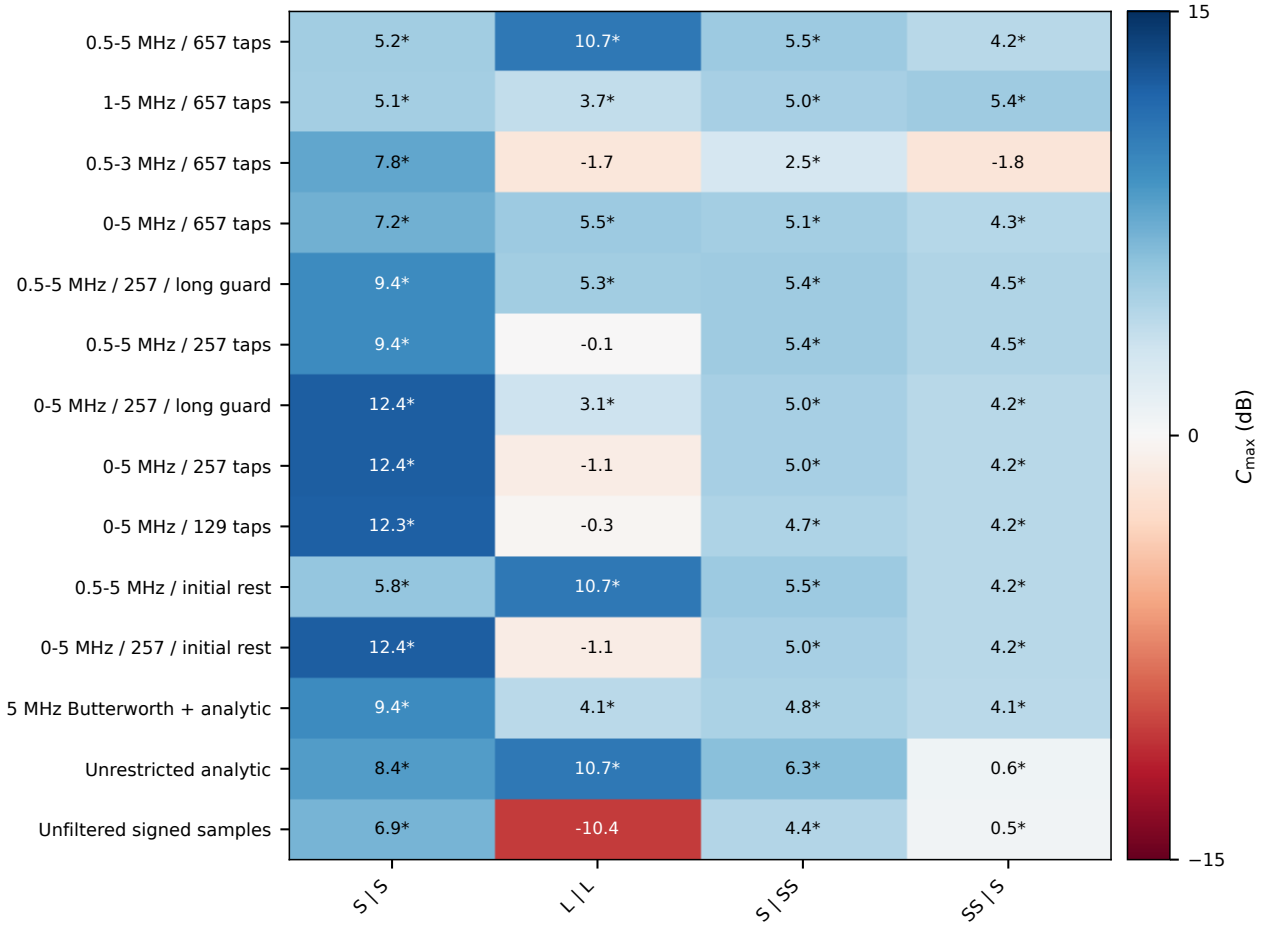


Figure A.12: All fourteen processing variants for the four full-weld paths at 0.2 mm spacing. Entries are LOF-only $C_{\max}$ in dB; asterisks mark global maxima within 0.5 mm of the flaw. Colours saturate at ±15 dB but values and arrays remain untruncated. The survey grid is coarser than the confirmation tables.

A.8.4 Explicit full-weld delays and background coincidences

For surface coordinate q and pixel $p = (x, z)$, define $d(q,p) = \sqrt{(x-q)^2 + z^2}$ and $b(q,p) = \sqrt{(x-q)^2 + (2h-z)^2}$, with $h = 25.4$ mm. The four delays are

$$T_{S|S} = t_0 + [d(x_s,p) + d(x_r,p)]/c_S, \qquad T_{L|L} = t_0 + [d(x_s,p) + d(x_r,p)]/c_L, \tag{A.14}$$

$$T_{S|SS} = t_0 + [d(x_s,p) + b(x_r,p)]/c_S, \qquad T_{SS|S} = t_0 + [b(x_s,p) + d(x_r,p)]/c_S. \tag{A.15}$$

The reflected branches enforce specular same-mode reflection at the backwall. A near-vertical embedded separation is not generally specular for a top-to-flaw-to-top route; tip-associated scattering can dominate the direct image. Backwall-assisted illumination motivates tandem views [10], but a bright reflected arc need not be a mapped flaw face.

All 1685 source–receiver pairs have unobstructed direct-family geometry for nuisance prediction. This does not isolate their measured modes. The recordings contain much stronger source-generated bulk/surface packets and boundary returns. For a symmetric pair separated by d, a direct nuisance wave of speed c_n can coincide with a direct imaging hypothesis of speed $c_i > c_n$ at

$$z_{\text{ghost}} = \frac{d}{2}\sqrt{(c_i/c_n)^2 - 1}. \tag{A.16}$$

With $d = 20$ mm, L focusing can map direct S motion to about 15.4 mm depth, while S focusing can map Rayleigh motion to about 4.1 mm. Finite pulses broaden these loci. A 5 mm grid minimum or a near-source receiver exclusion therefore cannot guarantee clean deeper images.

A.8.5 Complete confirmation register

Table A.7 lists every finer/wider, group and gate confirmation, including pristine and difference contrasts. Short names refer to the filter definitions above. Grid entries are pixel spacing in mm; "wide" changes the field to $x = \pm 45$ mm. All other unlisted controls remain those of the stated protocol. Missing contrasts remain undefined. The portable arrays retain support, coherence and screened fractions for all three cases.

Table A.7: Complete full-weld confirmations. BP denotes 0.5–5 MHz and LP 0–5 MHz; G328 retains the 3.28 μs guard. Rest uses known zero prehistory. L/P/D are LOF, pristine and difference $C_{\max}$ in dB; N is LOF near support. Src/rx denote source/receiver group reductions; weld and side identify the weld surface and side surfaces.

Filter	Path	Grid	Change	L	P	D	N (%)
BP657	$L \mid L$	0.1	weld src	–	–	–	0.0
BP657	$L \mid L$	0.1	all	8.85	-21.94	9.10	100.0
BP657	$L \mid L$	0.1	weld rx	-11.77	-31.02	-5.63	100.0
BP657	$L \mid L$	0.1	side rx	6.43	-23.43	7.70	100.0
BP657	$L \mid L$	0.1	shoulder src	8.85	-21.94	9.10	100.0
BP657	$L \mid L$	0.1	shoulder+side	6.43	-23.43	7.70	100.0
BP657	$L \mid L$	0.25	wide	6.90	-29.50	8.70	100.0
BP657	$S \mid S$	0.1	weld src	-51.58	-47.53	-47.87	9.8
BP657	$S \mid S$	0.1	all	6.47	-21.39	16.13	100.0
BP657	$S \mid S$	0.05	all	6.63	-21.26	16.13	100.0
BP657	$S \mid S$	0.1	weld rx	-2.72	-0.27	2.73	92.4
BP657	$S \mid S$	0.1	side rx	4.01	-23.52	15.00	100.0
BP657	$S \mid S$	0.1	shoulder src	8.05	-7.91	13.64	92.4
BP657	$S \mid S$	0.1	shoulder+side	7.19	-7.28	12.38	92.4
BP657	$S \mid S$	0.25	wide	3.22	-28.90	10.29	100.0
BP657	$S \mid SS$	0.1	weld src	-14.43	-9.48	-8.64	9.8

Continued from previous page

Filter	Path	Grid	Change	L	P	D	N (%)
BP657	$S \mid SS$	0.1	all	4.65	-4.38	8.34	100.0
BP657	$S \mid SS$	0.1	weld rx	–	–	–	0.0
BP657	$S \mid SS$	0.1	side rx	4.84	-4.18	8.37	100.0
BP657	$S \mid SS$	0.1	shoulder src	5.28	-2.04	9.24	92.4
BP657	$S \mid SS$	0.1	shoulder+side	5.28	-1.53	8.82	92.4
BP657	$S \mid SS$	0.25	wide	4.88	-5.31	8.07	100.0
BP657	$SS \mid S$	0.1	weld src	–	–	–	0.0
BP657	$SS \mid S$	0.1	all	4.37	-6.52	4.49	100.0
BP657	$SS \mid S$	0.1	weld rx	-2.47	-0.43	-0.24	100.0
BP657	$SS \mid S$	0.1	side rx	4.92	-6.82	4.41	100.0
BP657	$SS \mid S$	0.1	shoulder src	4.37	-6.52	4.49	100.0
BP657	$SS \mid S$	0.1	shoulder+side	4.92	-6.82	4.41	100.0
BP657	$SS \mid S$	0.25	wide	-17.80	-36.66	-6.98	100.0
BP257/G328	$L \mid L$	0.1	all	4.96	-34.09	8.40	100.0
BP257/G328	$S \mid S$	0.1	all	10.26	-10.37	16.04	100.0
BP257/G328	$S \mid SS$	0.1	all	4.80	-4.57	8.26	100.0
BP257/G328	$SS \mid S$	0.1	all	4.65	-6.46	4.61	100.0
LP257/G328	$L \mid L$	0.1	all	3.24	-36.97	5.97	100.0
LP257/G328	$S \mid S$	0.1	all	12.79	-5.90	16.84	100.0
LP257/G328	$S \mid S$	0.05	all	12.63	-6.57	16.68	100.0
LP257/G328	$S \mid SS$	0.1	all	4.97	-3.75	8.58	100.0
LP257/G328	$SS \mid S$	0.1	all	4.32	-6.00	4.65	100.0
LP257/G328	$S \mid S$	0.1	all	12.79	-5.90	16.84	100.0
LP257/G328	$S \mid S$	0.1	weld src	-20.66	-19.08	-38.32	9.8
LP257/G328	$S \mid S$	0.1	weld rx	-2.64	-0.32	2.72	92.4
LP257/G328	$S \mid S$	0.1	gate 15%	10.16	-4.12	15.10	100.0
LP257/G328	$S \mid S$	0.1	gate 25%	13.79	-14.49	17.40	100.0
LP257/G328	$S \mid S$	0.1	LP launch	12.79	-5.90	16.84	100.0
LP257/G328	$S \mid S$	0.1	side rx	11.83	-5.27	15.60	100.0
LP257/G328	$S \mid S$	0.1	shoulder src	10.57	-4.14	14.23	92.4
LP257/G328	$S \mid S$	0.1	shoulder+side	9.61	-3.51	12.99	92.4
LP257/G328	$S \mid S$	0.25	wide	10.62	-20.09	10.83	100.0

A.9 Partial weld: original path comparison

A.9.1 Direct S illumination with a reflected S return

The selected hypothesis is $S \mid SS$: one S leg from source to trial pixel, followed by an S return through one backwall reflection. Weld-surface sources can illuminate the lower weld without traversing the empty groove. The reflected receiving branch reaches a side surface with a different angle and later arrival than a direct branch. The worked example illustrates why this can be geometrically useful, while the following images test whether the actual scattered field supports it.

The conventional catalogue contains eight pure-S splits, eight pure-L splits and the direct conversion hypotheses $L \mid S$ and $S \mid L$. All use the same common 657-tap 0.5–5 MHz filter and declared selection rules. Some hypotheses are weak, some are contaminated, and some have insufficient support. Their complete images remain in Section A.13.2; showing only the selected path without those controls would overstate general success.

A.9.2 What is available from all five source shots?

With all sources and receivers, the LOF-only global maximum is 16.568 μm/s at $(2.8, 19.95)$ mm, 0.0458 mm from the prescribed segment. Maximum-background contrast is 4.36 dB, percentile

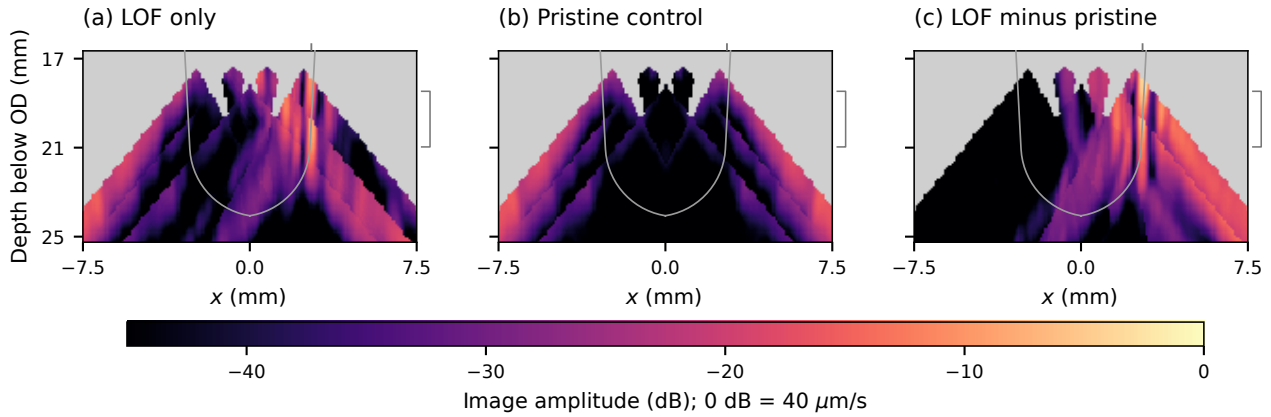


Figure A.13: All-source/all-receiver $S \mid SS$ reconstruction at 0.1 mm spacing, with pristine and waveform-difference controls. All panels use 0 dB = 40 μm/s over 45 dB. The first grid centre is 0.25 mm below the weld surface. Grey represents failure of the coverage rules, not zero recorded velocity. External marks indicate the prescribed separation without covering the reconstructed response.

contrast is 8.74 dB and near-flaw support is 95.8%. The near-flaw maximum therefore exceeds the strongest distant supported response, although the separation is less pronounced than a high-contrast isolated target might suggest.

The pristine control peaks at 6.820 μm/s at $(-6.9, 23.25)$ mm. The waveform difference peaks at 32.743 μm/s at $(2.8, 18.05)$ mm and has 11.09 dB contrast. The contrast and position changes demonstrate that introducing the LOF affects the measured field. They do not mean that the difference panel measures an additional flaw, or that a pristine reference must be used in the LOF-only reconstruction. Different signed interference and coherence after subtraction can move and strengthen the maximum.

A.10 Partial weld: original source and receiver comparisons

A.10.1 What whole-group tests reveal

The source and receiver play different physical roles. A weld-surface source can illuminate the lower weld directly, while a receiver on that same surface may encounter short-time contamination or an unfavourable return angle. A useful source position therefore need not be a useful receiving location for the same path.

Table A.8: Separate source or receiver group changes for $S \mid SS$, with the common processing and 0.1 mm grid. A dash denotes no near-flaw support under the declared qualification rules.

Acquisition	$C_{\max}$ (dB)	Near support (%)
All sources and receivers	4.36	95.8
Shoulder sources only	–	0.0
Weld-surface sources only	8.77	48.5
Side-surface receivers only	4.33	95.8
Weld-surface receivers only	-4.82	95.8
Shoulders and side-surface receivers	–	0.0

Shoulder generation alone provides no near-flaw image under the two-source qualification rule. This means insufficient admitted evidence for an image; it does not mean that the shoulder sources launch no flaw-interacting waves. Using only the three weld-surface sources with all receivers gives 8.77 dB contrast but 48.5% near support. The all-source case has lower contrast but substantially broader

support. Complementary source shots therefore remain useful even when their addition lowers a local contrast metric.

Keeping all sources and receiving only on the two side surfaces gives 4.33 dB and 95.8% near support, close to the all-receiver result. Receiving only on the weld surface instead gives negative maximum-background contrast despite high support. These comparisons motivate the joint combination of weld-surface generation and side-surface reception. They also explain why simply placing every optical measurement on the nearest available surface is not sufficient.

A.10.2 Left, right or both receiving sides?

The joint comparison fixes all three weld-surface sources S1–S3 and tests both side receiver groups, the left group alone and the right group alone. Each side has 157 positions at 0.25 mm pitch. The filter, path, thresholds and pixel spacing remain fixed. No individual receiver IDs are fitted to the known LOF.

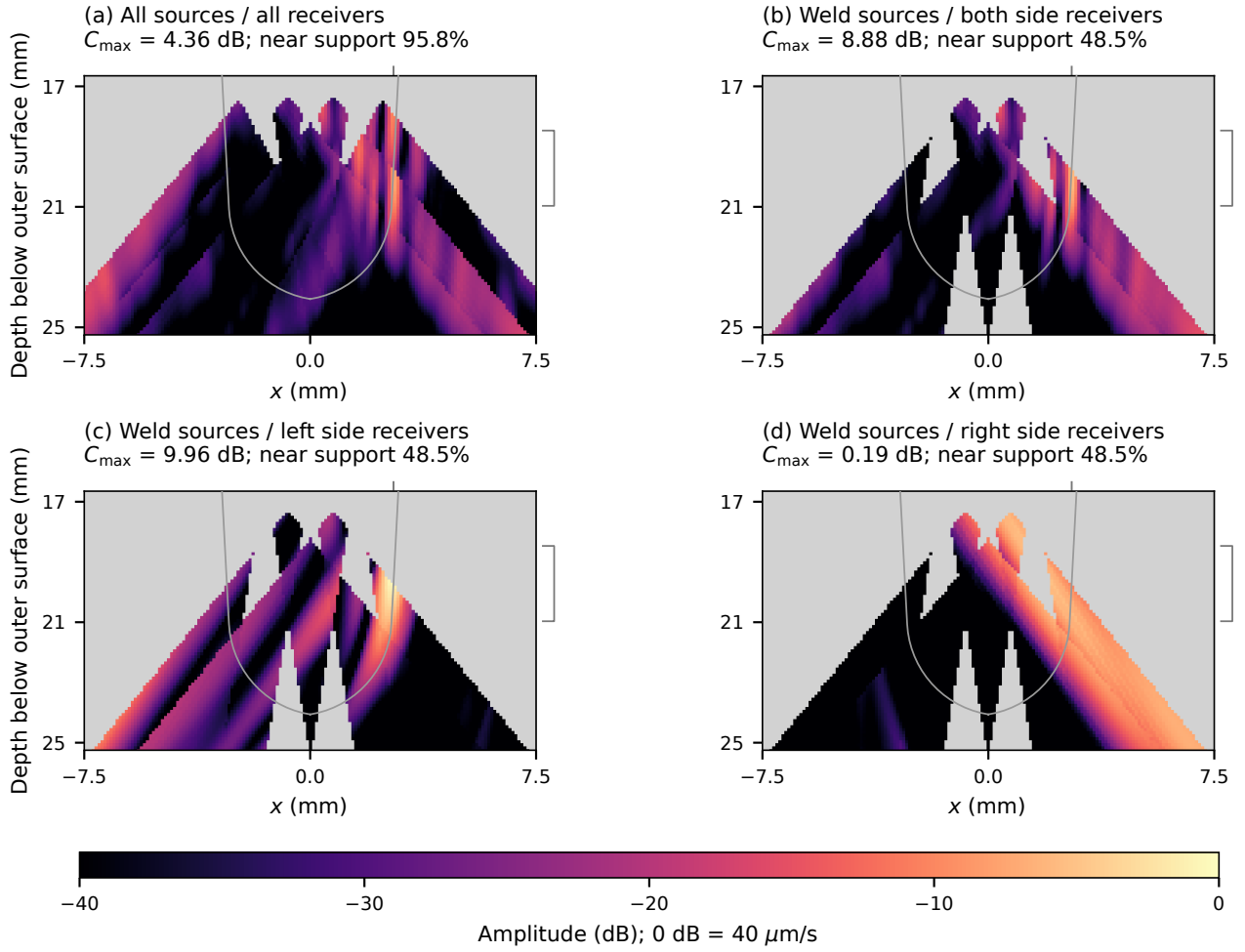


Figure A.14: Joint acquisition comparison for $S \mid SS$ with a shared 40 μm/s reference over 40 dB. The all-source image is compared with three weld-surface-source images using both, left or right side receivers. The clearer left-side indication comes with reduced coverage relative to the all-source image. Compare both the coloured response and the grey unsupported region.

Left-side reception gives 9.96 dB contrast and a global peak of 37.21 μm/s at $(2.7, 19.75)$ mm, 0.065 mm from the prescribed segment. Both sides together give 8.88 dB. Right-side reception alone gives only 0.19 dB, with its global maximum 0.664 mm from the segment, outside the 0.5 mm near-flaw

Table A.9: Joint acquisition metrics at 0.1 mm. C_{own} evaluates each image's supported field; C_{shared} uses their identical common supported subset. N denotes near-flaw support and d_{pk} the global-peak distance from the finite LOF.

Acquisition	C_{own} dB	N %	C_{shared} dB	d_{pk} mm
All sources / all receivers	4.36	95.8	7.02	0.046
Weld sources / both side surfaces	8.88	48.5	8.88	0.046
Weld sources / left side surface	9.96	48.5	9.96	0.065
Weld sources / right side surface	0.19	48.5	0.19	0.664

region. Left-side reception is therefore a useful reduced localizing view for this right-flank LOF and this path.

The improvement over both sides is 1.09 dB, not an order-of-magnitude change. Its physical plausibility comes from directional scattering and interference: the two sides observe different angles and different structural returns. Adding the second side can reduce coherent reinforcement of the useful signal or add coherent background. The present comparison does not uniquely separate those mechanisms, and it does not establish that left-side reception is best for a left-flank flaw or a different path.

A.10.3 The coverage cost must remain part of the result

All three weld-source acquisitions have the same supported pixels and only 48.5% near-flaw support. The all-source/all-receiver image supports 95.8%. Thus the 9.96 dB left-side score cannot be compared uncritically with the all-source 4.36 dB score: the latter also assesses a larger background and near-flaw region.

On the common supported set, the all-source result is 7.02 dB. The corresponding left-side advantage is 2.94 dB. This comparison isolates the change in reconstructed amplitude/background from the loss of available evidence. It still does not restore coverage to the omitted region. Figure A.15 displays that distinction directly.

For the left-side acquisition, the pristine contrast is -33.09 dB and the waveform-difference contrast is 9.86 dB. Both-side reception gives -10.36 and 7.95 dB for the corresponding controls. These support a flaw-related indication without introducing reference subtraction into the LOF-only image. They do not turn an unsupported region into an inspected region.

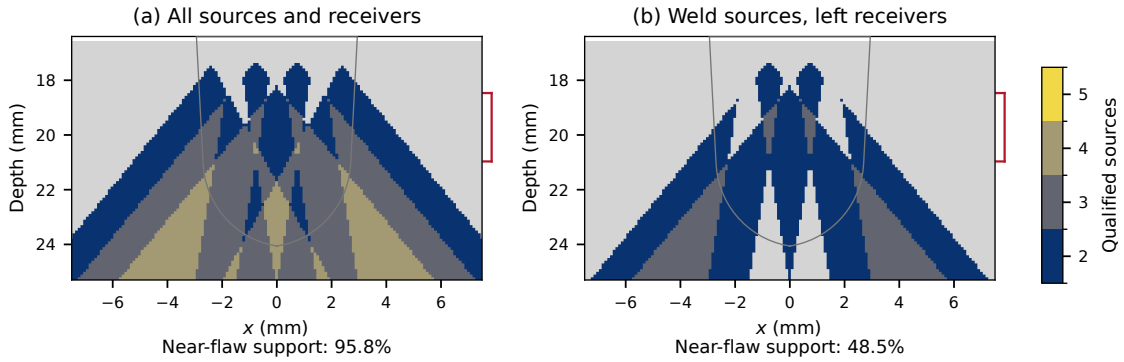


Figure A.15: Number of qualifying source views at each supported pixel for the all-source acquisition and the weld-source/left-receiver acquisition. The grey region has fewer than the required two qualified sources; white is air. Each qualified source separately satisfies the receiver-count and retained-fraction rules. This is a coverage map, not an amplitude image. Near-flaw support of 95.8% and 48.5% refers to area within 0.5 mm of the finite LOF, not the fraction of its length detected.

A.11 Partial weld: original sensitivity checks

A.11.1 Finer pixels and a wider field

At 0.05 mm spacing, left-side reception gives 10.26 dB and both-side reception 8.59 dB, with 47.8% near support. On the wider field at 0.25 mm spacing, they give 9.11 and 9.20 dB, with 43.4% near support and near-flaw global maxima. The useful localization survives both checks. The left-side contrast advantage does not persist in every field, so a universal claim of superiority would be too strong.

The all-source acquisition gives 4.36 dB at 0.05 mm and 6.43 dB in the wider field, with near-flaw maxima. These are separate coverage-rich controls. Their contrasts cannot be interpreted as gains from changing pixel size alone, because the evaluated background field and discrete support also change.

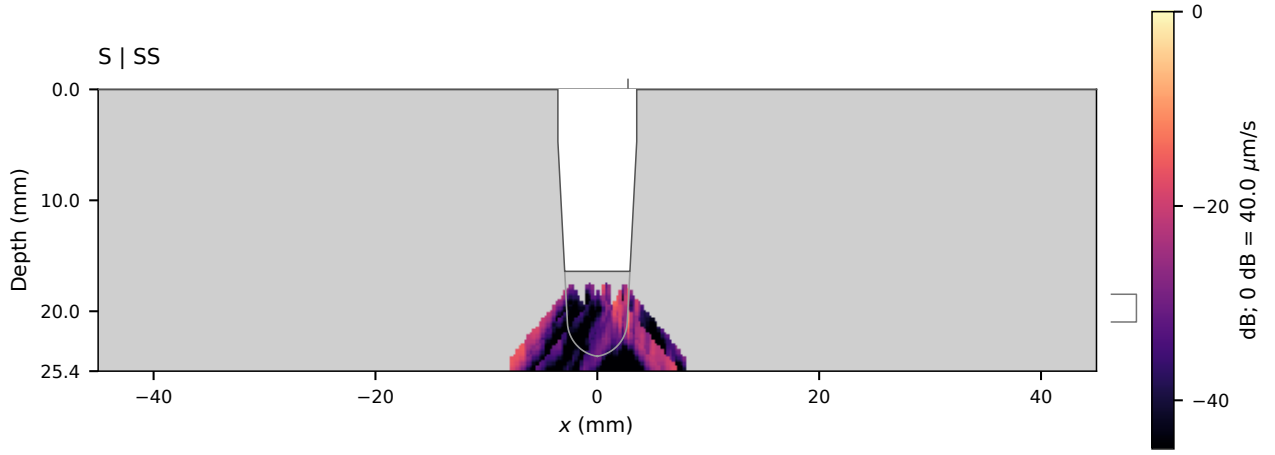


Figure A.16: All-source $S \mid SS$ control over the wider field at 0.25 mm spacing. The common reference is 40 μm/s over 45 dB. White identifies the empty groove, grey unsupported steel. This confirms a near-flaw global supported maximum for the broader-coverage acquisition; it does not imply inspection of every grey region.

A.11.2 Filtering and additional paths

For the all-source $S \mid SS$ confirmation, changing the common filter to 1–5 MHz gives 4.20 dB; shortening the 0.5–5 MHz kernel gives 4.29 dB; using the short 0–5 MHz low-pass gives 4.34 dB. The guard remains fixed and near support remains 95.8%. These small changes support keeping the common bandpass for the partial-weld recipe. They are not a full filter optimization repeated for the one-sided acquisition.

The eighteen-path joint survey with weld-surface sources and both receiving sides has only one route with positive maximum-background contrast and a near-flaw global maximum: $S \mid SS$. The left-only acquisition is a subsequent comparison for this route, not a complete one-sided survey of all eighteen paths. This limits the claim to a justified selected view rather than the best possible path/side combination.

More complicated paths can sample strong reverberations, lose usable receivers or accumulate timing-model error. Separate finite-bevel and corner hypotheses were also investigated. None improved the evidence under the tested model; the corner extension assumed a unit diffraction amplitude and therefore cannot rule out a better physical diffraction model. The conventional reconstruction already rejects rays through the groove. Failure of an additional path is not evidence that geometry was ignored, or that the associated elastic wave does not exist.

A.12 Partial weld: one-source and two-source requirements

A.12.1 A coverage rule, not a choice of a single laser

The minimum-source requirement applies separately at each pixel. Lowering it from two to one does not remove the other source shots: wherever two or three weld-surface sources qualify, their images are still combined exactly as before. It allows a pixel to be displayed when just one source has a sufficiently large, sufficiently coupled and sufficiently retained receiver aperture. The eight-receiver and 25% retained-fraction thresholds remain in force, as do geometry, filtering, pulse screening and receiver coherence.

For a newly admitted pixel with only one qualified source, Equation A.8 reduces to

$$I_P(p) = |C_s M_s|, \qquad K = 1. \tag{A.17}$$

There is still coherent combination across receivers; what is missing is another qualifying source view. The nominal two-source rule is a corroboration and coverage safeguard, not an elastic-law prohibition against imaging with one shot.

A.12.2 The controlled comparison

The test changes only the required number of sources within each acquisition. It includes all-source/all-receiver, weld-source/both-side and weld-source/left-side configurations at 0.1 mm, a finer left-side grid and wider left/both-side fields. Every configuration has LOF, pristine and waveform-difference controls. Two native replays reproduce the two-source baselines.

With weld-surface generation and left-side reception, near-flaw support rises from 48.5% to 95.8%. The indication is stronger in absolute amplitude, but background rises more: the peak becomes 46.15

Table A.10: One versus two required sources for partial-weld $S \mid SS$, at 0.1 mm with all other settings fixed. N is near-flaw support. Contrast uses each resulting supported field. All already-supported pixel amplitudes remain unchanged; newly admitted pixels have exactly one qualifying source.

Acquisition	Minimum sources	$C_{\max}$ dB	N %	d_{pk} mm
All sources / all receivers	2	4.36	95.8	0.046
All sources / all receivers	1	2.15	100.0	0.046
Weld sources / both sides	2	8.88	48.5	0.046
Weld sources / both sides	1	3.06	95.8	0.046
Weld sources / left side	2	9.96	48.5	0.065
Weld sources / left side	1	3.72	95.8	0.325

μm/s, while the strongest distant background becomes 30.07 μm/s. Consequently, $C_{\max}$ falls from 9.96 to 3.72 dB. The global maximum moves from $(2.7, 19.75)$ mm to $(2.9, 18.15)$ mm, 0.325 mm from the finite prescribed segment and above its upper tip. It remains inside the declared 0.5 mm evaluation neighbourhood, but is no longer the same interior indication.

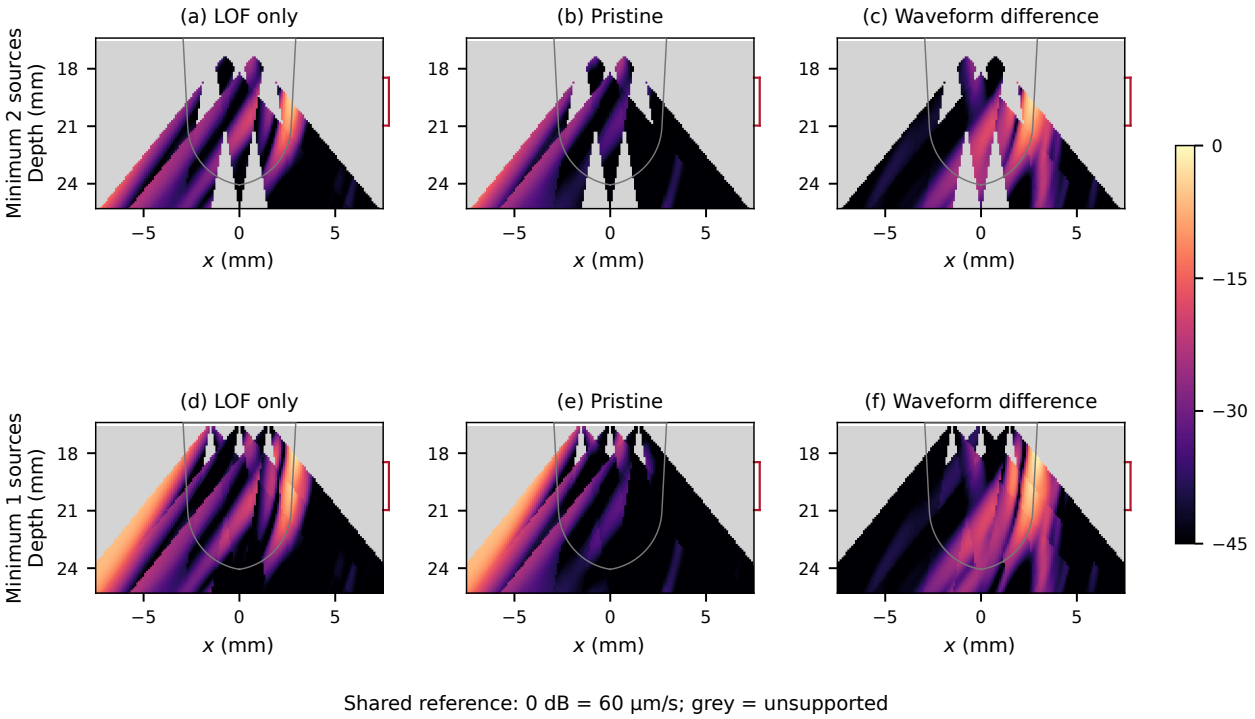


Figure A.17: Two-source and one-source admission for weld-surface generation and left-side reception. Columns show LOF-only, pristine and waveform-difference images. A shared 60 μm/s reference and 45 dB range keep every panel on the same amplitude scale without saturating the new peaks. Lowering the threshold fills previously grey pixels; it does not alter the values of the pixels already supported by two sources. The new regions include both a near-upper-tip response and substantial structural background.

With all sources and receivers, near-flaw support rises from 95.8% to 100%, while contrast falls from 4.36 to 2.15 dB. Its global peak and position remain unchanged. With the three weld-surface sources and both receiver sides, support rises from 48.5% to 95.8% and contrast falls from 8.88 to 3.06 dB, again without moving the global peak. Thus the large coverage recovery is especially relevant to weld-source-only imaging, but it is not a free gain in selectivity.

On the common supported pixels, every paired amplitude agrees with its two-source value within the numerical replay tolerance. The native checks also confirm unchanged source counts there, and exactly one source at every newly admitted pixel. This establishes the mechanism directly: the threshold changes which pixels are assessed, rather than changing the focusing operation at existing pixels. A matched-support contrast would therefore be unchanged and cannot by itself describe the value or risk of the recovered region.

A.12.3 Do the new pixels behave like useful additional evidence?

For the one-source left-side image, the pristine contrast is −37.53 dB and the difference contrast is 10.02 dB. The near-flaw response is therefore not accompanied by a comparably dominant pristine response at the same coordinates. However, the pristine image admits a 30.91 μm/s structural maximum elsewhere. This supports treating the recovered near-flaw response as informative while retaining the visible evidence of increased background.

At 0.05 mm spacing, one-source left reception gives 3.73 dB and 95.8% near support. Over the wider field it gives 3.22 dB and 94.3%, with a global maximum at $(2.75, 18.0)$ mm, 0.475 mm from the LOF. Both-side reception over that wider field gives 4.87 dB with the same near support and a maximum only 0.012 mm from the segment. Thus the left-only preference is not universal when the qualification rule and assessed field change. The upper-end response also approaches the boundary of the declared near-flaw neighbourhood; it should not be described as a precisely located flaw tip.

The useful interpretation is to retain the two-source image as the cleaner localizing view and provide the one-source image as an explicitly marked broader-coverage view. One-source pixels have valid receiver support, but less source-view corroboration. This is more informative than either hiding all of them or presenting their expanded coverage as equally selective evidence. The test does not establish that a physically single-source inspection would reproduce the result, since the accepted source can differ from pixel to pixel.

A.13 Partial weld: complete baseline comparisons

The main findings above use the confirmed 0.1 mm results and selected finer/wider checks. This section supplies the full reproducible record: the eighteen-path catalogue and all signal controls at 0.2 mm, the fourteen processing variants and their matrices, additional geometric hypotheses, timing estimates and all confirmation configurations. A path comparison, a filter comparison and a change of image field answer different questions; their grids and controls are stated separately.

A.13.1 Wall-response gate and source–receiver exclusion

The wall gate (5%) and source–receiver exclusion (1.5 mm) are disabled separately and jointly, with all other controls fixed. The audit includes the common-band path surveys and all confirmations: 52 full-weld and 46 partial-weld configurations. Three perturbations per configuration and three signal cases give 294 reconstructions and 882 paired image comparisons. Repeating the principal nominal settings checks the archived arrays.

Image change is measured before display clipping, over the intersection Ω of supported pixels:

$$\delta_I = 100 \frac{\|I_{\text{test}} - I_{\text{nominal}}\|_{2,\Omega}}{\|I_{\text{nominal}}\|_{2,\Omega}} \ \%. \tag{A.18}$$

Support changes are assessed separately. Table A.11 gives the principal LOF-only view; its support mask is identical in all four cases.

Table A.11: Independent and joint gate checks for the principal partial-weld $S \mid SS$ image at 0.1 mm spacing. Zero disables the corresponding gate. δ_I measures whole-image amplitude change on common support; it is not a change in detection probability.

Wall gate / exclusion	$C_{\max}$ (dB)	δ_I (%)	Near support (%)
5% / 1.5 mm	4.36	0.00	95.8
0% / 1.5 mm	4.36	0.00	95.8
5% / 0 mm	3.52	8.18	95.8
0% / 0 mm	3.52	8.18	95.8

Disabling the wall gate leaves the principal $S \mid SS$ image exactly unchanged. It is not redundant for the complete path family: on the 0.2 mm common-band grid, the LOF-only changes are 49.81% for $S \mid SSS$, 7.73% for $SS \mid SS$ and 39.84% for $SS \mid SSS$. These routes have more intermediate reflections, and the gate acts on the product of their predicted coefficients. The percentages describe changes in the reconstructed field, not gains in flaw detection. The same nominal wall gate is retained throughout the family; it does not explain the principal indication.

Disabling the separation rule changes the principal image by 8.18% and reduces $C_{\max}$ from 4.36 to 3.52 dB, with unchanged near support. Disabling both controls produces exactly the same principal image as disabling separation alone. The exclusion therefore has a measurable effect even though the focused flaw lies away from the source.

The 1.5 mm exclusion avoids the 1.425 mm effective source footprint, where directly applied loading complicates interpretation as a remotely received wave. It does not guarantee background removal. The exclusion distance and 5% wall threshold are declared engineering safeguards; neither has been established as a universal optimum.

A.13.2 The eighteen-path catalogue and all signal controls

For each pure mode M (L or S), the catalogue includes $M \mid M$, $M \mid MM$, $MM \mid M$, $M \mid MMM$, $MM \mid MM$, $MMM \mid M$, $MM \mid MMM$ and $MMM \mid MM$. In the four-leg split $MM \mid MM$, each branch has one backwall reflection. A three-leg branch has backwall and exposed-top reflections. The two direct conversion paths $L \mid S$ and $S \mid L$ complete the eighteen candidates. In the latter pair the mode changes at the pixel, not at a wall.

Figures A.18–A.20 show every primary path and signal case on the 0.2 mm survey grid using the same 40 μm/s reference and 55 dB range. The paired numerical table gives near amplitude, global-peak location relative to the flaw, contrast and support. A small supported region or a clean black image is not evidence of successful background elimination.

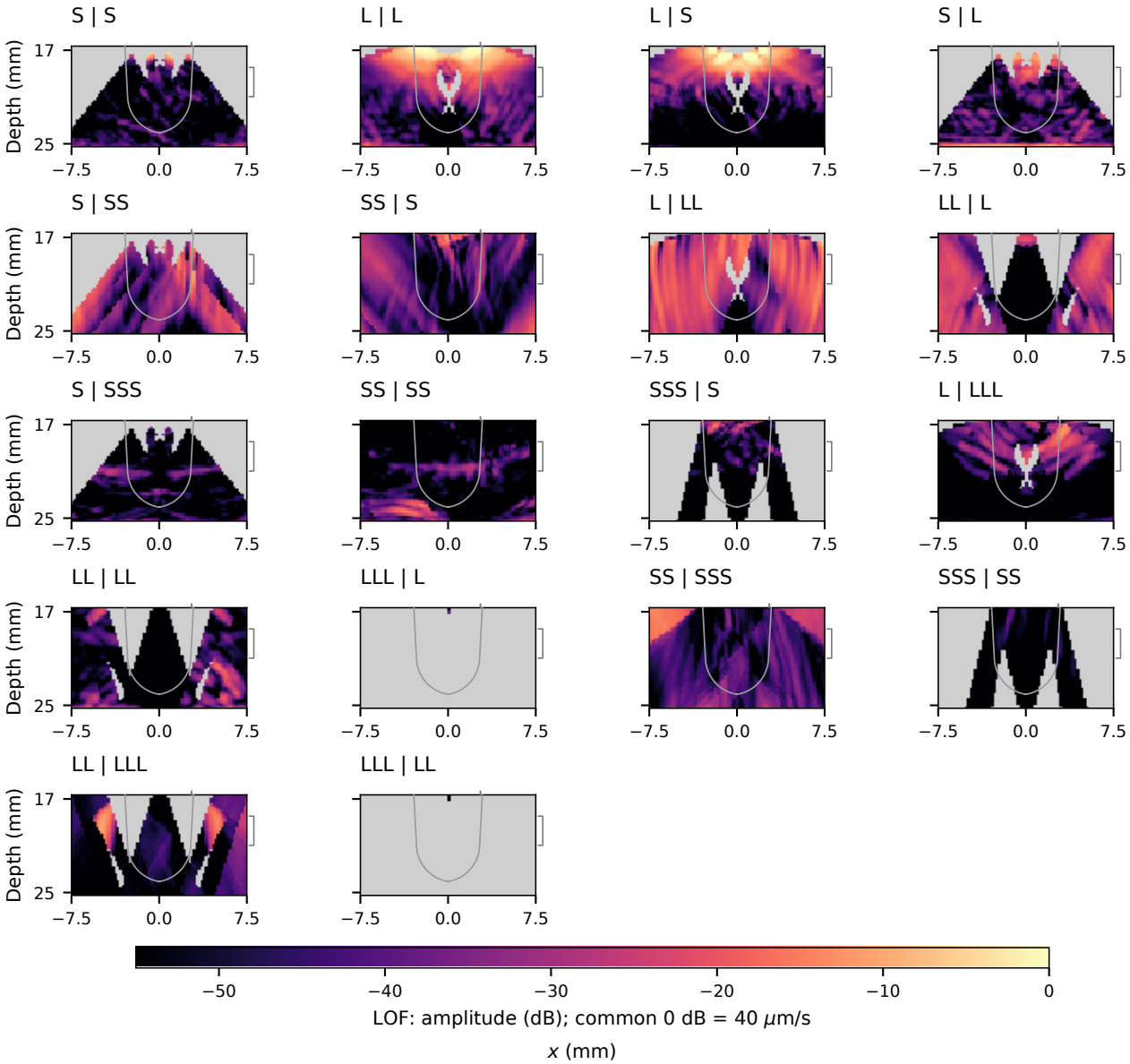


Figure A.18: All eighteen partial-weld LOF-only images under equal receiver averaging and the common bandpass, at 0.2 mm spacing. External marks locate the known flaw without covering the reconstructed signals. Air is white and unsupported material grey. Colour saturates above the common reference; numerical scores are untruncated.

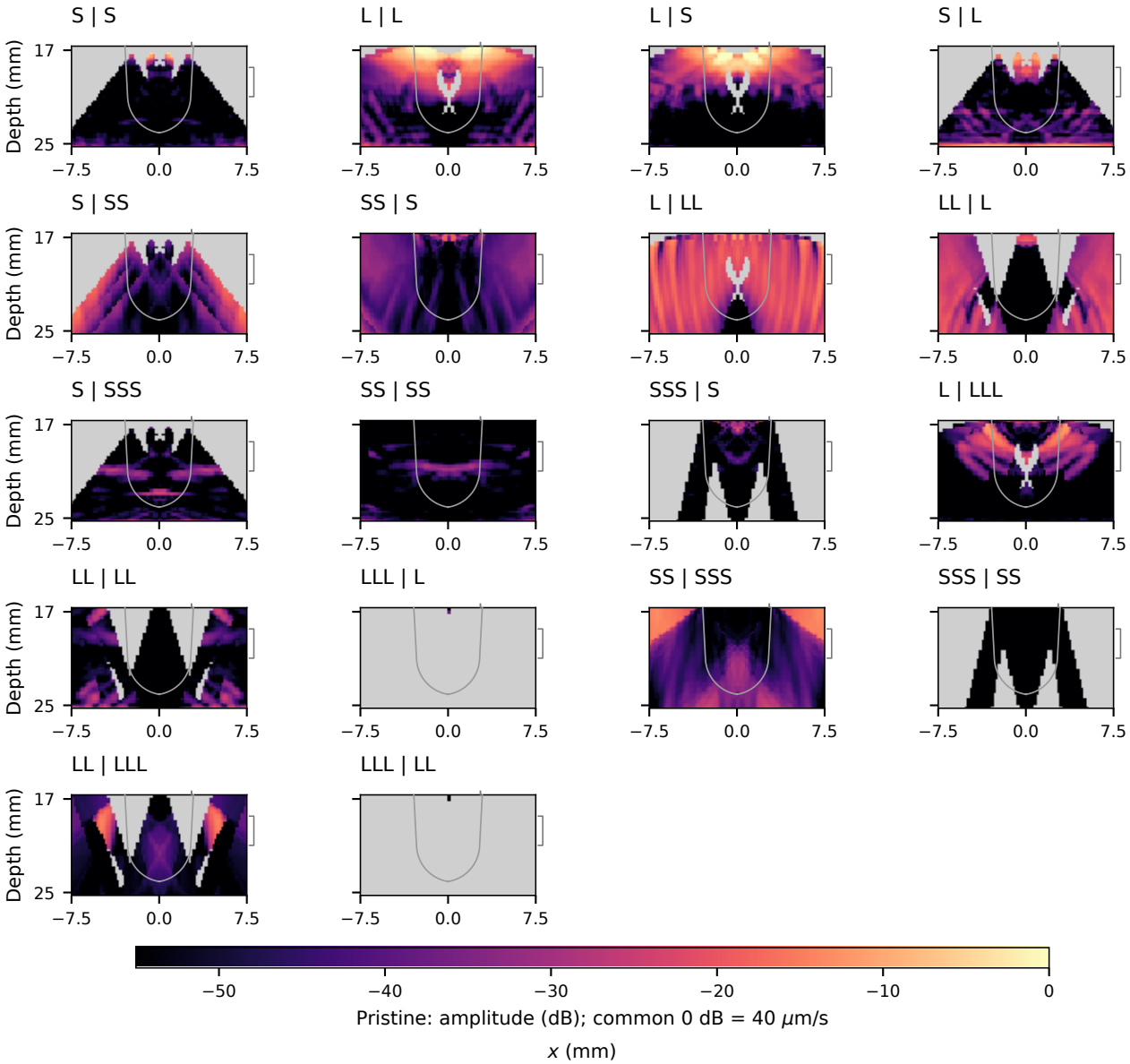


Figure A.19: All eighteen pristine-control images with the same controls, scale and grid as Figure A.18. The location marks are coordinate guides; these models contain no flaw. Structural response near the known LOF coordinates demonstrates why proximity alone is insufficient evidence.

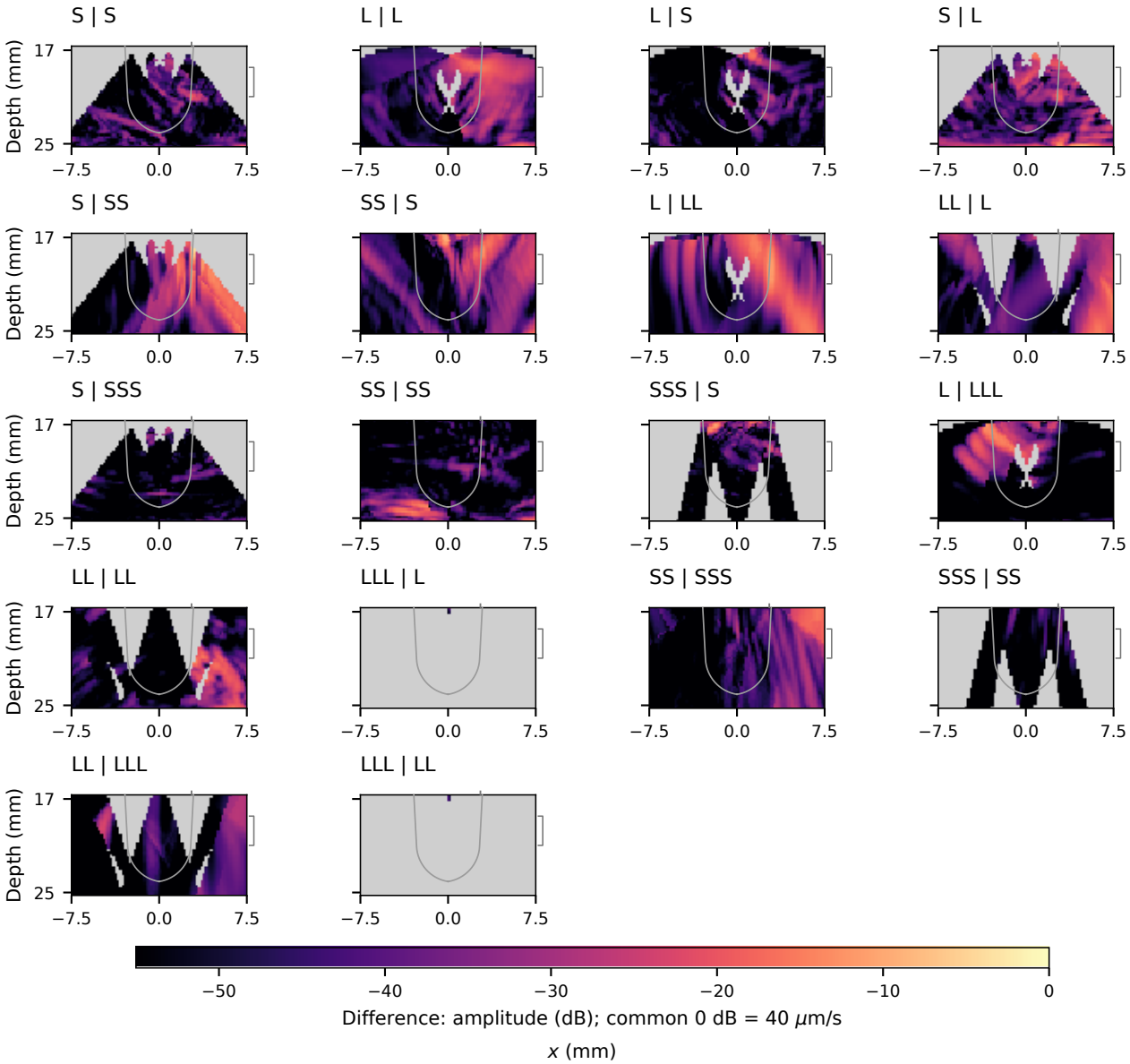


Figure A.20: All eighteen waveform-difference images on the same scale and grid. Signed waveforms are subtracted before nonlinear focusing. Changed reverberation and interference can contribute alongside direct flaw scattering. These are diagnostic controls, not the basis of the LOF-only screening procedure.

Table A.12: All eighteen primary partial-weld paths and three signal cases on the 0.2 mm survey grid. I_N is the near-region maximum in μm/s; distances are mm and contrasts dB. N is near support.

Path	Case	I_N	$C_{\max}$	C_{99}	N (%)	d_{pk}
$S \mid S$	LOF	1.74	-24.13	-7.86	95.2	2.362
$S \mid S$	Pristine	0.54	-33.23	-15.28	95.2	2.362
$S \mid S$	Difference	7.32	3.20	13.32	95.2	0.083
$L \mid L$	LOF	16.07	-10.45	-8.25	100.0	5.517
$L \mid L$	Pristine	13.32	-12.08	-9.88	100.0	1.456
$L \mid L$	Difference	6.46	-0.65	3.72	100.0	5.468
$L \mid S$	LOF	11.16	-15.99	-13.87	100.0	4.016
$L \mid S$	Pristine	10.89	-16.36	-14.36	100.0	1.839
$L \mid S$	Difference	2.11	-9.86	-0.95	100.0	1.736
$S \mid L$	LOF	4.13	-13.18	-10.99	95.2	4.324
$S \mid L$	Pristine	0.46	-32.08	-30.99	95.2	4.356
$S \mid L$	Difference	6.52	-7.42	-1.39	95.2	4.645
$S \mid SS$	LOF	13.34	3.65	6.90	95.2	0.038
$S \mid SS$	Pristine	3.04	-7.01	-5.78	95.2	9.868
$S \mid SS$	Difference	24.90	8.95	10.15	95.2	0.438
$SS \mid S$	LOF	3.20	-14.26	-5.10	100.0	4.151
$SS \mid S$	Pristine	0.42	-24.39	-16.89	100.0	4.698
$SS \mid S$	Difference	9.24	-8.07	3.45	100.0	3.972
$L \mid LL$	LOF	6.47	-8.89	-2.93	100.0	2.377
$L \mid LL$	Pristine	4.96	-13.61	-5.45	100.0	3.450
$L \mid LL$	Difference	12.46	4.17	5.11	100.0	0.049
$LL \mid L$	LOF	0.24	-29.18	-26.50	3.6	3.795
$LL \mid L$	Pristine	0.07	-40.03	-35.42	3.6	3.795
$LL \mid L$	Difference	0.40	-26.68	-24.48	3.6	5.314
$S \mid SSS$	LOF	0.92	-6.84	-2.50	95.2	6.807
$S \mid SSS$	Pristine	1.04	-10.34	-5.14	95.2	3.211
$S \mid SSS$	Difference	0.97	-6.05	0.25	95.2	2.412
$SS \mid SS$	LOF	1.59	-13.68	-8.06	100.0	7.287
$SS \mid SS$	Pristine	0.96	-8.90	-0.74	100.0	6.432
$SS \mid SS$	Difference	1.82	-12.75	-9.02	100.0	5.538
$SSS \mid S$	LOF	2.30	-21.57	-8.88	100.0	4.332
$SSS \mid S$	Pristine	0.10	-38.91	-29.62	100.0	1.867
$SSS \mid S$	Difference	4.37	-13.42	-6.02	100.0	2.253
$L \mid LLL$	LOF	9.48	-2.51	8.04	100.0	2.377
$L \mid LLL$	Pristine	9.75	-5.39	2.22	100.0	3.450
$L \mid LLL$	Difference	0.15	-35.86	-33.18	100.0	6.252
$LL \mid LL$	LOF	0.88	-16.36	-12.55	3.6	2.370
$LL \mid LL$	Pristine	0.01	-49.55	-46.53	3.6	8.378
$LL \mid LL$	Difference	7.49	0.84	3.79	3.6	0.522

Continued from previous page

Path	Case	I_N	$C_{\max}$	C_{99}	N (%)	d_{pk}
$LLL \mid L$	LOF	–	–	–	0.0	3.621
$LLL \mid L$	Pristine	–	–	–	0.0	3.621
$LLL \mid L$	Difference	–	–	–	0.0	3.117
$SS \mid SSS$	LOF	0.79	-20.47	-19.62	100.0	9.637
$SS \mid SSS$	Pristine	0.20	-32.90	-32.41	100.0	4.120
$SS \mid SSS$	Difference	1.01	-16.69	-15.22	100.0	4.120
$SSS \mid SS$	LOF	0.31	-3.53	-1.73	100.0	4.376
$SSS \mid SS$	Pristine	0.04	-6.18	-4.73	100.0	1.396
$SSS \mid SS$	Difference	0.71	-1.50	2.36	100.0	4.151
$LL \mid LLL$	LOF	0.04	-48.78	-46.00	3.6	7.670
$LL \mid LLL$	Pristine	0.03	-47.97	-46.34	3.6	2.096
$LL \mid LLL$	Difference	0.05	-36.01	-32.18	3.6	7.481
$LLL \mid LL$	LOF	–	–	–	0.0	3.257
$LLL \mid LL$	Pristine	–	–	–	0.0	3.450
$LLL \mid LL$	Difference	–	–	–	0.0	3.621

See the shared filter definitions and variants in Section A.5.1.

A.13.3 Complete filter/path comparison matrices

Figures A.21 and A.22 retain the full survey, including negative contrasts and unsupported routes. The common 0.2 mm grid isolates processing comparisons from a change in spatial sampling. The asterisk distinguishes a near-flaw global maximum from a positive contrast whose global maximum falls in the excluded guard region. The finer-grid conclusions appear in Section A.11; the complete confirmation register follows below.

On this survey grid, unrestricted analytic processing gives $S \mid SS$ 2.85 dB contrast, whereas signed unfiltered summation gives -0.14 dB and its global maximum lies outside the 0.5 mm near-flaw region. The five-leg $SSS \mid SS$ route gives weak positive contrasts of 1.59 and 1.18 dB for those respective controls. These outcomes show that changing the retained spectrum and phase representation can change the path ranking. They do not validate high-frequency propagation or establish a stronger alternative to the band-limited $S \mid SS$ result.

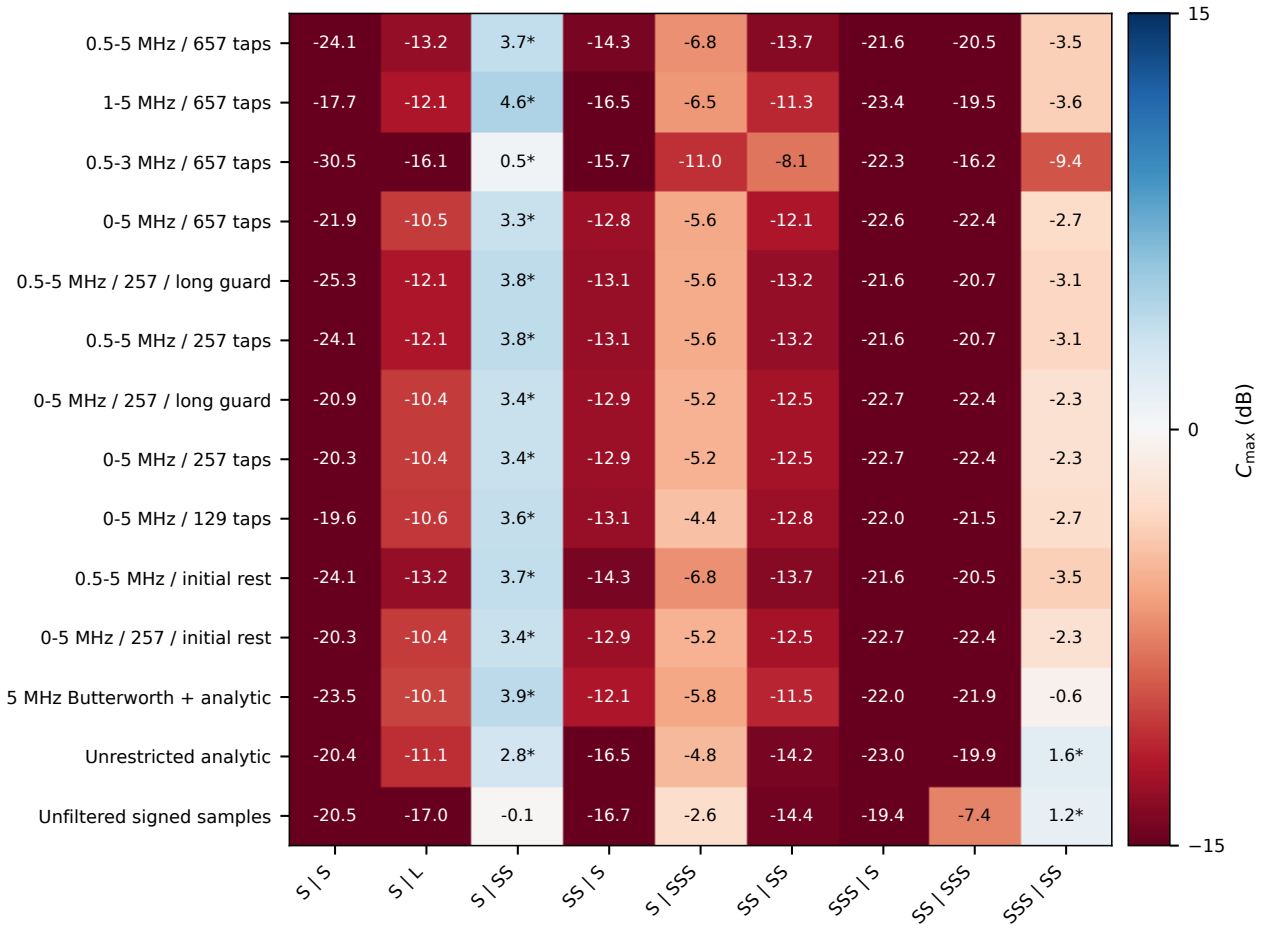


Figure A.21: All fourteen processing variants for incident-S partial-weld hypotheses at 0.2 mm spacing. Entries are LOF-only $C_{\max}$ in dB; an asterisk denotes a global maximum within 0.5 mm of the finite flaw. Colours saturate at ±15 dB, without truncating the printed values. Undefined contrast is shown by a dash.

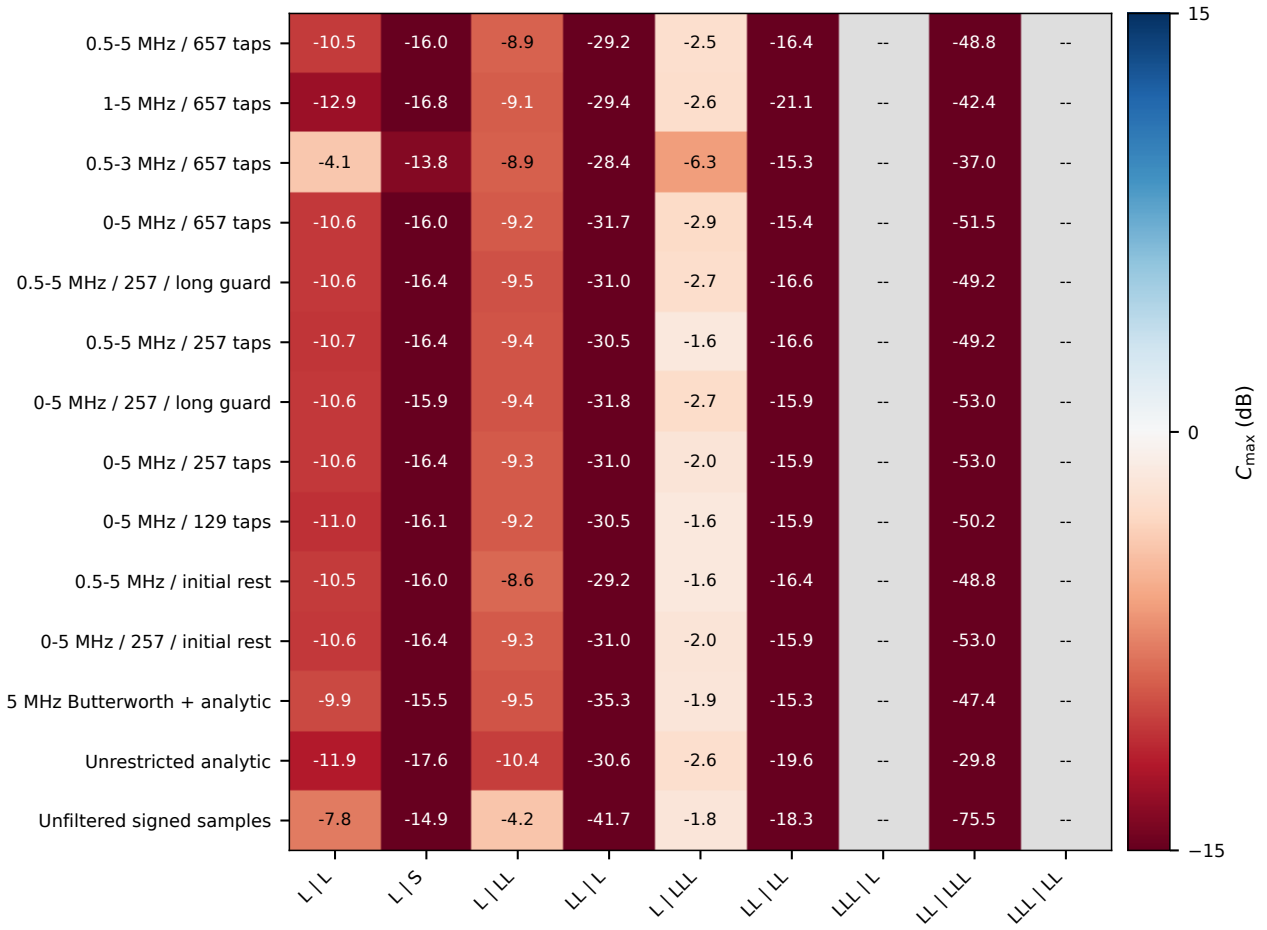


Figure A.22: Corresponding incident-L hypotheses under the same conventions. The signed unfiltered row changes signal representation as well as bandwidth; it is a diagnostic control rather than an envelope-equivalent reconstruction.

A.13.4 Additional bevel and corner hypotheses

Conventional imaging is geometry aware: actual endpoint heights set delays, finite exposed surfaces constrain reflections, and rays through the open groove are rejected. An additional catalogue tests 48 finite-bevel specular routes and 144 fixed-corner timing routes with equal receiver averaging at 0.2 mm spacing; six conventional controls use 0.1 mm spacing. Of the additional routes, 16 bevel and 60 corner paths have any supported pixels; only three bevel and 44 corner routes reach any near-flaw support. None places its global LOF-only maximum within 0.5 mm of the flaw or produces positive maximum-background contrast. Their failure therefore does not depend on ranking small differences against the finer-grid conventional control.

These negative outcomes describe the original equal-mean acquisition. The expanded weighted group survey in Section A.15 identifies useful corner-timing views under different controls, while retaining their calibration limitations. The extension used a unit corner-diffraction coefficient without a calibrated elastic amplitude or phase. Its negative result concerns those declared models, not every possible diffracted-wave method. Adding many geometric routes can create more opportunities to focus structural packets. Their inclusion in a colour blend would conceal this weakness unless each route's support and pristine control were first examined. The extension protocol, complete path metrics and verification summaries remain in the evidence archive.

A.13.5 Timing estimates and limits of direct-arrival screening

The LOF lies 2.0675–4.5675 mm below the floor. A floor-to-flaw echo may overlap local source motion or occur before the centred filter has complete support. For an illustrative colocated vertical pair, $T = t_0 + 2d/c_m$; the 3.28 μs guard gives $d_S \geq 5.262$ mm or $d_L \geq 9.658$ mm. The corresponding absolute depths are 21.662 and 26.058 mm. This is an explanation of the early-time problem, not a universal grid minimum: the actual protocol excludes colocated pairs and includes longer bistatic/reflected routes.

A floor–backwall round trip adds $2(9\text{ mm})/c_S = 5.559$ μs or $2(9\text{ mm})/c_L = 3.028$ μs. Such delays can move a desired echo into usable time support, but add reverberation and competing paths. Known zero prehistory supplies missing filter input before the initial time; it does not remove a physical source packet overlapping a flaw echo. These are distinct mechanisms.

Only 365 of the 1655 source–receiver pairs have unobstructed direct-family geometry for L/S/R nuisance screening: 2×157 shoulder/side-surface pairs and 3×17 weld-surface pairs. A Rayleigh guide requires a common unobstructed flat surface; it does not propagate a fictitious straight surface wave through the groove or around a corner. Unscreened reflections and diffraction can remain in every recording. Strong coherent background can survive receiver coherence and focus into arcs at pixels unrelated to the flaw.

A.13.6 Complete confirmation register

The following register includes all 28 confirmation configurations and their LOF, pristine and difference contrasts. Grid entries are spacing in mm; wide fields use $x = \pm 45$ mm. All other settings follow the named protocol and the stated acquisition reduction.

Table A.13: Complete partial-weld confirmations. BP denotes 0.5–5 MHz and LP 0–5 MHz; G328 retains the 3.28 μs guard. Rest uses known zero prehistory. L/P/D are LOF, pristine and difference $C_{\max}$ in dB; N is LOF near support. Src/rx denote source/receiver group reductions; weld and side identify the weld surface and side surfaces.

Filter	Path	Grid	Change	L	P	D	N (%)
1–5/657	$L \mid L$	0.1	all	-13.03	-14.27	-0.77	100.0
1–5/657	$S \mid S$	0.1	all	-16.88	-29.25	2.28	95.8
1–5/657	$S \mid SS$	0.1	all	4.20	-6.31	10.65	95.8
1–5/657	$SSS \mid SS$	0.1	all	-3.82	-6.04	-2.33	100.0
BP657	$L \mid L$	0.1	all	-10.60	-12.23	-0.65	100.0
BP657	$S \mid S$	0.1	all	-25.54	-35.30	2.54	95.8
BP657	$S \mid SS$	0.1	weld src	8.77	-9.91	8.08	48.5
BP657	$S \mid SS$	0.1	all	4.36	-6.87	11.09	95.8
BP657	$S \mid SS$	0.05	all	4.36	-6.48	11.21	95.8
BP657	$S \mid SS$	0.1	weld rx	-4.82	-2.68	1.38	95.8
BP657	$S \mid SS$	0.1	side rx	4.33	-7.27	11.34	95.8
BP657	$S \mid SS$	0.1	shoulder src	–	–	–	0.0
BP657	$S \mid SS$	0.1	shoulder+side	–	–	–	0.0
BP657	$S \mid SS$	0.25	wide	6.43	-6.40	10.81	94.3
BP657	$SSS \mid SS$	0.1	all	-4.02	-5.88	-0.94	100.0
BP257/G328	$L \mid L$	0.1	all	-10.79	-12.40	-0.52	100.0
BP257/G328	$S \mid S$	0.1	all	-27.18	-40.58	2.51	95.8
BP257/G328	$S \mid SS$	0.1	all	4.29	-6.58	11.18	95.8
BP257/G328	$SSS \mid SS$	0.1	all	-3.87	-6.62	-1.22	100.0
LP257/rest	$L \mid L$	0.1	all	-10.68	-11.79	-1.16	100.0
LP257/rest	$S \mid S$	0.1	all	-22.18	-47.73	3.08	95.8
LP257/rest	$S \mid SS$	0.1	all	4.34	-5.70	11.33	95.8
LP257/rest	$SSS \mid SS$	0.1	all	-2.72	-1.68	-1.07	100.0
LP257/G328	$L \mid L$	0.1	all	-10.68	-11.79	-1.40	100.0
LP257/G328	$S \mid S$	0.1	all	-22.69	-47.74	2.94	95.8
LP257/G328	$S \mid SS$	0.1	all	4.34	-5.70	11.33	95.8
LP257/G328	$S \mid SS$	0.05	all	4.32	-5.29	11.42	95.8
LP257/G328	$SSS \mid SS$	0.1	all	-2.72	-1.68	-1.07	100.0

A.13.7 Joint-acquisition path survey and confirmation register

The joint acquisition uses the three weld-surface sources and both side receiver groups for an eighteen-path survey at 0.2 mm spacing. Only $S \mid SS$ gives positive LOF-only maximum-background contrast with a global maximum inside the 0.5 mm near-flaw region. Its 7.63 dB contrast and 50.6% near support motivate the 0.1 mm receiving-side comparison. The left/right test is therefore conditional on this path; it is not an exhaustive search over every path-specific one-sided aperture.

The eighteen survey, three receiving-side and four finer/wider configurations form 25 scientific comparisons. One additional all-source native replay reproduces the existing 0.1 mm arrays in all three signal cases, with identical source counts and amplitude agreement within 2×10^{-9} relative tolerance (10^{-15} m s^{-1} absolute allowance). All 78 signal-case metrics are independently recomputed from the archived arrays. The protocol and loadable left-side settings accompany the evidence; no additional COMSOL solve is required.

Table A.14: All eighteen joint-acquisition paths at 0.2 mm spacing, with fixed common-bandpass processing. Contrasts are in dB; N is LOF-only near support (%). A dash denotes undefined contrast because a required supported region is absent.

Path	LOF	Pristine	Difference	N
$S \mid S$	-24.94	-50.90	1.77	50.6
$L \mid L$	-9.21	-10.83	-0.41	100.0
$L \mid S$	-14.74	-15.11	-8.10	100.0
$S \mid L$	-19.85	-40.11	-9.19	50.6
$S \mid SS$	7.63	-11.22	7.71	50.6
$SS \mid S$	-13.52	-15.50	0.72	86.7
$L \mid LL$	-7.64	-11.73	5.41	100.0
$LL \mid L$	–	–	–	0.0
$S \mid SSS$	-13.72	-34.14	-4.92	50.6
$SS \mid SS$	-38.18	-43.07	-37.83	86.7
$SSS \mid S$	–	–	–	0.0
$L \mid LLL$	-0.74	-3.58	-38.88	100.0
$LL \mid LL$	–	–	–	0.0
$LLL \mid L$	–	–	–	0.0
$SS \mid SSS$	-21.14	-34.18	-14.15	86.7
$SSS \mid SS$	–	–	–	0.0
$LL \mid LLL$	–	–	–	0.0
$LLL \mid LL$	–	–	–	0.0

Table A.15: Joint-acquisition confirmation controls for $S \mid SS$. Every row uses weld-surface generation. The first three rows use the standard 0.1 mm grid; "finer" is 0.05 mm and "wide" is $x = \pm 45$ mm, $z = 0.25$–25.25 mm at 0.25 mm. Contrast columns are in dB, and N is LOF-only near support (%).

Receiving aperture / grid	LOF	Pristine	Difference	N
Both sides / 0.1 mm	8.88	-10.36	7.95	48.5
Left side / 0.1 mm	9.96	-33.09	9.86	48.5
Right side / 0.1 mm	0.19	-8.53	-4.48	48.5
Left side / finer	10.26	-33.09	10.14	47.8
Left side / wide	9.11	-43.74	10.38	43.4
Both sides / finer	8.59	-10.60	7.99	47.8
Both sides / wide	9.20	-12.24	6.93	43.4

A.13.8 Complete one-source requirement controls

The source-requirement audit adds six scientific configurations and two two-source numerical replays, separate from the primary and joint-acquisition counts above. Each has three signal cases. All 24 paired checks confirm unchanged amplitudes and source counts on previously supported pixels, within 2×10^{-9} relative and 10^{-15} $\mathrm{m\,s^{-1}}$ absolute replay allowances. All newly supported pixels have exactly one qualifying source. The audit changes no filter, receiver threshold, gate or physical acquisition group within a pair. The reproducible protocol, arrays, metrics and loadable one-source

Table A.16: Complete one-source requirement controls. Standard grids use 0.1 mm, finer grids 0.05 mm, and wide fields 0.25 mm over $x = \pm 45$ mm and $z = 0.25$–25.25 mm. Contrasts use each image's full supported field; N is LOF-only near-flaw support.

Acquisition / field	LOF dB	Pristine dB	Difference dB	N %
All / standard	2.15	-10.11	9.38	100.0
Weld, both / standard	3.06	-7.40	11.24	95.8
Weld, left / standard	3.72	-37.53	10.02	95.8
Weld, left / finer	3.73	-36.85	10.02	95.8
Weld, left / wide	3.22	-42.78	9.50	94.3
Weld, both / wide	4.87	-8.49	10.60	94.3

left-side settings are retained in the source-requirement audit folder. The two-source settings remain available as the nominal comparison. No additional COMSOL solve is needed for this qualification change.

A.14 Partial weld: revised weighted result and grid checks

A.14.1 Reassessment: better contrast with broad near-flaw support

The preceding comparisons are the original equal-mean baseline. A broader review now tests response weighting, frequency band, angular and wall gates, source–receiver exclusion, direct and reflected screens, fixed mutes, coherence, receiver retention, source combination and acquisition groups. This appendix retains the complete comparisons. No new COMSOL solve is required: all changes reconstruct the same recorded wave fields.

For all five sources and all receivers, weighting alone improves the 0.1 mm common-band result from 4.36 to 8.05 dB with unchanged 95.8% near support. A 1–5 MHz band gives 8.59 dB. Lowering the angular gate from 20% to 15% recovers a small additional region without accepting the large background introduced at 10% or 5%. With the 15% gate and no fixed mute, the 0.025 mm result is 9.09 dB with 96.9% near support.

Adding only the narrow backwall-SS mute in Equation A.10 raises that matched-grid contrast to 14.45 dB while retaining 96.9% near support. The global peak is 58.64 μm/s at $(2.700, 19.625)$ mm, 0.071 mm from the LOF. The pristine near-region maximum is 2.53 μm/s and its global maximum is elsewhere. The gain is not simply a reduction of every background value: muting also changes phase cancellation and coherence, and the largest background rises from 10.73 to 11.11 μm/s. The reconstructed near peak rises from 30.53 to 58.64 μm/s. These remain processed velocity scores, not a doubling of physical flaw-scattered energy.

The SAFT grid matters to peak-based scores. The muted result is 15.77 dB at 0.05 mm, but 14.45

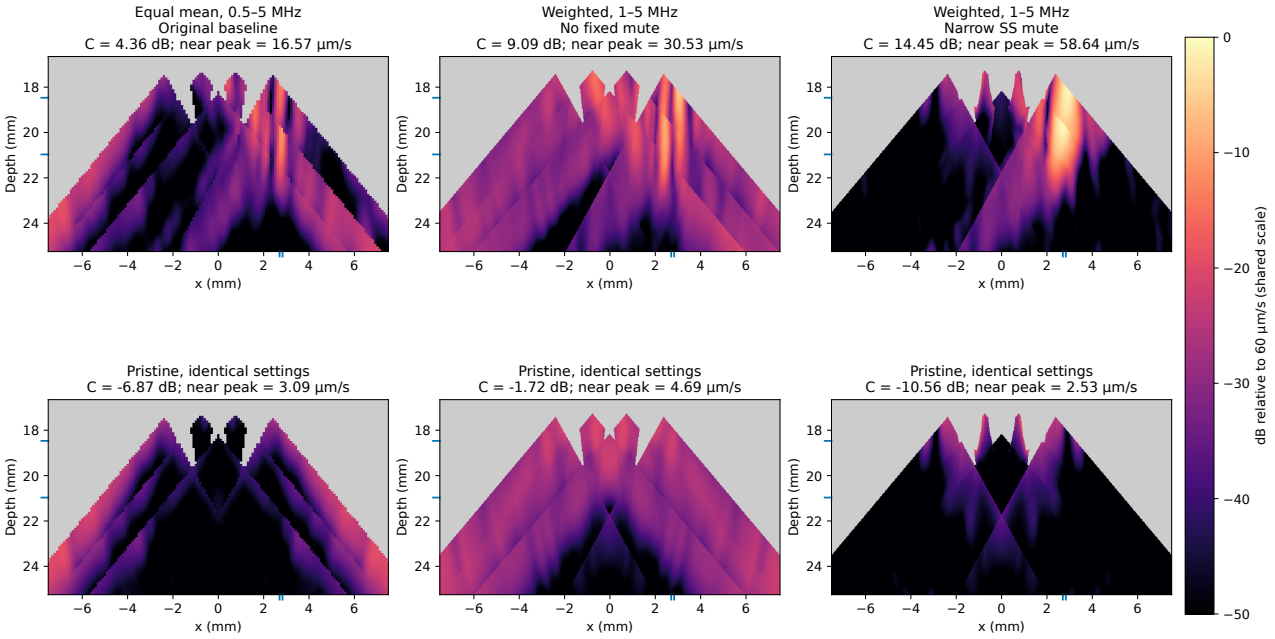


Figure A.23: Updated partial-weld $S \mid SS$ comparison. Columns show the original equal-mean baseline (0.1 mm), response weighting with 1–5 MHz and a 15% angular gate (0.025 mm), and the same weighted recipe with the narrow SS mute (0.025 mm). Rows show LOF-only and pristine controls on one absolute colour scale. Grey denotes unsupported material; external marks locate the prescribed segment. Only the last two columns form a matched-grid mute comparison.

dB at both 0.025 mm and a half-pixel-shifted 0.05 mm grid because those grids sample a stronger competing background peak. The more conservative 14.45 dB is therefore used here. The indication also survives the wider-field check, with its sampling stated in Table A.17. A finer image grid samples the focusing function more closely; it does not improve the finite-element solution or establish finer acoustic resolution.

Table A.17: Partial-weld grid checks. Local fields are $x = \pm 7.5$ mm and depths 16.65–25.25 mm; wide fields are $x = \pm 45$ mm and depths 0.25–25.25 mm. Entries state pixel spacing in mm. d_{pk} is the global-peak distance from the finite LOF. The shifted grid starts 0.025 mm later on both axes.

Recipe / grid	$C_{\max}$ (dB)	Near support (%)	d_{pk} (mm)
Unmuted, 0.025	9.09	96.9	0.291
SS mute, 0.025	14.45	96.9	0.071
Unmuted, shifted 0.05	9.07	96.9	0.310
SS mute, shifted 0.05	14.45	96.9	0.099
SS mute, 0.05	15.77	96.8	0.070
SS mute, wide 0.25	16.18	96.2	0.015
SS mute, wide 0.1	15.77	96.7	0.070

A.14.2 What limits support, and do right-side paths help?

The 25% rule applies to each source's geometrically and angularly eligible aperture, not to all 331 receivers. A 600-combination audit of receiver fraction, actual/effective count and source minimum shows that the receiver thresholds do not limit either original all-aperture $S \mid SS$ anchor. Their least retained actual and effective counts are 224 and 189.1. Lowering those thresholds cannot recover channels that already failed visibility or angular admission. The remaining support loss chiefly reflects source geometry and launch sensitivity. At the 20% gate, all 101 evaluated points on the

exact LOF segment already have two-source support; missing pixels in its 0.5 mm neighbourhood do not imply an equally missing portion of flaw length. Allowing only one qualifying source fills that neighbourhood but admits more background.

Right-side receivers do contain useful information. A fixed shot at the right weld position S3, $(1.5, 16.4)$ mm, with the 157 right receivers gives a useful $S \mid LL$ view: direct S illumination, conversion at the trial point and a backwall-reflected L return. At 0.05 mm it gives 3.02 dB unmuted, or 5.93 dB after a narrow predicted-LL mute, with 96.8% near support. This is a genuinely single-shot acquisition, unlike an image that permits one of several shots to qualify at each pixel. It is a complementary view rather than evidence that right reception is generally inferior to left reception.

Some longer paths look promising only in a small field. For example, left-weld S1 with right receivers and $SSL \mid SS$ gives 9.24 dB locally after raising the lower cutoff to 1.5 MHz, but -25.24 dB in the wider field, where its global maximum is 21.14 mm from the flaw. Rejecting full reflected-packet windows can instead remove all near support for $S \mid LL$. Useful local information, isolated physical-path identification and robust wide-field localization must therefore remain separate claims.

The extended review covers all 324 implemented path hypotheses, all seven nonempty physical receiver-group combinations and nine fixed source views. It also retains failures and pristine controls. Parameters were examined using this known flaw, so the revised recipe is a tested working choice for this dataset, not a validated optimum or a probability-of-detection result.

The wider search also finds useful reduced acquisitions beyond right-side reception. A single S3 shot with left receivers gives $S \mid SS$ 11.49 dB locally and 11.03 dB in the wider field, with approximately 96–97% near support. S1 with left and weld-surface receivers gives $SLS \mid S$ 15.80 dB locally and 4.01 dB wider, with 98.6% and 100% support respectively. These values use 0.05 and 0.25 mm grids. Conversely, the initially highest-scoring $SLS \mid L$ view is nearly reproduced by the pristine specimen. Path, aperture, grid and pristine controls therefore all affect which indications are credible.

A.15 Partial weld: expanded path and settings investigation

This section documents the review that changes the interpretation of the partial-weld result. The earlier equal-mean and reduced-aperture images remain valid baseline calculations. They do not exhaust the information in the recordings. In particular, the left-side preference was conditional on $S \mid SS$, its original filter and its selected source group. It cannot be generalized to every path or single weld-source shot.

A.15.1 Declared search scope

The broad path survey holds the weighted 1–5 MHz band, 15% angular gate, 5% wall gate, 1.5 mm source exclusion, direct L/S/R screens, CF exponent one and RMS source rule fixed. Its grid is 0.2 mm over $x = \pm 7.5$ mm and depths 16.65–25.25 mm. Three disjoint receiver groups are exported separately for each of 324 paths: 132 conventional mode sequences, 48 finite-bevel reflection hypotheses and 144 fixed-corner timing hypotheses. The 972 native configurations each contain LOF, pristine and waveform-difference cases.

For each source, disjoint-group sums Q_s, W_s, $\sum w_{sr}^2$, P_s and accepted/eligible counts are added *before* evaluating coherence and qualification. Averaging already formed group images would be

incorrect. All seven nonempty group combinations are evaluated with five single-source views, three weld sources requiring one or two, and all five sources requiring one or two. This gives 20,412 fixed acquisition/path views and 61,236 signal-case images. Every choice is fixed for the whole image; there is no pixelwise selection of the best path or brightest source.

The survey identifies 53 conventional paths (227 fixed views), 10 corner paths (61 fixed views), 0 wall paths (0 fixed views) with positive contrast, at least 20% near support and a global maximum within 0.5 mm of the flaw. These are local-field screening results; they are not all independently confirmed candidates. The complete register below and CSV expose the acquisition choices and coverage behind each score.

A.15.2 Following up the strongest path-survey indications

Fifteen fixed views receive native 0.05 mm local and 0.25 mm wider-field checks. Selection includes high local scores, broad support, clear LOF/pristine separation and deliberately retained pristine lookalikes. Table A.18 prevents the best coarse-grid score from becoming the conclusion without these checks.

Table A.18: All fifteen atlas follow-ups. Subscripts f/w mean 0.05 mm local / 0.25 mm wide field; C is contrast (dB), N near support (%), R_f the fine-grid LOF/pristine near-peak ratio (dB), and d_w the wider-field global-peak distance (mm). A high C with R_f near zero can be a pristine lookalike.

Path; acquisition	C_f	N_f	R_f	C_w	N_w	d_w
$SLS \mid L$; left/source1	15.83	76.3	0.36	1.58	73.6	0.50
$SLS \mid S$; left/source1	17.33	76.3	-4.69	-26.37	73.6	21.45
$LSS \mid LL$; left_weld/source1	21.04	37.0	1.43	-12.57	41.5	14.51
$LSS \mid L$; left_weld/source1	19.98	37.0	0.32	0.32	41.5	0.24
$SSS \mid L$; left/weld_min2	8.40	53.6	1.11	-6.47	52.8	34.95
$LS \mid SL$; sides/source4	5.54	100.0	0.70	15.27	100.0	0.05
$LSS \mid S$; weld/source1	17.10	29.7	28.31	14.26	35.8	0.08
$SLL \mid S$; weld/source1	9.38	57.3	17.78	10.12	60.4	0.21
$SLS \mid S$; left_weld/source1	15.80	98.6	28.48	4.01	100.0	0.08
$S \mid SS$; left/source3	11.49	96.8	50.09	11.03	96.2	0.27
$SL \mid SS$; sides/source4	7.27	100.0	15.94	3.60	100.0	0.53
$SS \mid S$; left_weld/source1	3.78	99.0	35.22	3.62	100.0	0.33
$SL \mid S$; left_weld/source4	7.15	100.0	23.77	3.74	100.0	0.18
Corner 5 L rx; right/source4	16.94	82.6	2.16	-33.97	77.4	34.87
Corner 4 S tx; left_weld/source0	6.19	100.0	22.82	7.90	100.0	0.18

The largest coarse-grid score, S1 with left receivers and $SLS \mid L$, drops from 31.76 to 15.83 dB on the finer grid. Its near-flaw amplitude is only 0.36 dB above the pristine control, and the wide-field peak lies just outside the near region. It is not persuasive evidence of a uniquely flaw-generated indication. Likewise, S4 with both-side receivers and $LS \mid SL$ has 100% near support but only 0.70 dB LOF/pristine separation on the fine grid. High localization contrast alone is insufficient when a similar response exists without a flaw.

There are useful alternatives. S3 with left receivers and unmuted $S \mid SS$ retains 11.49/11.03 dB in the fine/wide fields, approximately 96–97% near support and about 50 dB LOF/pristine near-peak separation. S1 with left and weld receivers and $SLS \mid S$ gives 15.80/4.01 dB and 98.6/100%

support. The same source/group combination with $SS \mid S$ gives 3.78/3.62 dB and 99.0/100% support. The latter views provide complementary information without declaring that the reconstructed delay uniquely identifies the scattered packet.

Some narrower views also survive the wider field: S1 with weld receivers gives $LSS \mid S$ 17.10/14.26 dB but only 29.7/35.8% near support. A corner-timing route from S0 through corner 4 gives 6.19/7.90 dB with complete near support, but retains an uncalibrated corner scattering coefficient. It warrants further physical modeling rather than replacement of the conventional working recipe. None of the finite-bevel routes meets the survey's combined positive-contrast, local-peak and 20% support criteria.

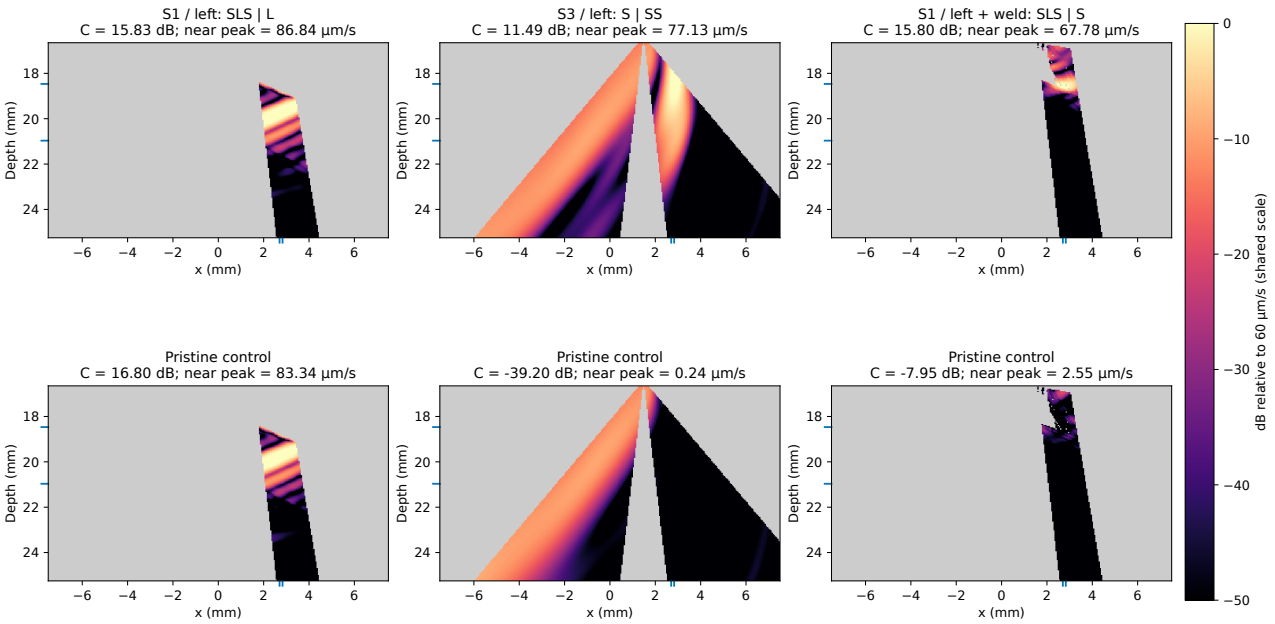


Figure A.24: Three follow-up examples at 0.05 mm: a high-scoring pristine lookalike ($SLS \mid L$, S1/left), a useful single-shot shear view ($S \mid SS$, S3/left), and a broader complementary path ($SLS \mid S$, S1/left+weld). LOF-only and pristine controls share one absolute scale. The complete follow-up table also reports wider-field failures.

The group recombination is checked against 16 independently requested native reconstructions (48 signal cases). The complete atlas also checks every original single-group image against its reconstructed statistics. The source-subset follow-up evaluates all 31 nonempty fixed subsets and every feasible source-count minimum, giving 80 views per signal case on the 0.05 mm candidate grid. A reduced set can retain a high near-flaw score while losing much of the total assessed field. The all-five-source candidate remains the principal working choice; the source-subset register exposes this coverage cost.

A.15.3 Settings matrix and interpretation

The 252 native settings configurations comprise 63 variants on four anchors: all-source/all-receiver $S \mid SS$; three weld sources with left receivers for $S \mid SS$; S3 with right receivers for $S \mid LL$; and S1 with right receivers for $SSL \mid SS$. The latter two require one source because only one shot is selected. Table A.19 states the controls covered. These are representative one-factor sensitivities plus all reflected-screen combinations, not every possible Cartesian combination of continuous settings or arbitrary receiver masks.

Table A.19: Controls examined in the partial-weld review. Baseline values are included where not repeated in the variation list.

Control	Values and purpose
Frequency band	0.5–3, 0.5–5, 1–3, 1–5, 1.5–5 and 2–5 MHz; historical fourteen-filter comparison retained separately.
Angular gate	0, 5, 10, 15, 20 and 30% launch/reception magnitudes.
Wall gate / exclusion	Wall response 0, 2.5, 5 and 10%; source–receiver exclusion 0, 0.75, 1.5, 3 and 5 mm.
Response / component	Equal or complex response-weighted mean; normal v_y, with tangential v_x as a diagnostic outside the assumed normal optical measurement.
Direct screens	All eight combinations of L, S and Rayleigh; tail levels 20, 30, 40 dB and margins 0, 0.05, 0.1, 0.2 μs.
Reflected screens	All 16 combinations of LL, LS/SL, SS and repeated SSSS packet screening.
Fixed arrival mutes	Each of R, L, S, LL, LS/SL and SS separately, and all together. Width checks include 0.1–0.8 and 1.2 μs.
Physical / relative aperture	All seven combinations of left, weld and right surfaces; receiver subsets relative to the trial pixel (left/right); interleaved even/odd receiver IDs.
Source choices	Five individual shots, weld and all-shot views; all 31 fixed subsets with every feasible source-count minimum for the candidate.
Retention guards	Fraction 0, 0.1, 0.25, 0.5, 1; actual floors 1, 4, 8, 16, 32; effective floors 0, 4, 8, 16, 32; relaxed guards and source-count/fixed-source policies.
Coherence / combination	CF powers 0, 0.5, 1, 2; RMS versus coherent source averaging. These alter the score, not the acquired acoustic information.
Record repairs / frozen aperture	Case-local and paired repair rejection; an aperture frozen at $(0, 20)$ mm, independent of the flaw coordinates. Paired repair metadata are a reference-assisted diagnostic.
Image grid / field	0.2, 0.1, 0.05, 0.025 mm; half-pixel shift at 0.05 mm; wider $x = \pm 45$ mm fields. Display reference and clipping never enter numerical metrics.

The full settings matrix below reports each native variation, including failures. Offline qualification and coherence variations reuse sufficient statistics; they are distinct from new propagation or filtering calculations. The evidence bundle retains their complete metrics, selections and checks. Thread count and colour-display controls do not define a different physical reconstruction and are not treated as tuning variables.

Table A.20: Complete native settings matrix at 0.2 mm: each cell is contrast in dB / near support in percent. Dashes indicate undefined contrast. Labels follow the reproducible protocol; mute0–5 correspond to R, L, S, LL, LS/SL and SS. Reflected bit strings use the stored order (least significant first): LL, LS/SL, SS, SSSS.

Variant	All $S \mid SS$	Left $S \mid SS$	S3 $S \mid LL$	S1 $SSL \mid SS$
baseline	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
band0.5_3	1.46 / 100	0.01 / 81	-1.78 / 96	-1.22 / 93
band0.5_5	8.46 / 95	7.78 / 54	2.47 / 95	3.45 / 100
band1_3	1.77 / 100	3.72 / 77	-1.37 / 96	-1.16 / 95
band1.5_5	8.72 / 95	11.68 / 54	1.16 / 95	8.95 / 100
band2_5	8.44 / 95	10.17 / 53	-3.52 / 95	4.48 / 100
angular0	2.13 / 100	5.14 / 100	1.35 / 100	0.79 / 100
angular0.05	2.13 / 100	5.07 / 100	1.35 / 99	0.79 / 100
angular0.1	0.86 / 100	5.05 / 98	1.35 / 99	1.10 / 100
angular0.2	8.80 / 95	10.13 / 51	3.03 / 95	8.04 / 75
angular0.3	8.64 / 93	9.67 / 22	5.76 / 69	– / 0
wall0	8.72 / 95	10.72 / 54	3.03 / 95	3.34 / 100
wall0.025	8.72 / 95	10.72 / 54	3.03 / 95	3.34 / 100
wall0.1	8.72 / 95	10.72 / 54	3.03 / 95	3.20 / 100
offset0	8.66 / 95	10.72 / 54	3.03 / 95	3.15 / 100
offset0.75	8.67 / 95	10.72 / 54	3.03 / 95	3.15 / 100
offset3	8.83 / 95	10.72 / 54	3.03 / 95	3.15 / 100
offset5	8.87 / 95	10.72 / 54	3.03 / 95	3.15 / 100
equal	4.45 / 95	11.93 / 54	-0.54 / 95	-10.47 / 100
tangential	-0.98 / 95	-8.62 / 54	-8.59 / 95	-23.91 / 100
screen_off	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
screen_L	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
screen_S	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
screen_R	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
screen_LS	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
screen_LR	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
screen_SR	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
reflect0001	8.83 / 95	10.72 / 54	– / 0	3.15 / 100
reflect0010	– / 0	0.72 / 5	– / 0	3.15 / 100
reflect0011	– / 0	0.72 / 5	– / 0	3.15 / 100
reflect0100	– / 0	– / 0	3.03 / 95	3.15 / 100
reflect0101	– / 0	– / 0	– / 0	3.15 / 100
reflect0110	– / 0	– / 0	– / 0	3.15 / 100
reflect0111	– / 0	– / 0	– / 0	3.15 / 100
reflect1000	8.72 / 95	10.72 / 54	3.03 / 95	3.30 / 100
reflect1001	8.83 / 95	10.72 / 54	– / 0	3.30 / 100
reflect1010	– / 0	0.72 / 5	– / 0	3.30 / 100
reflect1011	– / 0	0.72 / 5	– / 0	3.30 / 100
reflect1100	– / 0	– / 0	3.03 / 95	3.30 / 100
reflect1101	– / 0	– / 0	– / 0	3.30 / 100
reflect1110	– / 0	– / 0	– / 0	3.30 / 100
reflect1111	– / 0	– / 0	– / 0	3.30 / 100
tail20	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
tail40	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100

Variant	All $S \mid SS$	Left $S \mid SS$	S3 $S \mid LL$	S1 $SSL \mid SS$
margin0	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
margin0.05	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
margin0.2	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
mute0	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
mute1	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
mute2	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
mute3	8.72 / 95	10.72 / 54	-11.64 / 70	3.15 / 100
mute4	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
mute5	15.28 / 88	12.48 / 54	3.03 / 95	3.15 / 100
mute_all	15.28 / 88	12.48 / 54	-1.99 / 70	3.15 / 100
mute_all_width0.3	15.77 / 95	10.67 / 54	7.94 / 95	3.15 / 100
mute_all_width1.2	7.41 / 51	2.53 / 12	– / 0	3.15 / 100
aperture_left	10.04 / 95	10.72 / 54	– / 0	– / 0
aperture_right	-3.24 / 95	– / 0	3.03 / 95	3.15 / 100
aperture_even	8.78 / 95	10.66 / 54	2.84 / 95	5.70 / 100
aperture_odd	8.65 / 95	10.70 / 54	3.32 / 95	4.72 / 100
clean_case	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
clean_paired	8.72 / 95	10.72 / 54	3.03 / 95	3.15 / 100
frozen_centre	3.93 / 5	0.66 / 5	3.40 / 95	– / 0

The 25% retention threshold is an engineering guard, not a literature-derived detection criterion. In the original all-aperture anchors it is inactive, even when raised to 100% of the already eligible receivers. In contrast, broad reflected-pulse exclusion can make retention or time support restrictive. It is therefore incorrect either to blame the 25% rule for the original missing region or to assume it is irrelevant for every new path and screen. The receiver-fraction audit, source-count audit and angular audit answer different questions.

A.15.4 Why a short arrival mute helps, and what it does not prove

The fixed mute suppresses samples near a predicted specimen-only backwall arrival that could otherwise be focused under a flaw-scattering delay. It is applied to every trial pixel by its predicted time, without a flaw-location window. The result nevertheless depends on a width selected using the known example. Its improvement can involve less destructive cancellation and higher receiver coherence as well as reduced nuisance contributions. Accordingly, a larger output peak is not interpreted as greater measured energy or calibrated signal-to-noise ratio.

For all-source $S \mid SS$, widths 0.1–0.5 μs give about 14.8–15.8 dB on the 0.1 mm grid. Width 0.6 μs reduces near support to 87.9%, and 0.8 μs to 68.8%. The 0.3 μs choice therefore lies in a useful tested range rather than at a single isolated sharp optimum. Even/odd receiver subsets retain 15.65/15.78 dB with 96.7% near support. The narrower single-shot $S \mid LL$ view is more sensitive: widths 0.1–0.3 μs give 5.92 dB, while 0.6 μs gives -1.97 dB and only 70.6% support.

Increasing the coherence exponent to two also raises local contrast, but the updated principal recipe retains exponent one. A more nonlinear display score is not additional evidence of independent scattering. Similarly, increasing the lower frequency cutoff can improve local $SSL \mid SS$ contrast

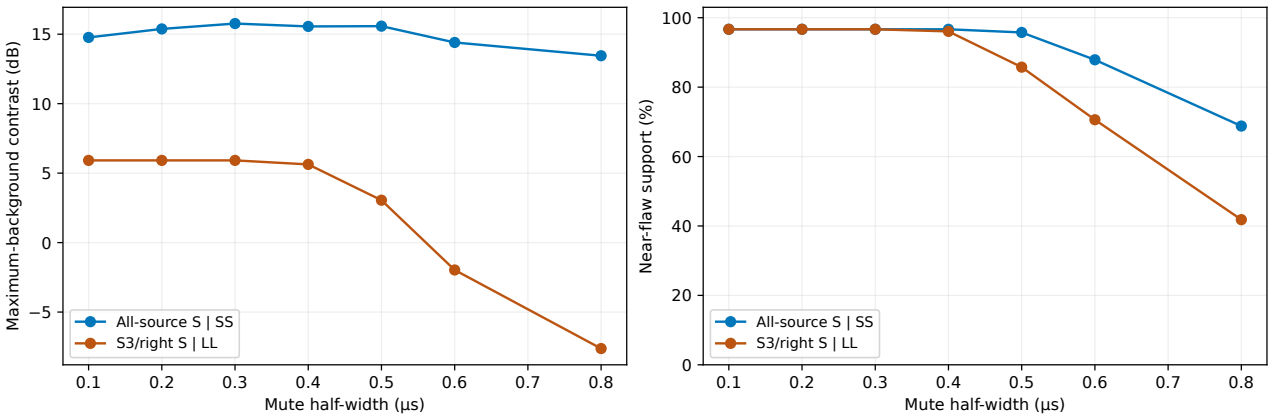


Figure A.25: Fixed-mute width sensitivity at 0.1 mm. These sweeps activate all six predicted-arrival mute families; their narrower 0.3 μs result is then simplified to SS-only for $S \mid SS$ and LL-only for the single-shot $S \mid LL$ view. The independent family checks and finer/wider images test that simplification. Support drops as the excluded interval expands. A smooth curve between sampled widths is not assumed.

while leaving a dominant remote false indication. These controls are useful precisely because they prevent every numerical contrast gain from being called an imaging improvement.

A.15.5 Single weld shots and right-side receivers

The 157 right receivers span approximately $x = 4.5534$–43.5534 mm. For S3 and $S \mid LL$, the retained signal is much stronger in absolute amplitude than the multi-bounce S1 $SSL \mid SS$ candidate. The earlier right-side survey includes all 132 conventional sequences and 20 short sequences on near (below 15 mm), middle (15–35 mm) and far (above 35 mm) right-receiver windows. All controls use the declared fixed window throughout the image.

The $S \mid LL$ near response is not cleanly separable from every predicted reflected packet. Activating the full measured reflected-pulse screens removes its near support. The short LL mute retains information by excluding less time, but cannot by itself prove an isolated flaw-converted wave. The correct conclusion is that this fixed acquisition contains useful defect-associated focusing information, supported by pristine/difference controls and a wider field. It is a candidate supplementary acquisition, not a demonstrated mode-pure detection channel.

A.15.6 Grid, source coverage and remaining uncertainty

The finest matched mute comparison uses 207,345 image pixels at 0.025 mm. The shifted 0.05 mm comparison gives 14.45 dB muted and 9.07 dB unmuted, consistent with the finer-grid conclusions. The change from 15.77 dB on the original 0.05 mm grid illustrates the sensitivity of a maximum-background statistic to narrow peaks and changing apertures. These checks do not claim complete asymptotic grid convergence. Wider-field sampling is stated separately, and it is not evidence of finite-element mesh convergence.

The principal two-source recipe supports only 3.76% of the entire 90 mm by 25 mm wider field at 0.1 mm. Its wider-field contrast therefore concerns assessed pixels, not verified inspection coverage across the plate. This is compatible with high support around the prescribed flaw and must not be hidden by cropping grey areas.

The known defect coordinates enter retrospective metrics and external figure guides, not travel-time calculation, channel gates, the frozen-aperture anchor or the fixed-mute centres. Nonetheless, select-

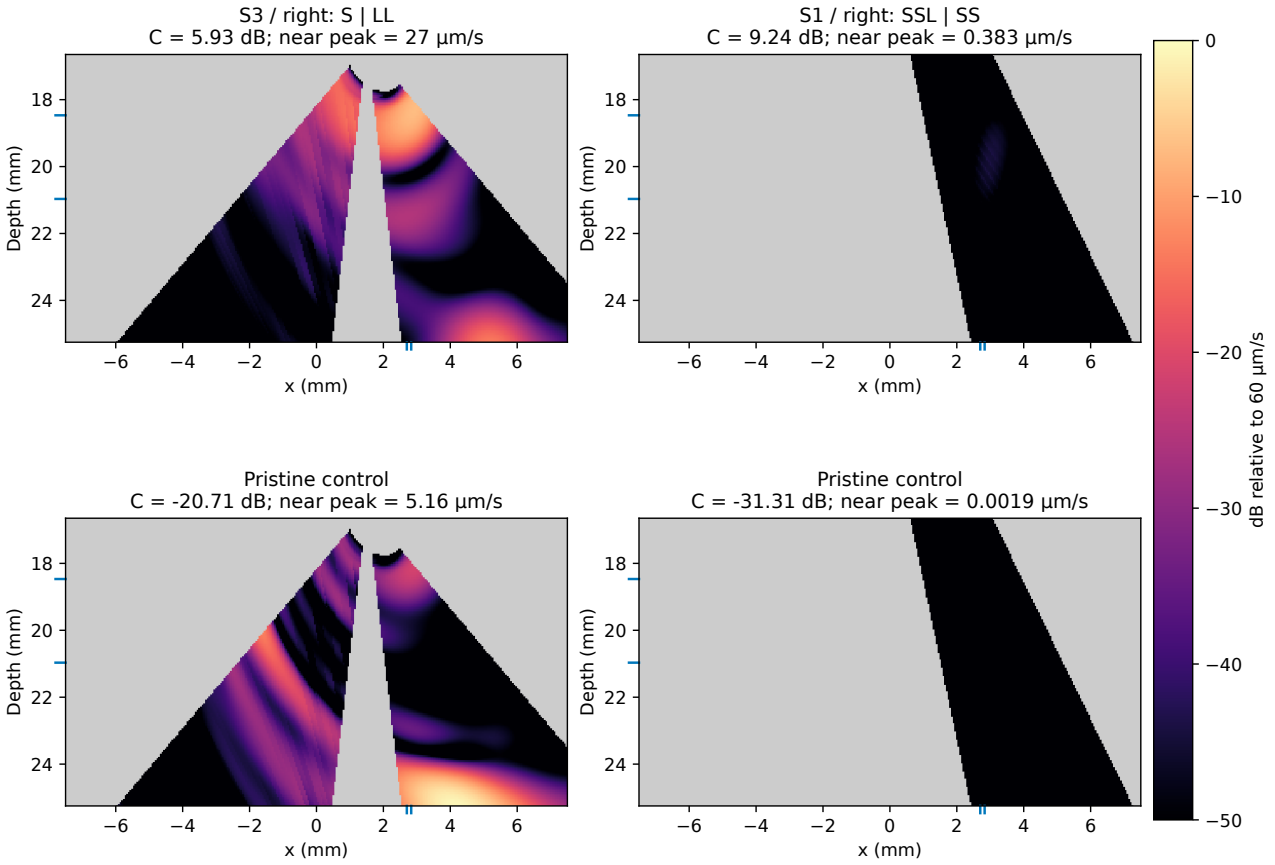


Figure A.26: Two fixed weld-source/right-receiver alternatives. The useful S3 $S \mid LL$ view uses 1–5 MHz and the narrow LL mute. S1 $SSL \mid SS$ uses 1.5–5 MHz without a fixed mute. LOF and pristine panels use the same absolute scale, making their very different amplitudes visible. Both are at 0.05 mm. Neither delay label uniquely identifies a physical packet in the mixed recordings.

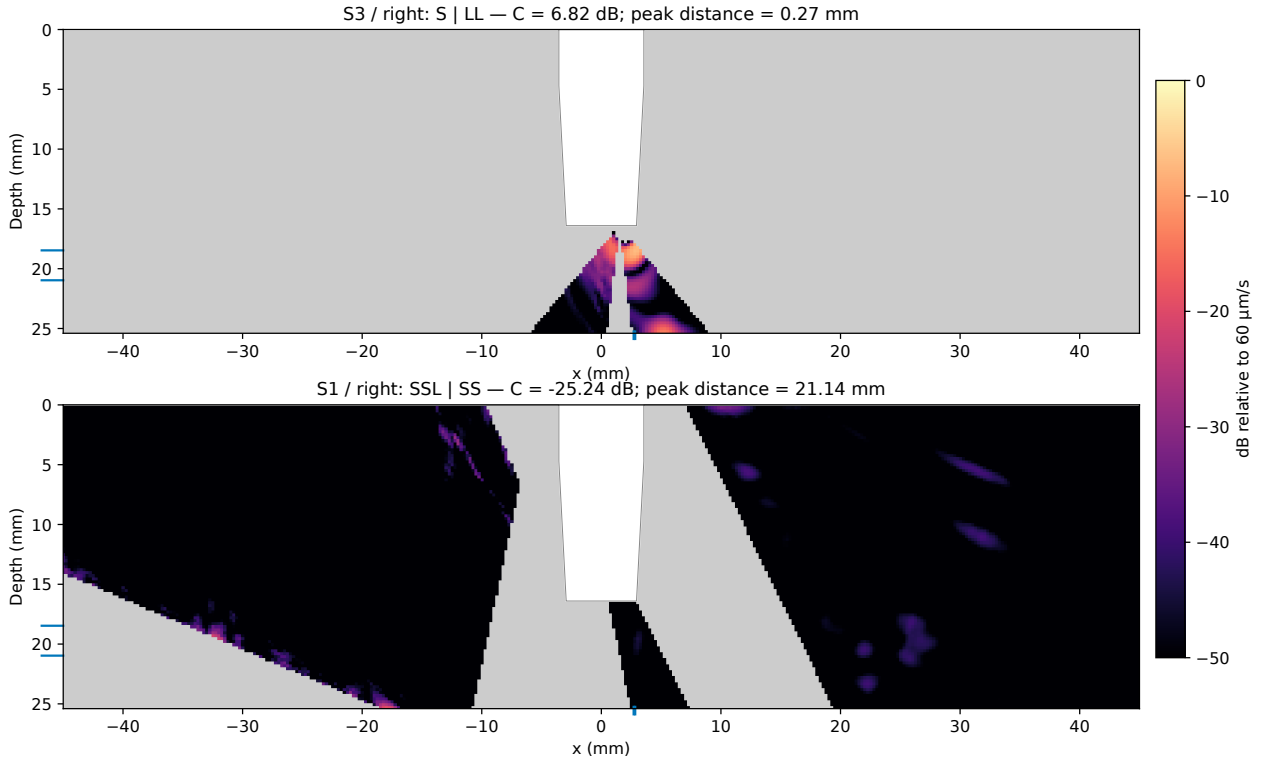


Figure A.27: The same alternatives on the wider field at 0.25 mm. The $S \mid LL$ indication survives with 6.82 dB contrast; the apparently improved local $SSL \mid SS$ image fails with −25.24 dB. Common absolute scaling is retained, rather than independently brightening the weaker route.

ing among many configurations using one known flaw introduces selection bias. Independent flaw locations, dimensions, noise, material variation and model convergence are still needed before choosing a robust instrument prescription. No probability of detection, reliable flaw sizing or experimental energy sensitivity follows from the current contrasts.

A.15.7 Complete path and source registers

The following compact path register gives the best observed fixed view per path among views with at least 20% near support. It is an exploratory summary, not a proposed per-pixel combination or a claim of optimal acquisition. A dash means no view meets that coverage floor. The machine-readable atlas retains all views, all three signal cases, amplitudes, support, pristine ratio and peak locations. Corner timing models retain uncalibrated corner amplitude and phase.

Table A.21: All 324 path hypotheses, ordered as in the native catalogue. Best fixed view with at least 20% near support: C in dB, N in percent and peak distance in mm. This retrospective ranking is not a per-pixel fusion rule. Exact identifiers and all unselected views are retained in atlas_metrics.csv.

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
1	$L \mid L$	right / source2	-5.54	100	1.65
2	$L \mid S$	sides / weld_min1	-13.27	100	1.84
3	$S \mid L$	all / all_min2	-11.60	95	4.32
4	$S \mid S$	sides / source4	-5.46	100	4.45
5	$LL \mid L$	right / source0	-4.43	100	4.40
6	$LL \mid S$	left / source4	-5.94	72	3.04
7	$LS \mid L$	left / weld_min1	4.39	75	0.11
8	$LS \mid S$	left_weld / source1	-0.93	99	0.91

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
9	$SL \mid L$	weld / source4	-0.17	100	2.36
10	$SL \mid S$	left / source1	14.18	75	0.52
11	$SS \mid L$	weld / weld_min1	-6.09	100	1.12
12	$SS \mid S$	left_weld / source1	7.45	99	0.33
13	$L \mid LL$	right / source3	-0.10	72	1.07
14	$L \mid LS$	weld / all_min1	1.47	99	0.33
15	$L \mid SL$	left / source4	-1.76	100	4.95
16	$L \mid SS$	all / source4	3.73	100	0.29
17	$S \mid LL$	weld_right / source3	3.42	95	0.23
18	$S \mid LS$	sides / source4	-3.21	100	4.30
19	$S \mid SL$	sides / source4	0.57	100	0.09
20	$S \mid SS$	left / source2	12.92	54	0.07
21	$LL \mid LL$	weld / source4	-0.81	83	4.29
22	$LL \mid LS$	weld / source4	6.55	82	0.44
23	$LL \mid SL$	weld / weld_min1	-1.83	51	3.03
24	$LL \mid SS$	weld / source4	1.79	83	0.15
25	$LS \mid LL$	weld / source1	-1.43	100	9.20
26	$LS \mid LS$	weld_right / source0	6.53	100	0.31
27	$LS \mid SL$	sides / source4	16.46	100	0.10
28	$LS \mid SS$	all / source4	1.51	100	0.51
29	$SL \mid LL$	weld / source1	-1.14	100	8.18
30	$SL \mid LS$	weld / source4	2.59	99	0.62
31	$SL \mid SL$	weld / weld_min1	-1.22	64	6.30
32	$SL \mid SS$	sides / source4	7.80	100	0.48
33	$SS \mid LL$	weld / source1	-3.16	100	8.00
34	$SS \mid LS$	weld / source4	-6.66	99	1.42
35	$SS \mid SL$	weld / weld_min1	-2.38	67	5.71
36	$SS \mid SS$	sides / source4	2.34	100	0.41
37	$L \mid LLL$	all / source2	4.02	100	0.48
38	$L \mid LLS$	left_weld / source4	-5.61	100	4.73
39	$L \mid LSL$	weld / source3	0.28	72	0.05
40	$L \mid LSS$	weld / source3	0.63	55	0.11
41	$L \mid SLL$	weld / weld_min2	4.53	34	0.10
42	$L \mid SLS$	sides / source2	-22.19	100	3.80
43	$L \mid SSL$	weld / weld_min2	0.77	41	0.10
44	$L \mid SSS$	weld / source4	2.88	82	0.49
45	$S \mid LLL$	left_weld / all_min2	3.29	95	0.28
46	$S \mid LLS$	sides / weld_min2	-26.48	54	5.12
47	$S \mid LSL$	weld / source3	5.15	75	0.11
48	$S \mid LSS$	weld / source3	4.44	51	0.53
49	$S \mid SLL$	weld / source3	0.84	92	0.23
50	$S \mid SLS$	all / all_min2	-11.50	95	4.32
51	$S \mid SSL$	weld / source3	-0.48	92	2.13
52	$S \mid SSS$	weld / source4	9.32	82	0.53
53	$LLL \mid L$	left / all_min1	-11.08	75	3.48
54	$LLL \mid S$	left / source0	0.11	75	0.52

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
55	$LLS \mid L$	left / source0	-16.53	75	3.48
56	$LLS \mid S$	left / source0	1.54	75	0.52
57	$LSL \mid L$	left / source0	19.16	75	0.52
58	$LSL \mid S$	left_weld / all_min1	5.41	100	0.49
59	$LSS \mid L$	left_weld / source1	20.30	36	0.13
60	$LSS \mid S$	weld / source1	16.17	30	0.08
61	$SLL \mid L$	left_weld / source1	0.42	89	0.99
62	$SLL \mid S$	weld / source1	15.58	57	0.50
63	$SLS \mid L$	left / source1	31.76	75	0.26
64	$SLS \mid S$	left / source1	23.25	75	0.11
65	$SSL \mid L$	left / weld_min2	3.83	20	0.91
66	$SSL \mid S$	left_weld / source2	11.67	23	0.10
67	$SSS \mid L$	left / weld_min2	18.61	52	0.11
68	$SSS \mid S$	left_weld / source2	5.57	54	0.09
69	$LL \mid LLL$	left / source4	-4.65	83	4.32
70	$LL \mid LLS$	left / source4	-3.63	83	3.78
71	$LL \mid LSL$	weld / source4	-0.55	51	3.92
72	$LL \mid LSS$	weld / source0	-2.47	55	1.13
73	$LL \mid SLL$	sides / source0	-2.56	100	1.22
74	$LL \mid SLS$	sides / source0	0.57	100	0.91
75	$LL \mid SSL$	weld / source1	5.52	31	0.28
76	$LL \mid SSS$	all / weld_min1	-8.77	59	4.35
77	$LS \mid LLL$	weld / source4	6.64	99	0.48
78	$LS \mid LLS$	left / source0	-1.78	100	1.28
79	$LS \mid LSL$	weld / source0	6.35	30	0.38
80	$LS \mid LSS$	weld / source4	0.30	55	0.70
81	$LS \mid SLL$	weld / source0	0.92	94	0.27
82	$LS \mid SLS$	weld_right / source0	-4.76	100	1.06
83	$LS \mid SSL$	left / source4	-0.06	100	1.54
84	$LS \mid SSS$	left / source0	0.42	100	0.66
85	$SL \mid LLL$	weld / source4	-0.01	99	1.05
86	$SL \mid LLS$	right / source4	-0.22	100	0.70
87	$SL \mid LSL$	right / source1	-0.09	100	2.19
88	$SL \mid LSS$	weld / source0	0.81	55	0.32
89	$SL \mid SLL$	weld / source4	-11.58	94	2.38
90	$SL \mid SLS$	right / weld_min2	-20.44	93	4.57
91	$SL \mid SSL$	weld / source4	0.76	95	0.67
92	$SL \mid SSS$	weld / source4	4.85	82	0.28
93	$SS \mid LLL$	weld / source4	-7.94	99	4.57
94	$SS \mid LLS$	right / source4	-23.96	100	4.32
95	$SS \mid LSL$	weld / all_min1	-8.42	33	8.38
96	$SS \mid LSS$	weld / source0	-2.66	55	6.48
97	$SS \mid SLL$	sides / source4	1.26	100	0.63
98	$SS \mid SLS$	right / source1	-3.46	100	9.29
99	$SS \mid SSL$	left / source0	-2.62	100	1.44
100	$SS \mid SSS$	sides / source4	-0.86	100	1.88

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
101	$LLL \mid LL$	left_weld / source0	2.62	100	0.91
102	$LLL \mid LS$	all / source0	4.16	100	0.89
103	$LLL \mid SL$	left / source0	-13.87	100	3.53
104	$LLL \mid SS$	left_weld / source0	-3.11	100	5.47
105	$LLS \mid LL$	left_weld / source0	-3.95	100	1.25
106	$LLS \mid LS$	left_weld / source0	0.01	100	0.86
107	$LLS \mid SL$	weld / source0	-5.87	100	3.62
108	$LLS \mid SS$	weld / source0	-2.71	100	4.73
109	$LSL \mid LL$	weld / all_min1	-2.89	100	5.66
110	$LSL \mid LS$	weld / source0	-1.06	99	1.69
111	$LSL \mid SL$	left / source0	-2.77	100	3.28
112	$LSL \mid SS$	weld / all_min1	-8.18	100	4.74
113	$LSS \mid LL$	left_weld / source1	21.37	36	0.49
114	$LSS \mid LS$	all / source1	11.45	36	0.08
115	$LSS \mid SL$	right / source1	-0.42	36	1.85
116	$LSS \mid SS$	sides / source1	1.96	36	0.67
117	$SLL \mid LL$	weld_right / source1	0.63	89	0.21
118	$SLL \mid LS$	all / source1	0.34	89	0.70
119	$SLL \mid SL$	weld / source1	-6.50	25	4.16
120	$SLL \mid SS$	left_weld / source0	0.92	100	0.23
121	$SLS \mid LL$	all / source1	-7.56	99	4.28
122	$SLS \mid LS$	sides / source1	5.36	99	0.08
123	$SLS \mid SL$	left / source0	2.00	100	0.51
124	$SLS \mid SS$	all / source1	2.08	99	0.29
125	$SSL \mid LL$	weld_right / weld_min2	-0.08	42	0.53
126	$SSL \mid LS$	weld / source0	2.28	99	0.33
127	$SSL \mid SL$	weld / source1	2.00	46	0.14
128	$SSL \mid SS$	weld_right / weld_min2	8.59	42	0.11
129	$SSS \mid LL$	weld / source1	-1.98	100	1.91
130	$SSS \mid LS$	weld / source0	2.86	99	0.63
131	$SSS \mid SL$	all / weld_min1	2.37	100	0.26
132	$SSS \mid SS$	weld / all_min1	1.85	100	0.33
133	`exp_corner_1_S_tx_direct_1`	–	–	–	–
134	`exp_corner_1_S_tx_direct_2`	–	–	–	–
135	`exp_corner_1_S_tx_id_1`	–	–	–	–
136	`exp_corner_1_S_tx_id_2`	–	–	–	–
137	`exp_corner_1_S_tx_pixel-id_1`	–	–	–	–
138	`exp_corner_1_S_tx_pixel-id_2`	–	–	–	–
139	`exp_corner_1_S_rx_direct_1`	–	–	–	–
140	`exp_corner_1_S_rx_direct_2`	–	–	–	–

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
141	exp_corner_1_S_rx_id_1	–	–	–	–
142	exp_corner_1_S_rx_id_2	–	–	–	–
143	exp_corner_1_S_rx_pixel-id_1	–	–	–	–
144	exp_corner_1_S_rx_pixel-id_2	–	–	–	–
145	exp_corner_1_L_tx_direct_1	–	–	–	–
146	exp_corner_1_L_tx_direct_2	–	–	–	–
147	exp_corner_1_L_tx_id_1	–	–	–	–
148	exp_corner_1_L_tx_id_2	–	–	–	–
149	exp_corner_1_L_tx_pixel-id_1	–	–	–	–
150	exp_corner_1_L_tx_pixel-id_2	–	–	–	–
151	exp_corner_1_L_rx_direct_1	–	–	–	–
152	exp_corner_1_L_rx_direct_2	–	–	–	–
153	exp_corner_1_L_rx_id_1	–	–	–	–
154	exp_corner_1_L_rx_id_2	–	–	–	–
155	exp_corner_1_L_rx_pixel-id_1	–	–	–	–
156	exp_corner_1_L_rx_pixel-id_2	–	–	–	–
157	exp_corner_2_S_tx_direct_1	–	–	–	–
158	exp_corner_2_S_tx_direct_2	–	–	–	–
159	exp_corner_2_S_tx_id_1	–	–	–	–
160	exp_corner_2_S_tx_id_2	–	–	–	–
161	exp_corner_2_S_tx_pixel-id_1	–	–	–	–
162	exp_corner_2_S_tx_pixel-id_2	–	–	–	–
163	exp_corner_2_S_rx_direct_1	–	–	–	–
164	exp_corner_2_S_rx_direct_2	–	–	–	–
165	exp_corner_2_S_rx_id_1	–	–	–	–
166	exp_corner_2_S_rx_id_2	–	–	–	–
167	exp_corner_2_S_rx_pixel-id_1	–	–	–	–
168	exp_corner_2_S_rx_pixel-id_2	–	–	–	–

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
169	exp_corner_2_L_tx_direct_1	–	–	–	–
170	exp_corner_2_L_tx_direct_2	–	–	–	–
171	exp_corner_2_L_tx_id_1	–	–	–	–
172	exp_corner_2_L_tx_id_2	–	–	–	–
173	exp_corner_2_L_tx_pixel-id_1	–	–	–	–
174	exp_corner_2_L_tx_pixel-id_2	–	–	–	–
175	exp_corner_2_L_rx_direct_1	–	–	–	–
176	exp_corner_2_L_rx_direct_2	–	–	–	–
177	exp_corner_2_L_rx_id_1	–	–	–	–
178	exp_corner_2_L_rx_id_2	–	–	–	–
179	exp_corner_2_L_rx_pixel-id_1	–	–	–	–
180	exp_corner_2_L_rx_pixel-id_2	–	–	–	–
181	exp_corner_3_S_tx_direct_1	left / source0	-13.03	75	3.16
182	exp_corner_3_S_tx_direct_2	right / source0	-2.36	100	5.82
183	exp_corner_3_S_tx_id_1	left / source4	-5.12	75	4.18
184	exp_corner_3_S_tx_id_2	weld / source4	-5.65	100	1.78
185	exp_corner_3_S_tx_pixel-id_1	left_weld / source0	-6.70	100	3.96
186	exp_corner_3_S_tx_pixel-id_2	sides / source0	-1.43	100	1.80
187	exp_corner_3_S_rx_direct_1	left / source4	-8.61	100	5.86
188	exp_corner_3_S_rx_direct_2	left / weld_min2	0.09	100	0.90
189	exp_corner_3_S_rx_id_1	weld / source4	-0.26	100	2.63
190	exp_corner_3_S_rx_id_2	weld / source0	-0.00	100	1.69
191	exp_corner_3_S_rx_pixel-id_1	left / source3	-12.46	95	3.27
192	exp_corner_3_S_rx_pixel-id_2	left / source4	-19.94	100	4.32
193	exp_corner_3_L_tx_direct_1	sides / source0	-5.82	100	2.10
194	exp_corner_3_L_tx_direct_2	right / source0	-26.48	100	4.24
195	exp_corner_3_L_tx_id_1	left / source4	-18.54	75	4.45
196	exp_corner_3_L_tx_id_2	sides / source4	-4.39	100	3.38

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
197	exp_corner_3_L_tx_pixel-id_1	right / source0	-4.45	100	4.40
198	exp_corner_3_L_tx_pixel-id_2	weld / source0	-18.70	100	5.25
199	exp_corner_3_L_rx_direct_1	left / source4	-0.32	100	4.73
200	exp_corner_3_L_rx_direct_2	left / all_min1	-0.39	100	1.15
201	exp_corner_3_L_rx_id_1	left_weld / source4	-0.01	100	0.92
202	exp_corner_3_L_rx_id_2	all / source4	0.00	83	0.79
203	exp_corner_3_L_rx_pixel-id_1	left / source2	-0.00	100	8.61
204	exp_corner_3_L_rx_pixel-id_2	left / source0	-0.02	100	1.50
205	exp_corner_4_S_tx_direct_1	sides / source4	-8.88	100	9.97
206	exp_corner_4_S_tx_direct_2	sides / source4	0.75	100	0.36
207	exp_corner_4_S_tx_id_1	left_weld / source0	6.76	100	0.28
208	exp_corner_4_S_tx_id_2	sides / source0	1.27	100	0.40
209	exp_corner_4_S_tx_pixel-id_1	left / source4	-2.28	75	1.93
210	exp_corner_4_S_tx_pixel-id_2	sides / source4	-2.20	100	6.46
211	exp_corner_4_S_rx_direct_1	right / source4	-0.03	100	1.24
212	exp_corner_4_S_rx_direct_2	right / source0	-0.00	100	5.87
213	exp_corner_4_S_rx_id_1	weld / all_min2	5.08	95	0.76
214	exp_corner_4_S_rx_id_2	left / source1	-0.00	100	2.94
215	exp_corner_4_S_rx_pixel-id_1	right / source2	-4.13	54	5.47
216	exp_corner_4_S_rx_pixel-id_2	right / source0	-3.01	100	1.33
217	exp_corner_4_L_tx_direct_1	left / source4	-13.90	75	5.35
218	exp_corner_4_L_tx_direct_2	left / source4	-0.12	100	5.23
219	exp_corner_4_L_tx_id_1	sides / source0	-20.81	100	7.54
220	exp_corner_4_L_tx_id_2	left / source0	-8.86	100	4.33
221	exp_corner_4_L_tx_pixel-id_1	left / source4	-1.51	75	4.96
222	exp_corner_4_L_tx_pixel-id_2	weld / source4	-3.18	100	2.38
223	exp_corner_4_L_rx_direct_1	right / source1	-1.61	100	2.52

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
224	exp_corner_4_L_rx_direct_2	right / all_min1	-3.97	100	5.27
225	exp_corner_4_L_rx_id_1	left / source2	0.00	100	0.88
226	exp_corner_4_L_rx_id_2	left_weld / source4	-0.01	83	4.65
227	exp_corner_4_L_rx_pixel-id_1	right / source2	-0.01	100	3.93
228	exp_corner_4_L_rx_pixel-id_2	right / source0	-0.00	100	2.60
229	exp_corner_5_S_tx_direct_1	–	–	–	–
230	exp_corner_5_S_tx_direct_2	–	–	–	–
231	exp_corner_5_S_tx_id_1	right / source4	3.90	49	0.11
232	exp_corner_5_S_tx_id_2	right / source4	-8.88	49	5.01
233	exp_corner_5_S_tx_pixel-id_1	–	–	–	–
234	exp_corner_5_S_tx_pixel-id_2	–	–	–	–
235	exp_corner_5_S_rx_direct_1	right / source4	-0.94	49	3.63
236	exp_corner_5_S_rx_direct_2	right / weld_min1	-0.22	49	1.22
237	exp_corner_5_S_rx_id_1	right / source4	-10.48	49	4.02
238	exp_corner_5_S_rx_id_2	right / source4	-2.33	49	4.28
239	exp_corner_5_S_rx_pixel-id_1	right / source2	-0.31	54	4.10
240	exp_corner_5_S_rx_pixel-id_2	right / source2	0.02	100	0.18
241	exp_corner_5_L_tx_direct_1	left / source4	-14.34	33	4.87
242	exp_corner_5_L_tx_direct_2	weld / source4	-5.66	49	1.71
243	exp_corner_5_L_tx_id_1	–	–	–	–
244	exp_corner_5_L_tx_id_2	–	–	–	–
245	exp_corner_5_L_tx_pixel-id_1	left / source4	-5.17	75	1.53
246	exp_corner_5_L_tx_pixel-id_2	left / source4	-9.16	100	5.01
247	exp_corner_5_L_rx_direct_1	right / source2	-0.01	49	1.07
248	exp_corner_5_L_rx_direct_2	right / source0	-0.08	49	1.76
249	exp_corner_5_L_rx_id_1	right / source1	-6.39	49	3.68
250	exp_corner_5_L_rx_id_2	right / all_min2	0.78	45	0.08
251	exp_corner_5_L_rx_pixel-id_1	right / weld_min1	-0.00	100	4.25

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
252	exp_corner_5_L_rx_pixel-id_2	right / source4	20.01	83	0.47
253	exp_corner_6_S_tx_direct_1	–	–	–	–
254	exp_corner_6_S_tx_direct_2	–	–	–	–
255	exp_corner_6_S_tx_id_1	sides / source4	-0.65	46	2.47
256	exp_corner_6_S_tx_id_2	sides / source4	0.92	46	0.16
257	exp_corner_6_S_tx_pixel-id_1	–	–	–	–
258	exp_corner_6_S_tx_pixel-id_2	–	–	–	–
259	exp_corner_6_S_rx_direct_1	–	–	–	–
260	exp_corner_6_S_rx_direct_2	–	–	–	–
261	exp_corner_6_S_rx_id_1	right / source3	-5.03	41	5.01
262	exp_corner_6_S_rx_id_2	right / source4	-1.50	46	3.49
263	exp_corner_6_S_rx_pixel-id_1	–	–	–	–
264	exp_corner_6_S_rx_pixel-id_2	–	–	–	–
265	exp_corner_6_L_tx_direct_1	–	–	–	–
266	exp_corner_6_L_tx_direct_2	–	–	–	–
267	exp_corner_6_L_tx_id_1	–	–	–	–
268	exp_corner_6_L_tx_id_2	–	–	–	–
269	exp_corner_6_L_tx_pixel-id_1	–	–	–	–
270	exp_corner_6_L_tx_pixel-id_2	–	–	–	–
271	exp_corner_6_L_rx_direct_1	–	–	–	–
272	exp_corner_6_L_rx_direct_2	–	–	–	–
273	exp_corner_6_L_rx_id_1	right / all_min1	-0.16	46	1.94
274	exp_corner_6_L_rx_id_2	right / source0	-0.03	46	1.25
275	exp_corner_6_L_rx_pixel-id_1	–	–	–	–
276	exp_corner_6_L_rx_pixel-id_2	–	–	–	–
277	exp_wall_2_S_tx_direct_1	–	–	–	–
278	exp_wall_2_S_tx_direct_2	–	–	–	–
279	exp_wall_2_S_tx_id_1	–	–	–	–
280	exp_wall_2_S_tx_id_2	–	–	–	–

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
281	exp_wall_2_S_tx_pixel-id_1	–	–	–	–
282	exp_wall_2_S_tx_pixel-id_2	–	–	–	–
283	exp_wall_2_S_rx_direct_1	–	–	–	–
284	exp_wall_2_S_rx_direct_2	–	–	–	–
285	exp_wall_2_S_rx_id_1	–	–	–	–
286	exp_wall_2_S_rx_id_2	–	–	–	–
287	exp_wall_2_S_rx_pixel-id_1	–	–	–	–
288	exp_wall_2_S_rx_pixel-id_2	–	–	–	–
289	exp_wall_2_L_tx_direct_1	–	–	–	–
290	exp_wall_2_L_tx_direct_2	–	–	–	–
291	exp_wall_2_L_tx_id_1	–	–	–	–
292	exp_wall_2_L_tx_id_2	–	–	–	–
293	exp_wall_2_L_tx_pixel-id_1	–	–	–	–
294	exp_wall_2_L_tx_pixel-id_2	–	–	–	–
295	exp_wall_2_L_rx_direct_1	–	–	–	–
296	exp_wall_2_L_rx_direct_2	–	–	–	–
297	exp_wall_2_L_rx_id_1	–	–	–	–
298	exp_wall_2_L_rx_id_2	–	–	–	–
299	exp_wall_2_L_rx_pixel-id_1	–	–	–	–
300	exp_wall_2_L_rx_pixel-id_2	–	–	–	–
301	exp_wall_4_S_tx_direct_1	–	–	–	–
302	exp_wall_4_S_tx_direct_2	–	–	–	–
303	exp_wall_4_S_tx_id_1	–	–	–	–
304	exp_wall_4_S_tx_id_2	–	–	–	–
305	exp_wall_4_S_tx_pixel-id_1	–	–	–	–
306	exp_wall_4_S_tx_pixel-id_2	–	–	–	–
307	exp_wall_4_S_rx_direct_1	–	–	–	–
308	exp_wall_4_S_rx_direct_2	–	–	–	–
309	exp_wall_4_S_rx_id_1	–	–	–	–
310	exp_wall_4_S_rx_id_2	–	–	–	–
311	exp_wall_4_S_rx_pixel-id_1	–	–	–	–
312	exp_wall_4_S_rx_pixel-id_2	–	–	–	–
313	exp_wall_4_L_tx_direct_1	–	–	–	–
314	exp_wall_4_L_tx_direct_2	–	–	–	–

No.	Path identifier	Receiver group / source view	C	N	d_{pk}
315	`exp_wall_4_L_tx_id_1`	–	–	–	–
316	`exp_wall_4_L_tx_id_2`	–	–	–	–
317	`exp_wall_4_L_tx_pixel-id_1`	–	–	–	–
318	`exp_wall_4_L_tx_pixel-id_2`	–	–	–	–
319	`exp_wall_4_L_rx_direct_1`	–	–	–	–
320	`exp_wall_4_L_rx_direct_2`	–	–	–	–
321	`exp_wall_4_L_rx_id_1`	–	–	–	–
322	`exp_wall_4_L_rx_id_2`	–	–	–	–
323	`exp_wall_4_L_rx_pixel-id_1`	–	–	–	–
324	`exp_wall_4_L_rx_pixel-id_2`	–	–	–	–

Table A.22: All 80 fixed source-subset/count views for the SS-mute candidate at 0.05 mm. Source IDs S0–S4 follow the geometry figure. Scores on this grid need not equal the finer 0.025 mm maximum-background score.

Fixed source IDs	Minimum	$C_{\max}$ (dB)	Near (%)	Whole field (%)
0	1	–	0.0	47.2
1	1	2.02	4.1	41.8
2	1	15.87	51.1	42.7
3	1	14.64	96.8	41.8
4	1	1.12	100.0	47.2
0,1	1	2.02	4.1	63.4
0,1	2	–	0.0	25.5
0,2	1	15.87	51.1	69.8
0,2	2	–	0.0	20.1
0,3	1	14.64	96.8	72.8
0,3	2	–	0.0	16.1
0,4	1	1.12	100.0	89.0
0,4	2	–	0.0	5.4
1,2	1	17.30	51.1	56.8
1,2	2	9.31	4.1	27.7
1,3	1	14.64	96.8	62.8
1,3	2	3.80	4.1	20.8
1,4	1	1.12	100.0	72.8
1,4	2	8.03	4.1	16.1
2,3	1	14.64	96.8	56.8
2,3	2	15.66	51.1	27.7
2,4	1	12.36	100.0	69.8
2,4	2	15.12	51.1	20.1
3,4	1	10.70	100.0	63.4
3,4	2	15.00	96.8	25.5
0,1,2	1	17.30	51.1	72.5
0,1,2	2	9.31	4.1	44.9

Fixed source IDs	Minimum	$C_{\max}$ (dB)	Near (%)	Whole field (%)
0,1,2	3	–	0.0	14.2
0,1,3	1	14.64	96.8	78.5
0,1,3	2	3.80	4.1	42.0
0,1,3	3	–	0.0	10.2
0,1,4	1	1.12	100.0	93.2
0,1,4	2	8.03	4.1	38.8
0,1,4	3	–	0.0	4.1
0,2,3	1	14.64	96.8	80.0
0,2,3	2	15.66	51.1	39.5
0,2,3	3	–	0.0	12.2
0,2,4	1	12.36	100.0	93.0
0,2,4	2	15.12	51.1	42.6
0,2,4	3	–	0.0	1.5
0,3,4	1	10.70	100.0	93.2
0,3,4	2	15.00	96.8	38.8
0,3,4	3	–	0.0	4.1
1,2,3	1	14.64	96.8	65.1
1,2,3	2	16.96	51.1	46.1
1,2,3	3	8.35	4.1	15.0
1,2,4	1	12.36	100.0	80.0
1,2,4	2	16.56	51.1	39.5
1,2,4	3	8.86	4.1	12.2
1,3,4	1	10.70	100.0	78.5
1,3,4	2	15.00	96.8	42.0
1,3,4	3	4.38	4.1	10.2
2,3,4	1	11.56	100.0	72.5
2,3,4	2	15.09	96.8	44.9
2,3,4	3	15.09	51.1	14.2
0,1,2,3	1	14.64	96.8	80.3
0,1,2,3	2	16.96	51.1	55.1
0,1,2,3	3	8.35	4.1	31.1
0,1,2,3	4	–	0.0	6.8
0,1,2,4	1	12.36	100.0	95.2
0,1,2,4	2	16.56	51.1	53.2
0,1,2,4	3	8.86	4.1	29.7
0,1,2,4	4	–	0.0	0.8
0,1,3,4	1	10.70	100.0	94.2
0,1,3,4	2	15.00	96.8	60.7
0,1,3,4	3	4.38	4.1	20.1
0,1,3,4	4	–	0.0	2.8
0,2,3,4	1	11.56	100.0	95.2
0,2,3,4	2	15.09	96.8	53.2
0,2,3,4	3	15.09	51.1	29.7
0,2,3,4	4	–	0.0	0.8
1,2,3,4	1	11.56	100.0	80.3
1,2,3,4	2	15.77	96.8	55.1

Fixed source IDs	Minimum	$C_{\max}$ (dB)	Near (%)	Whole field (%)
1,2,3,4	3	16.09	51.1	31.1
1,2,3,4	4	8.03	4.1	6.8
0,1,2,3,4	1	11.56	100.0	95.5
0,1,2,3,4	2	15.77	96.8	63.1
0,1,2,3,4	3	16.09	51.1	44.0
0,1,2,3,4	4	8.03	4.1	17.9
0,1,2,3,4	5	–	0.0	0.0

A.16 Electronic results, settings and reproduction

The electronic supplement accompanies the LaTeX chapter. It retains the complete numerical records behind the printed comparisons, rather than just successful images. Paths below are relative to the supplement's `evidence/` directory.

Table A.23: Electronic evidence supplied with this appendix. Each file is listed with its checksum in the package manifest.

Directory	Contents
`baseline/`	Full/partial dataset descriptions, common reconstruction protocol, selected recipes, acquisition comparisons and implementation checks.
`filter_study_20261101/`	Full filter-comparison metrics, exact filter evidence, admission guards and filter verification.
`gate_ablation_20261101/`	Angular, wall-response and source–receiver exclusion comparisons.
`partial_acquisition_20261101/`	Receiver-group and weld-source comparisons, including finer/wider checks.
`partial_source_requirement_20261101/`	One-/two-source comparisons, replay checks and the corresponding settings.
`partial_geometry_paths_20261030/`	Original bevel/corner protocols, catalogue, metrics and regression checks.
`thermoelastic_source_calibration/`	Source calibration comparison and independent thermal/mechanical checks.
`review/review/`	Full 324-path survey, settings matrices, confirmations, refinement, source subsets and final checks.
`review/earlier/`	The 25% retention/source audit and the dedicated right-receiver path investigation.

The main survey file is `review/review/atlas_metrics.csv`. It contains all 20,412 fixed acquisition/path rows and the metrics for all three signal cases. `review/review/protocol.json` defines the path catalogue, source views, receiver groups and fixed controls. The settings review contains both native and offline metrics; offline group recombination sums the underlying channel statistics before applying coherence and source qualification.

The principal updated partial-weld settings are supplied as

- `review/review/settings/saft_partial_weighted_SS_mute_025.json`: the reported 0.025 mm image;
- `review/review/settings/saft_partial_weighted_SS_mute_05.json`: the quicker aligned 0.05 mm image;
- `review/review/settings/saft_partial_right_S_LL_mute_05.json`: the complementary S3/right-receiver view.

The 0.05 mm principal preset gives 15.77 dB on its aligned grid. It should not be substituted silently for the more conservative 14.45 dB fine/shifted-grid result. Saved settings retain their original dataset fingerprints so that a different recording set is not mistaken for the reported one.

The document can be compiled from its supplied figures without COMSOL. Recomputing the wave fields requires COMSOL and the model inputs; repeating the reconstruction requires the corresponding receiver recordings and analysis code. Large raw histories and reconstruction arrays are not embedded in the appendix's portable document bundle. The accompanying clean simulation repository identifies the required inputs, generated results and reproduction commands separately.

A.16.1 Checks completed and limits of the evidence

The source calibration, native/independent reconstruction checks, group recombination checks and finer/wider image tests address different questions. None substitutes for production-mesh, internal time-step or absorbing-boundary convergence. The finite-element design frequency is a mesh setting, and the recorded sampling rate is an output property; neither alone establishes an accurate imaging bandwidth.

At fixed duration and approximately fixed elements per wavelength, a uniformly refined two-dimensional explicit calculation has a useful work estimate proportional to f^3: roughly f^2 more spatial unknowns and f more stable time steps. This is a scaling estimate, not a measured runtime law for these production models. Local refinement, minimum element size, solver overhead and hardware can change the actual factor.

The printed and electronic results describe the tested configurations, including failures. They are not an exhaustive search of every continuous parameter or arbitrary receiver subset. The known flaw location enters retrospective metrics and display guides. Reusing it to select among settings limits generalization, even when the focusing equations themselves do not use that location. Independent flaws, material variation and measurement noise are required before claiming detection probability, reliable flaw sizing or experimental sensitivity.

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